Program overview

19-May-2025 12:07

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Schakelprogramma's EWI 2024	**Bridgingprogram EWI 2024**		
Bridging programme AM for TUD students 2024			
Programme AM for TUD BSc CT/AES/WB/MT/LR			
AM2020	Optimization	6	
AM2080	Introduction to Statistics	6	
TW1-11TWN	Problem Solving and Proofs	3	
TW1-22	Analysis 1	5	
Choose			
AM2040	Complex Function Theory	6	
TW1-31	Discrete Mathematics	5	
Choose			
AM2070	Partial Differential Equations	6	
WI3150TU	Partial Differential Equations A	3	
WI3151TU	Partial Differential Equations B	3	
Programme AM for TUD BSc EE/TN/NB			
AM2020	Optimization	6	
AM2060	Numerical Methods 1	6	
AM2080	Introduction to Statistics	6	
TW1-11TWN	Problem Solving and Proofs	3	
TW1-22	Analysis 1	5	
Choose			
AM2040	Complex Function Theory	6	
TW1-31	Discrete Mathematics	5	
Choose			
AM2070	Partial Differential Equations	6	
WI3150TU	Partial Differential Equations A	3	
WI3151TU	Partial Differential Equations B	3	
Bridging programme AM for BSc Econometrics 2024			
AM2060	Numerical Methods 1	6	
TW1-11TWN	Problem Solving and Proofs	3	
TW1-22	Analysis 1	5	
Choose			
AM2040	Complex Function Theory	6	
TW1-31	Discrete Mathematics	5	
Choose			
AM2070	Partial Differential Equations	6	
WI3150TU	Partial Differential Equations A	3	
WI3151TU	Partial Differential Equations B	3	
Bridging programme CESE for HBO TI 2024			
CSE1110	Software Quality and Testing	5	
CSE1205	Linear Algebra	5	
CSE1300	Reasoning and Logic	5	
CSE2310	Algorithm Design	5	
CSE2315	Automata, Computability and Complexity	5	
CSE2420	Digital Systems	5	
CSE2430	Operating Systems	5	
EE3D11	Computer Architecture and Organisation	5	
IFEEMCS010500	Probability and Statistics	3	
IFEEMCS011100	Calculus for Science, part 1	3	
IFEEMCS011200	Calculus for Science, part 2	3	
Bridging programme CESE for HBO EE 2024			
CSE1100	Introduction to Programming	5	
CSE1405	Computer Networks	5	
EE2S2	Systems and Control	5	
EE3D11	Computer Architecture and Organisation	5	
IFEEMCS010400	Linear Algebra	5	
IFEEMCS010500	Probability and Statistics	3	
IFEEMCS011100	Calculus for Science, part 1	3	
IFEEMCS011200	Calculus for Science, part 2	3	
TN2545	Systems and Signals	6	
WI1909TH	Differential Equations	3	

Bridging programme CESE for TUD BSc ME/AE/AP 2024

CSE2420	Digital Systems	5
CSE2425	Embedded Software	5
EE3115TU	Digital Communication Systems	4
EE3125TU	Advanced Electronics for Robotics	5
EE3130TU	Marsrover project	5
ET3033TU	Circuit Analysis	3
ET3604LR	Electronic Circuits	3

Bridging programme CS for HBO TI 2024

CSE1110	Software Quality and Testing	5
CSE1200	Calculus	5
CSE1205	Linear Algebra	5
CSE1210	Probability Theory and Statistics	5
CSE1300	Reasoning and Logic	5
CSE1505	Information and Data Management	5
CSE2310	Algorithm Design	5
CSE2315	Automata, Computability and Complexity	5
CSE2510	Machine Learning	5

Bridging programme CS for WO TUD BSc TW and EE 2024

CSE1100	Introduction to Programming	5
CSE1105	Collaborative Software Engineering Project	5
CSE1110	Software Quality and Testing	5
CSE1305	Algorithms and Data Structures	5
CSE1405	Computer Networks	5
CSE1500	Web and Database Technology	5
CSE1505	Information and Data Management	5
CSE2115	Software Engineering Methods	5
CSE2310	Algorithm Design	5
CSE2315	Automata, Computability and Complexity	5
CSE2510	Machine Learning	5

Bridging Programme DSAIT for WO MSc 2024

CSE1100	Introduction to Programming	5
CSE1105	Collaborative Software Engineering Project	5
CSE1110	Software Quality and Testing	5
CSE1400	Computer Organisation	5
CSE1405	Computer Networks	5
CSE1505	Information and Data Management	5
CSE2115	Software Engineering Methods	5
CSE2310	Algorithm Design	5
CSE2315	Automata, Computability and Complexity	5

Bridging Programme DSAIT for WO BSc TW and EE 2024

CSE1100	Introduction to Programming	5
CSE1105	Collaborative Software Engineering Project	5
CSE1110	Software Quality and Testing	5
CSE1305	Algorithms and Data Structures	5
CSE1405	Computer Networks	5
CSE1500	Web and Database Technology	5
CSE1505	Information and Data Management	5
CSE2115	Software Engineering Methods	5
CSE2310	Algorithm Design	5
CSE2315	Automata, Computability and Complexity	5
CSE2510	Machine Learning	5

Bridging programme EE for HBO EE 2024

EE1E1	Electrical Energy Fundamentals	5
EE1M1	Calculus	5
EE1M2	Calculus and Linear Algebra	5
EE1M3	Linear Algebra and Differential Equations	5
EE1P1	Electricity and Magnetism	5
EE2C1	Transistor Circuits	5
EE2M1	Probability and statistics	5
EE2T1	Telecommunication and Sensing	5
WB2235	Signal Analysis	6

Bridging programme EE for TUD BSc ME/AE/AP 2024

CSE2420	Digital Systems	5
EE3115TU	Digital Communication Systems	4
EE3120TU	Guided and Wireless EM Transfer	5
EE3125TU	Advanced Electronics for Robotics	5
EE3130TU	Marsrover project	5
ET3033TU	Circuit Analysis	3
ET3604LR	Electronic Circuits	3

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Schakelprogramma's EWI 2024	
Introduction 1	More information on the following webpage: https://www.tudelft.nl/studenten/faculteiten/ewi-studentenportal/onderwijs/studeren-bij-ewi/schakelprogrammas-ewi/

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging programme AM for TUD students 2024

Introduction 1	The AM bridging programme for students with a B.Sc. degree in Aerospace Engineering, Applied Earth Sciences, Applied Physics, Civil Engineering, Electrical Engineering, Marine Technology, Mechanical Engineering, or Nanobiology consists of the courses listed below.

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Programme AM for TUD BSc CT/AES/WB/MT/LR

AM2020	Optimization	6
Responsible Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	0/4/0/0 hc 0/4/0/0 wc	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear Algebra 1 (AM1030), Analysis 2 (AM1070), Mathematical Structures (AM1010)	
Course Contents	Introduction to optimization, convexity. Modelling (Mixed) Integer Linear Programming problems. Linear programming: Simplex method, duality. Network optimization: shortest paths, max flow. Integer linear programming: branch & bound, Gomory cutting planes. Nonlinear optimization: Newton method, steepest descent. Complexity: NP-hardness. Approximation algorithms.	
Study Goals	The students are able to formulate discrete optimization problems as (mixed integer) linear programming problems. The students are able to execute basic algorithms for solving linear programming, integer linear programming and nonlinear programming problems, to understand the outcomes of the algorithms and to apply duality. The students know basic theorems of linear and nonlinear optimization and are able to prove variations of these theorems. The students can apply basic algorithms for network optimization problems (shortest path, max flow, travelling salesman problem). The students are able to design reductions between optimization problems and to prove correctness of those reductions.	
Education Method	4 hours lecture and 4 hours of exercise class per week, plus feedback exercises and homework exercises	
Literature and Study Materials	Lecture notes	
Assessment	Written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	3 hours	
Permitted Materials during Tests	A non-graphical calculator	
Judgement	The exam grade is the final grade	
Co-Instructor	Dr.ir. L.J.J. van Iersel	

AM2080	Introduction to Statistics	6
Responsible Instructor	Prof.dr.ir. G. Jongbloed	
Instructor	Dr. A.F.F. Derumigny	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Introduction to Probability (first year probability course)	
Course Contents	Statistical Models; Descriptive Statistics, Estimation theory; Testing theory; Confidence regions; Regression	
Study Goals	By the end of this course: 1. You know the concept of statistical model and you are able to formulate such a model for standard datasets. 2. You know several numerical and graphical summaries of a dataset and their relation with the probability distribution in the statistical model for the corresponding dataset and you are able to produce them with the statistical package R. 3. You know the concepts of Mean Squared Error and unbiasedness for an estimator and you are able to determine them for simple estimators. 4. You know the Maximum Likelihood, Method of Moments, and Bayesian approach towards estimation and you are able to construct estimators according to these principles and compare their performance either by hand or by means of a simulation with the package R. 5. You know the basic concepts involved in hypothesis testing and you are able to setup and conduct a statistical test in simple statistical models. 6. You know the setup of the Student-t test for one sample, for two paired samples, and two independent samples, and you are able to perform these tests either by hand or by means of the package R. 7. You know the setup of likelihood ratio tests and you are able to perform these tests either by hand or by means of the package R. 8.You know the concept of confidence interval and you are able to construct these intervals from pivots or near-pivots for simple statistical models. 9. You know the formulation of the linear regression model and the maximum likelihood estimators for the unknown parameters and you can fit these models by means of the statistical package R. 10. You know several concepts to check the assumptions and assess the fit of a linear regression model and you are able to determine these by means of the package R. 11. You know the content of important results, such as Theorem 4.29 and its proof, Theorem 4.43 and Theorem 5.9. 12. You know the definition of score function and Fisher information and their properties formulated in Lemma 5.10.	
Education Method	Lectures, exercise classes, and computer assignments.	
Computer Use	Statistical analyses are performed by means of the (free-ware) package R. See www.r-project.org	
Literature and Study Materials	"An Introduction to Mathematical Statistics", 1st edition. Fetsje Bijma, Marianne Jonker and Aad van der Vaart. Slides of the lectures are made available through Brightspace. Weekly written and/or computer assignments Statistical Software package R and the program R-studio	
Assessment	Mandatory R-assignments that have to be completed with a PASS. Written exam in week 10 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	3 hour written exam	
Permitted Materials during Tests	When the exam is held on campus, then the exam is a "closed book" exam. You are only allowed to use sheets with information on probability distributions and R-commands. You are not allowed to use any books or notes. When the exam is held online, then book and personal notes are permitted. NOT permitted is contact with third parties or getting information from the internet during the exam.	
Judgement	Grade of the written exam (or the resit). The course is completed only with a sufficient grade for the written exam and if all R-assignments have been completed with a PASS	
Co-Instructor	Dr. rer. nat. F. Mies	

TW1-11TWN	Problem Solving and Proofs	3
Responsible Instructor	Dr. B. van den Dries	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	See Dutch version	

TW1-22	Analysis 1	5
Responsible Instructor	F. Bartolucci	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	This course builds upon the courses Problem Solving and Proofs (TW1-11) and Calculus (TW1-12) and their contents are closely linked and interwoven.	
Course Contents	This is an introduction to analysis for functions of one real variable. The following topics are treated: - sequences - series - continuity - differentiation - integration.	
Study Goals	At the end of the course, you should be able to: LO1) restate the definitions and give examples of the notions covered during the course LO1.1: convergent, bounded, divergent and Cauchy sequence, subsequence, supremum, infimum, maximum, minimum, (strictly) monotone sequence, limes superior and inferior LO1.2: convergent and absolutely convergent series LO1.3: continuous, bounded and uniformly continuous function, continuous limit LO1.4: differentiable function, (strictly) monotone function LO1.5: Riemann integral, uniform convergence, improper integral LO2) restate the statements of the results covered in the lectures on the notions listed in LO1 and explain their proofs LO3) apply the definitions together with the theorems treated during the course to prove results for sequences, series, continuous functions, differentiable functions, integrals and sequences of functions	
Education Method	The course consists of frontal lectures where the theory is explained. These are complemented with exercise classes.	
Reader	Lecture Notes Analysis 1 (available on Brightspace).	
Assessment	There are two written partial exams. The midterm exam takes 1 hour and counts 25% of the final grade while the final exam takes 3 hours and counts 75% of the final grade. The resit exam fully determines the final grade (the partial exams will not contribute to the final grade in this case). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Permitted Materials during Tests	No course materials are permitted.	
Judgement	There are two ways to obtain a final mark for the course Analysis 1, TW1-22: 1. The final grade is determined by the grades of the partial exams, the first counting 25% and the second counting 75% of the final grade. OR 2. The final grade is the grade of the resit. Earlier partial exam results do not count towards the final mark. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Choose

AM2040	Complex Function Theory	6

Responsible Instructor	Dr.ir. W.G.M. Groenevelt
Contact Hours / Week x/x/x/x	0/0/0/4 hc: 0/0/0/4 wc
Education Period	4
Start Education	4
Exam Period	4 5
Course Language	Dutch
Expected prior knowledge	AM1010 Mathematical Structures, AM1020 Kaleidoscope (Complex Numbers), AM1040 Analysis 1, AM1070 Analysis 2
Course Contents	This course offers an introduction to the theory of analytic functions of one variable. An analytic function is a complex-valued function defined on an open set in the complex plane that is complex differentiable on its domain. Unlike functions in a real variable, the mere existence of a complex derivative has strong implications for the properties of the function. At the center of the course is the Cauchy integral formula for analytic functions. This formula expresses the value of an analytic function inside a circle in terms of the values of the function on the circle. The integral formula allows us to derive numerous beautiful properties of analytic functions. The general goals of the course are: - Being able to extend the basic notions from real-valued function of a real variable to complex-valued functions of a complex variable and having an understanding of the similarities and differences between real and complex analysis. - Acquiring analytic skills that enable one to choose an applicable methods from Complex Analysis for problems from various fields such as mathematical physics, system theory, signal analysis and the skills to turn this choice into a solution of the given problem. The following topics will be covered: 1. The basic complex function and their properties: the exponential function, trigonometric functions and logarithms 2. Analytic functions, the Cauchy-Riemann equations, harmonic functions 3. The Cauchy integral formula with applications, such as Liouville's theorem and the maximum-modulus principle. 4. Power and Laurent Series 5. Classification of isolated singularities 6. The residue theorem with applications, such as the argument principle and Rouché's theorem.
Study Goals	By the end of the course you should be able to 1. perform basic mathematical operations (arithmetics, powers, roots) with complex numbers in Cartesian and polar forms, and solve elementary complex equations; 2. describe mapping behaviour of complex functions, including multi-valuedness and branching; 3. find the Taylor and Laurent series of a function and determine its domain of convergence; 4. determine the isolated singularities of a function, as well as the type of the singularities; 5. evaluate contour integrals using parametrization, primitive functions, Cauchys integral formula and the Residue theorem; 6. state and apply fundamental properties of analytic functions, including the Cauchy-Riemann equations, Liouvilles theorem, the maximum-modulus principle, the Identity theorem, Rouchés theorem; 7. apply various techniques to evaluate (improper) integrals, including contour integration, ML-estimates, Jordan's lemma, shrinking path lemma; 8. explain concepts, state and prove theorems and properties involving the above topics.
Education Method	Lectures and exercise classes
Literature and Study Materials	Nakhlé H. Asmar, Loukas Grafakos - Complex Analysis with applications, Undergraduate Texts in Mathematics. Springer, 2018, ISBN 978-3-319-94062-5. A digital version of this book is accessible online for TU-students via the TU library.
Assessment	Written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Judgement	The overall grade is the grade for the exam or for the resit.
Co-Instructor	Dr. J.A.M. de Groot

TW1-31	Discrete Mathematics	5
Responsible Instructor	Dr. C.E. Groenland	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	Dutch (on request English)	
Required for	Probability, Optimalisation, Graph Theory	
Expected prior knowledge	From Problems and Proofs: notion of sets (complement, intersection, union), proof by induction.	
Course Contents	This course provides an introduction to discrete mathematics. We study discrete objects such as integers, sets, and graphs. Examples of graphs include social networks or road networks. In the first half of the course, we delve into various counting principles, such as binomial and multinomial coefficients, double counting, inclusion-exclusion, and recurrence relations. We also cover properties of integers such as modular arithmetic and the uniqueness of prime factorization. In the second half of the course, we explore basic concepts of graph theory such as Eulerian walks, trees, matching, planar graphs, and graph coloring. Throughout the course, emphasis is also placed on applying 'discrete thinking' to solve problems and on algorithmic aspects, including Euclid's algorithm, breadth-first search, bipartite matching, and greedy algorithms.	
Study Goals	1) Use the language of discrete mathematics to formalise mathematical and real-life problems. 2) Formulate rigorous mathematical proofs. 3) Solve counting problems using techniques such as double counting, binomial coefficients, bijections, inclusion-exclusion and recurrence relations. 4) State the main definitions and theorems and explain the proofs of these theorems. 5) Prove new mathematical statements by combining concepts, examples and theorems covered in the course. 6) Apply algorithms from the course such as Euclidean's algorithm for finding the greatest common divisor, augmenting path algorithm for finding a maximum matching and tree algorithms such as depth-first search.	
Education Method	4h lectures and 4h tutorials/instructions per week	
Books	Discrete Mathematics Elementary and Beyond from L. Lovász, J. Pelikán and K. Vesztergombi	
Prerequisites	Problems and Proofs	
Assessment	Written exam (midterm 40%, endterm 60%)	
Co-Instructor	Prof.dr. D.C. Gijswijt	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Choose

AM2070	Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	0/0/4/4 hc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course Ordinary Differential Equations AM2030	
Course Contents	Quasi-linear first order PDEs, and a simple model describing traffic flow.Types of second order PDEs : parabolic, elliptic, and hyperbolic PDEs. Derivation of the wave equation. Initial and initial-boundary value problems. Waves and reflctions. Fourier series and the method of separation of variables. Sturm-Liouville problems. derivation of the heat equation. Boundary value problems. Delta functions and distributions. Green's functions for heat, wave, and Laplace equations. Fourier and Laplace transform methods.	
Study Goals	At the end of the course, the student should be able to: 1. solve initial value problems and initial-boundary value problems for wave, transport, and heat equations. 2. solve boundary value problems for Laplace and Helmholz equations. 3. make the right choice for the method to solve the problems listed in 1 and 2. 4. give a physical interpretation of the the constructed solutions. 5. visualize the results by using the formula manipulation package MAXIMA.	
Education Method	lecture, exercise class, and practical work	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	Home-work exercises, practical work, and a written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. "The final mark for the course is the mark for the exam, but it is only valid when all practical work have been completed with a pass. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	pen, pencil, passer, geo-triangle (tabels will be handed out during the exam)	
Special Information	the study guide information may change depending on the developments around the coronavirus.	
Judgement	Final grading. 1) The practical work is mandatory, and as long as the practical work is not completed no final grading will be given. 2) Twice per year a written exam will be held. For the written exam in the fourth teaching quarter the following rules hold. The final grading is equal to the maximum of { the written exam grading, the average of the written exam grading and the home-work grading (where the exam grading has to be at least 5.0)}.For the written exam in the other period the final grading is equal to the written exam grading.	
Co-Instructor	Ir. W. van Horssen	

WI3150TU	Partial Differential Equations A	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, ordinary differential equations	
Course Contents	I: (Wi3150TU) Introduction. Types of second order equations. Initial and initial boundary value problems. Fourier series. Quasi-linear, first order partial differential equations. Waves and reflections of waves. Separation of variables. Sturm-Liouville problems. Parabolic, elliptic and hyperbolic equations. Maximum principle. Diffusion and heat transport problems. Lectures (3 ECTS). II: (Wi3151TU) Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972.	
Assessment	Written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. The resit consists of a written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. H.M. Schuttelaars	
Co-Instructor	Dr.ir. W.T. van Horssen	

WI3151TU	Partial Differential Equations B	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	WI3150TU	
Course Contents	Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures and a feedback assignment.	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	The feedback assignment and a written exam. The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The resit consists of a written exam (100%). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Judgement	The feedback assignment is mandatory, and as long as the assignment is not completed no final grading will be given.	
Co-Instructor	Dr. H.M. Schuttelaars	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Programme AM for TUD BSc EE/TN/NB

AM2020	Optimization	6
Responsible Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	0/4/0/0 hc 0/4/0/0 wc	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear Algebra 1 (AM1030), Analysis 2 (AM1070), Mathematical Structures (AM1010)	
Course Contents	Introduction to optimization, convexity. Modelling (Mixed) Integer Linear Programming problems. Linear programming: Simplex method, duality. Network optimization: shortest paths, max flow. Integer linear programming: branch & bound, Gomory cutting planes. Nonlinear optimization: Newton method, steepest descent. Complexity: NP-hardness. Approximation algorithms.	
Study Goals	The students are able to formulate discrete optimization problems as (mixed integer) linear programming problems. The students are able to execute basic algorithms for solving linear programming, integer linear programming and nonlinear programming problems, to understand the outcomes of the algorithms and to apply duality. The students know basic theorems of linear and nonlinear optimization and are able to prove variations of these theorems. The students can apply basic algorithms for network optimization problems (shortest path, max flow, travelling salesman problem). The students are able to design reductions between optimization problems and to prove correctness of those reductions.	
Education Method	4 hours lecture and 4 hours of exercise class per week, plus feedback exercises and homework exercises	
Literature and Study Materials	Lecture notes	
Assessment	Written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	3 hours	
Permitted Materials during Tests	A non-graphical calculator	
Judgement	The exam grade is the final grade	
Co-Instructor	Dr.ir. L.J.J. van Iersel	

AM2060	Numerical Methods 1	6
Responsible Instructor	mr.ir.drs. V.N.S.R. Dwarka	
Contact Hours / Week x/x/x/x	0/0/2/2 hc; 0/0/2/4 wc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	Further numerical courses, modelling and BSc-project	
Expected prior knowledge	Analysis I and II, Linear Algebra I and II	
Summary	Introduction to numerical mathematics, rounding/truncation errors, interpolation, numerical differentiation, numerical time integration for initial value problems, boundary value problems (finite differences), heat equation, numerical integration (quadrature), non-linear equations.	
Course Contents	Numerical mathematics is the study of methods that give numerical approximations to mathematical problems. In most cases, the exact solution cannot be obtained (easily). During this course, numerical methods are treated for the solution of ordinary differential equations, and a relatively simple partial differential equation (the time-dependent heat equation). Further several numerical methods are treated which can also be applied to problems that do not involve differential equations. Such methods are: interpolation, numerical integration, and solution strategies for non-linear equations. The use of numerical software (MATLAB/Python) is an essential part of the course material.	
Study Goals	After having completed the course, the student will be able to design, analyse and implement numerical methods for ordinary differential equations. At the end of the coarse, the student will be able to: -design numerical methods for ordinary differential equations -analyse properties of the numerical methods (for example stability and consistency) -understand the numerical treatment of time-dependent and stationary ordinary differential equations and/or a combination of both -distinguish between the numerical treatment of a linear and non-linear problem -implement the numerical methods (for example using python or matlab)	
Education Method	Two hours of lectures and an afternoon of lab work/instruction per week: -During the instruction hour, you will be able to work on homework sets. These homework sets are for practice purposes and are not-graded. -During the lab instruction, you will be able to work on the lab assignment (programming).	
Literature and Study Materials	Numerical Methods for Ordinary Differential Equations, C. Vuik, F.J. Vermolen, M.B. van Gijzen, en M.J. Vuik (DAP / VSSD, 2015)	
Assessment	Lab assignment (pass/fail which grants access to exam) and a written test. Only a pass from the previous study study year will be valid. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Pen, non-programmable calculator and workpaper (which will be provided on the spot)	
Judgement	The final grade is determined as follows: 100% written test. The final grade will only be awarded provided the lab report is satisfactory.	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

AM2080	Introduction to Statistics	6
Responsible Instructor	Prof.dr.ir. G. Jongbloed	
Instructor	Dr. A.F.F. Derumigny	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Introduction to Probability (first year probability course)	
Course Contents	Statistical Models; Descriptive Statistics, Estimation theory; Testing theory; Confidence regions; Regression	
Study Goals	By the end of this course: 1. You know the concept of statistical model and you are able to formulate such a model for standard datasets. 2. You know several numerical and graphical summaries of a dataset and their relation with the probability distribution in the statistical model for the corresponding dataset and you are able to produce them with the statistical package R. 3. You know the concepts of Mean Squared Error and unbiasedness for an estimator and you are able to determine them for simple estimators. 4. You know the Maximum Likelihood, Method of Moments, and Bayesian approach towards estimation and you are able to construct estimators according to these principles and compare their performance either by hand or by means of a simulation with the package R. 5. You know the basic concepts involved in hypothesis testing and you are able to setup and conduct a statistical test in simple statistical models. 6. You know the setup of the Student-t test for one sample, for two paired samples, and two independent samples, and you are able to perform these tests either by hand or by means of the package R. 7. You know the setup of likelihood ratio tests and you are able to perform these tests either by hand or by means of the package R. 8.You know the concept of confidence interval and you are able to construct these intervals from pivots or near-pivots for simple statistical models. 9. You know the formulation of the linear regression model and the maximum likelihood estimators for the unknown parameters and you can fit these models by means of the statistical package R. 10. You know several concepts to check the assumptions and assess the fit of a linear regression model and you are able to determine these by means of the package R. 11. You know the content of important results, such as Theorem 4.29 and its proof, Theorem 4.43 and Theorem 5.9. 12. You know the definition of score function and Fisher information and their properties formulated in Lemma 5.10.	
Education Method	Lectures, exercise classes, and computer assignments.	
Computer Use	Statistical analyses are performed by means of the (free-ware) package R. See www.r-project.org	
Literature and Study Materials	"An Introduction to Mathematical Statistics", 1st edition. Fetsje Bijma, Marianne Jonker and Aad van der Vaart. Slides of the lectures are made available through Brightspace. Weekly written and/or computer assignments Statistical Software package R and the program R-studio	
Assessment	Mandatory R-assignments that have to be completed with a PASS. Written exam in week 10 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	3 hour written exam	
Permitted Materials during Tests	When the exam is held on campus, then the exam is a "closed book" exam. You are only allowed to use sheets with information on probability distributions and R-commands. You are not allowed to use any books or notes. When the exam is held online, then book and personal notes are permitted. NOT permitted is contact with third parties or getting information from the internet during the exam.	
Judgement	Grade of the written exam (or the resit). The course is completed only with a sufficient grade for the written exam and if all R-assignments have been completed with a PASS	
Co-Instructor	Dr. rer. nat. F. Mies	

TW1-11TWN	Problem Solving and Proofs	3
Responsible Instructor	Dr. B. van den Dries	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	See Dutch version	

TW1-22	Analysis 1	5
Responsible Instructor	F. Bartolucci	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	This course builds upon the courses Problem Solving and Proofs (TW1-11) and Calculus (TW1-12) and their contents are closely linked and interwoven.	
Course Contents	This is an introduction to analysis for functions of one real variable. The following topics are treated: - sequences - series - continuity - differentiation - integration.	
Study Goals	At the end of the course, you should be able to: LO1) restate the definitions and give examples of the notions covered during the course LO1.1: convergent, bounded, divergent and Cauchy sequence, subsequence, supremum, infimum, maximum, minimum, (strictly) monotone sequence, limes superior and inferior LO1.2: convergent and absolutely convergent series LO1.3: continuous, bounded and uniformly continuous function, continuous limit LO1.4: differentiable function, (strictly) monotone function LO1.5: Riemann integral, uniform convergence, improper integral LO2) restate the statements of the results covered in the lectures on the notions listed in LO1 and explain their proofs LO3) apply the definitions together with the theorems treated during the course to prove results for sequences, series, continuous functions, differentiable functions, integrals and sequences of functions	
Education Method	The course consists of frontal lectures where the theory is explained. These are complemented with exercise classes.	
Reader	Lecture Notes Analysis 1 (available on Brightspace).	
Assessment	There are two written partial exams. The midterm exam takes 1 hour and counts 25% of the final grade while the final exam takes 3 hours and counts 75% of the final grade. The resit exam fully determines the final grade (the partial exams will not contribute to the final grade in this case). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Permitted Materials during Tests	No course materials are permitted.	
Judgement	There are two ways to obtain a final mark for the course Analysis 1, TW1-22: 1. The final grade is determined by the grades of the partial exams, the first counting 25% and the second counting 75% of the final grade. OR 2. The final grade is the grade of the resit. Earlier partial exam results do not count towards the final mark. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Choose

AM2040	Complex Function Theory	6
Responsible Instructor	Dr.ir. W.G.M. Groenevelt	
Contact Hours / Week x/x/x/x	0/0/0/4 hc: 0/0/0/4 wc	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	AM1010 Mathematical Structures, AM1020 Kaleidoscope (Complex Numbers), AM1040 Analysis 1, AM1070 Analysis 2	
Course Contents	This course offers an introduction to the theory of analytic functions of one variable. An analytic function is a complex-valued function defined on an open set in the complex plane that is complex differentiable on its domain. Unlike functions in a real variable, the mere existence of a complex derivative has strong implications for the properties of the function. At the center of the course is the Cauchy integral formula for analytic functions. This formula expresses the value of an analytic function inside a circle in terms of the values of the function on the circle. The integral formula allows us to derive numerous beautiful properties of analytic functions. The general goals of the course are: - Being able to extend the basic notions from real-valued function of a real variable to complex-valued functions of a complex variable and having an understanding of the similarities and differences between real and complex analysis. - Acquiring analytic skills that enable one to choose an applicable methods from Complex Analysis for problems from various fields such as mathematical physics, system theory, signal analysis and the skills to turn this choice into a solution of the given problem. The following topics will be covered: 1. The basic complex function and their properties: the exponential function, trigonometric functions and logarithms 2. Analytic functions, the Cauchy-Riemann equations, harmonic functions 3. The Cauchy integral formula with applications, such as Liouville's theorem and the maximum-modulus principle. 4. Power and Laurent Series 5. Classification of isolated singularities 6. The residue theorem with applications, such as the argument principle and Rouché's theorem.	
Study Goals	By the end of the course you should be able to 1. perform basic mathematical operations (arithmetics, powers, roots) with complex numbers in Cartesian and polar forms, and solve elementary complex equations; 2. describe mapping behaviour of complex functions, including multi-valuedness and branching; 3. find the Taylor and Laurent series of a function and determine its domain of convergence; 4. determine the isolated singularities of a function, as well as the type of the singularities; 5. evaluate contour integrals using parametrization, primitive functions, Cauchys integral formula and the Residue theorem; 6. state and apply fundamental properties of analytic functions, including the Cauchy-Riemann equations, Liouvilles theorem, the maximum-modulus principle, the Identity theorem, Rouchés theorem; 7. apply various techniques to evaluate (improper) integrals, including contour integration, ML-estimates, Jordan's lemma, shrinking path lemma; 8. explain concepts, state and prove theorems and properties involving the above topics.	
Education Method	Lectures and exercise classes	
Literature and Study Materials	Nakhlé H. Asmar, Loukas Grafakos - Complex Analysis with applications, Undergraduate Texts in Mathematics. Springer, 2018, ISBN 978-3-319-94062-5. A digital version of this book is accessible online for TU-students via the TU library.	
Assessment	Written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	The overall grade is the grade for the exam or for the resit.	
Co-Instructor	Dr. J.A.M. de Groot	

TW1-31	Discrete Mathematics	5
Responsible Instructor	Dr. C.E. Groenland	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	Dutch (on request English)	
Required for	Probability, Optimalisation, Graph Theory	
Expected prior knowledge	From Problems and Proofs: notion of sets (complement, intersection, union), proof by induction.	
Course Contents	This course provides an introduction to discrete mathematics. We study discrete objects such as integers, sets, and graphs. Examples of graphs include social networks or road networks. In the first half of the course, we delve into various counting principles, such as binomial and multinomial coefficients, double counting, inclusion-exclusion, and recurrence relations. We also cover properties of integers such as modular arithmetic and the uniqueness of prime factorization. In the second half of the course, we explore basic concepts of graph theory such as Eulerian walks, trees, matching, planar graphs, and graph coloring. Throughout the course, emphasis is also placed on applying 'discrete thinking' to solve problems and on algorithmic aspects, including Euclid's algorithm, breadth-first search, bipartite matching, and greedy algorithms.	
Study Goals	1) Use the language of discrete mathematics to formalise mathematical and real-life problems. 2) Formulate rigorous mathematical proofs. 3) Solve counting problems using techniques such as double counting, binomial coefficients, bijections, inclusion-exclusion and recurrence relations. 4) State the main definitions and theorems and explain the proofs of these theorems. 5) Prove new mathematical statements by combining concepts, examples and theorems covered in the course. 6) Apply algorithms from the course such as Euclidean's algorithm for finding the greatest common divisor, augmenting path algorithm for finding a maximum matching and tree algorithms such as depth-first search.	
Education Method	4h lectures and 4h tutorials/instructions per week	
Books	Discrete Mathematics Elementary and Beyond from L. Lovász, J. Pelikán and K. Vesztergombi	
Prerequisites	Problems and Proofs	
Assessment	Written exam (midterm 40%, endterm 60%)	
Co-Instructor	Prof.dr. D.C. Gijswijt	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Choose

AM2070	Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	0/0/4/4 hc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course Ordinary Differential Equations AM2030	
Course Contents	Quasi-linear first order PDEs, and a simple model describing traffic flow.Types of second order PDEs : parabolic, elliptic, and hyperbolic PDEs. Derivation of the wave equation. Initial and initial-boundary value problems. Waves and reflctions. Fourier series and the method of separation of variables. Sturm-Liouville problems. derivation of the heat equation. Boundary value problems. Delta functions and distributions. Green's functions for heat, wave, and Laplace equations. Fourier and Laplace transform methods.	
Study Goals	At the end of the course, the student should be able to: 1. solve initial value problems and initial-boundary value problems for wave, transport, and heat equations. 2. solve boundary value problems for Laplace and Helmholz equations. 3. make the right choice for the method to solve the problems listed in 1 and 2. 4. give a physical interpretation of the the constructed solutions. 5. visualize the results by using the formula manipulation package MAXIMA.	
Education Method	lecture, exercise class, and practical work	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	Home-work exercises, practical work, and a written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. "The final mark for the course is the mark for the exam, but it is only valid when all practical work have been completed with a pass. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	pen, pencil, passer, geo-triangle (tabels will be handed out during the exam)	
Special Information	the study guide information may change depending on the developments around the coronavirus.	
Judgement	Final grading. 1) The practical work is mandatory, and as long as the practical work is not completed no final grading will be given. 2) Twice per year a written exam will be held. For the written exam in the fourth teaching quarter the following rules hold. The final grading is equal to the maximum of { the written exam grading, the average of the written exam grading and the home-work grading (where the exam grading has to be at least 5.0)}.For the written exam in the other period the final grading is equal to the written exam grading.	
Co-Instructor	Ir. W. van Horssen	

WI3150TU	Partial Differential Equations A	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, ordinary differential equations	
Course Contents	I: (Wi3150TU) Introduction. Types of second order equations. Initial and initial boundary value problems. Fourier series. Quasi-linear, first order partial differential equations. Waves and reflections of waves. Separation of variables. Sturm-Liouville problems. Parabolic, elliptic and hyperbolic equations. Maximum principle. Diffusion and heat transport problems. Lectures (3 ECTS). II: (Wi3151TU) Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972.	
Assessment	Written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. The resit consists of a written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. H.M. Schuttelaars	
Co-Instructor	Dr.ir. W.T. van Horssen	

WI3151TU	Partial Differential Equations B	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	WI3150TU	
Course Contents	Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures and a feedback assignment.	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	The feedback assignment and a written exam. The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The resit consists of a written exam (100%). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Judgement	The feedback assignment is mandatory, and as long as the assignment is not completed no final grading will be given.	
Co-Instructor	Dr. H.M. Schuttelaars	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging programme AM for BSc Econometrics 2024	
Introduction 1	The 36-EC AM bridging programme for students with a B.Sc. degree in Econometrics consists of the courses listed below and one 2nd or 3rd year elective course from the TU Delft Applied Mathematics B.Sc. programme (see https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=28473 and https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=28476).

AM2060	Numerical Methods 1	6
Responsible Instructor	mr.ir.drs. V.N.S.R. Dwarka	
Contact Hours / Week x/x/x/x	0/0/2/2 hc; 0/0/2/4 wc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	Further numerical courses, modelling and BSc-project	
Expected prior knowledge	Analysis I and II, Linear Algebra I and II	
Summary	Introduction to numerical mathematics, rounding/truncation errors, interpolation, numerical differentiation, numerical time integration for initial value problems, boundary value problems (finite differences), heat equation, numerical integration (quadrature), non-linear equations.	
Course Contents	Numerical mathematics is the study of methods that give numerical approximations to mathematical problems. In most cases, the exact solution cannot be obtained (easily). During this course, numerical methods are treated for the solution of ordinary differential equations, and a relatively simple partial differential equation (the time-dependent heat equation). Further several numerical methods are treated which can also be applied to problems that do not involve differential equations. Such methods are: interpolation, numerical integration, and solution strategies for non-linear equations. The use of numerical software (MATLAB/Python) is an essential part of the course material.	
Study Goals	After having completed the course, the student will be able to design, analyse and implement numerical methods for ordinary differential equations. At the end of the coarse, the student will be able to: -design numerical methods for ordinary differential equations -analyse properties of the numerical methods (for example stability and consistency) -understand the numerical treatment of time-dependent and stationary ordinary differential equations and/or a combination of both -distinguish between the numerical treatment of a linear and non-linear problem -implement the numerical methods (for example using python or matlab)	
Education Method	Two hours of lectures and an afternoon of lab work/instruction per week: -During the instruction hour, you will be able to work on homework sets. These homework sets are for practice purposes and are not-graded. -During the lab instruction, you will be able to work on the lab assignment (programming).	
Literature and Study Materials	Numerical Methods for Ordinary Differential Equations, C. Vuik, F.J. Vermolen, M.B. van Gijzen, en M.J. Vuik (DAP / VSSD, 2015)	
Assessment	Lab assignment (pass/fail which grants access to exam) and a written test. Only a pass from the previous study study year will be valid. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Pen, non-programmable calculator and workpaper (which will be provided on the spot)	
Judgement	The final grade is determined as follows: 100% written test. The final grade will only be awarded provided the lab report is satisfactory.	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

TW1-11TWN	Problem Solving and Proofs	3
Responsible Instructor	Dr. B. van den Dries	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	See Dutch version	

TW1-22	Analysis 1	5
Responsible Instructor	F. Bartolucci	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	This course builds upon the courses Problem Solving and Proofs (TW1-11) and Calculus (TW1-12) and their contents are closely linked and interwoven.	
Course Contents	This is an introduction to analysis for functions of one real variable. The following topics are treated: - sequences - series - continuity - differentiation - integration.	
Study Goals	At the end of the course, you should be able to: LO1) restate the definitions and give examples of the notions covered during the course LO1.1: convergent, bounded, divergent and Cauchy sequence, subsequence, supremum, infimum, maximum, minimum, (strictly) monotone sequence, limes superior and inferior LO1.2: convergent and absolutely convergent series LO1.3: continuous, bounded and uniformly continuous function, continuous limit LO1.4: differentiable function, (strictly) monotone function LO1.5: Riemann integral, uniform convergence, improper integral LO2) restate the statements of the results covered in the lectures on the notions listed in LO1 and explain their proofs LO3) apply the definitions together with the theorems treated during the course to prove results for sequences, series, continuous functions, differentiable functions, integrals and sequences of functions	
Education Method	The course consists of frontal lectures where the theory is explained. These are complemented with exercise classes.	
Reader	Lecture Notes Analysis 1 (available on Brightspace).	
Assessment	There are two written partial exams. The midterm exam takes 1 hour and counts 25% of the final grade while the final exam takes 3 hours and counts 75% of the final grade. The resit exam fully determines the final grade (the partial exams will not contribute to the final grade in this case). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Permitted Materials during Tests	No course materials are permitted.	
Judgement	There are two ways to obtain a final mark for the course Analysis 1, TW1-22: 1. The final grade is determined by the grades of the partial exams, the first counting 25% and the second counting 75% of the final grade. OR 2. The final grade is the grade of the resit. Earlier partial exam results do not count towards the final mark. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Choose

AM2040	Complex Function Theory	6
Responsible Instructor	Dr.ir. W.G.M. Groenevelt	
Contact Hours / Week x/x/x/x	0/0/0/4 hc: 0/0/0/4 wc	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	AM1010 Mathematical Structures, AM1020 Kaleidoscope (Complex Numbers), AM1040 Analysis 1, AM1070 Analysis 2	
Course Contents	This course offers an introduction to the theory of analytic functions of one variable. An analytic function is a complex-valued function defined on an open set in the complex plane that is complex differentiable on its domain. Unlike functions in a real variable, the mere existence of a complex derivative has strong implications for the properties of the function. At the center of the course is the Cauchy integral formula for analytic functions. This formula expresses the value of an analytic function inside a circle in terms of the values of the function on the circle. The integral formula allows us to derive numerous beautiful properties of analytic functions. The general goals of the course are: - Being able to extend the basic notions from real-valued function of a real variable to complex-valued functions of a complex variable and having an understanding of the similarities and differences between real and complex analysis. - Acquiring analytic skills that enable one to choose an applicable methods from Complex Analysis for problems from various fields such as mathematical physics, system theory, signal analysis and the skills to turn this choice into a solution of the given problem. The following topics will be covered: 1. The basic complex function and their properties: the exponential function, trigonometric functions and logarithms 2. Analytic functions, the Cauchy-Riemann equations, harmonic functions 3. The Cauchy integral formula with applications, such as Liouville's theorem and the maximum-modulus principle. 4. Power and Laurent Series 5. Classification of isolated singularities 6. The residue theorem with applications, such as the argument principle and Rouché's theorem.	
Study Goals	By the end of the course you should be able to 1. perform basic mathematical operations (arithmetics, powers, roots) with complex numbers in Cartesian and polar forms, and solve elementary complex equations; 2. describe mapping behaviour of complex functions, including multi-valuedness and branching; 3. find the Taylor and Laurent series of a function and determine its domain of convergence; 4. determine the isolated singularities of a function, as well as the type of the singularities; 5. evaluate contour integrals using parametrization, primitive functions, Cauchys integral formula and the Residue theorem; 6. state and apply fundamental properties of analytic functions, including the Cauchy-Riemann equations, Liouvilles theorem, the maximum-modulus principle, the Identity theorem, Rouchés theorem; 7. apply various techniques to evaluate (improper) integrals, including contour integration, ML-estimates, Jordan's lemma, shrinking path lemma; 8. explain concepts, state and prove theorems and properties involving the above topics.	
Education Method	Lectures and exercise classes	
Literature and Study Materials	Nakhlé H. Asmar, Loukas Grafakos - Complex Analysis with applications, Undergraduate Texts in Mathematics. Springer, 2018, ISBN 978-3-319-94062-5. A digital version of this book is accessible online for TU-students via the TU library.	
Assessment	Written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	The overall grade is the grade for the exam or for the resit.	
Co-Instructor	Dr. J.A.M. de Groot	

TW1-31	Discrete Mathematics	5
Responsible Instructor	Dr. C.E. Groenland	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	Dutch (on request English)	
Required for	Probability, Optimalisation, Graph Theory	
Expected prior knowledge	From Problems and Proofs: notion of sets (complement, intersection, union), proof by induction.	
Course Contents	This course provides an introduction to discrete mathematics. We study discrete objects such as integers, sets, and graphs. Examples of graphs include social networks or road networks. In the first half of the course, we delve into various counting principles, such as binomial and multinomial coefficients, double counting, inclusion-exclusion, and recurrence relations. We also cover properties of integers such as modular arithmetic and the uniqueness of prime factorization. In the second half of the course, we explore basic concepts of graph theory such as Eulerian walks, trees, matching, planar graphs, and graph coloring. Throughout the course, emphasis is also placed on applying 'discrete thinking' to solve problems and on algorithmic aspects, including Euclid's algorithm, breadth-first search, bipartite matching, and greedy algorithms.	
Study Goals	1) Use the language of discrete mathematics to formalise mathematical and real-life problems. 2) Formulate rigorous mathematical proofs. 3) Solve counting problems using techniques such as double counting, binomial coefficients, bijections, inclusion-exclusion and recurrence relations. 4) State the main definitions and theorems and explain the proofs of these theorems. 5) Prove new mathematical statements by combining concepts, examples and theorems covered in the course. 6) Apply algorithms from the course such as Euclidean's algorithm for finding the greatest common divisor, augmenting path algorithm for finding a maximum matching and tree algorithms such as depth-first search.	
Education Method	4h lectures and 4h tutorials/instructions per week	
Books	Discrete Mathematics Elementary and Beyond from L. Lovász, J. Pelikán and K. Vesztergombi	
Prerequisites	Problems and Proofs	
Assessment	Written exam (midterm 40%, endterm 60%)	
Co-Instructor	Prof.dr. D.C. Gijswijt	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Choose

AM2070	Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	0/0/4/4 hc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course Ordinary Differential Equations AM2030	
Course Contents	Quasi-linear first order PDEs, and a simple model describing traffic flow.Types of second order PDEs : parabolic, elliptic, and hyperbolic PDEs. Derivation of the wave equation. Initial and initial-boundary value problems. Waves and reflctions. Fourier series and the method of separation of variables. Sturm-Liouville problems. derivation of the heat equation. Boundary value problems. Delta functions and distributions. Green's functions for heat, wave, and Laplace equations. Fourier and Laplace transform methods.	
Study Goals	At the end of the course, the student should be able to: 1. solve initial value problems and initial-boundary value problems for wave, transport, and heat equations. 2. solve boundary value problems for Laplace and Helmholz equations. 3. make the right choice for the method to solve the problems listed in 1 and 2. 4. give a physical interpretation of the the constructed solutions. 5. visualize the results by using the formula manipulation package MAXIMA.	
Education Method	lecture, exercise class, and practical work	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	Home-work exercises, practical work, and a written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. "The final mark for the course is the mark for the exam, but it is only valid when all practical work have been completed with a pass. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	pen, pencil, passer, geo-triangle (tabels will be handed out during the exam)	
Special Information	the study guide information may change depending on the developments around the coronavirus.	
Judgement	Final grading. 1) The practical work is mandatory, and as long as the practical work is not completed no final grading will be given. 2) Twice per year a written exam will be held. For the written exam in the fourth teaching quarter the following rules hold. The final grading is equal to the maximum of { the written exam grading, the average of the written exam grading and the home-work grading (where the exam grading has to be at least 5.0)}.For the written exam in the other period the final grading is equal to the written exam grading.	
Co-Instructor	Ir. W. van Horssen	

WI3150TU	Partial Differential Equations A	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, ordinary differential equations	
Course Contents	I: (Wi3150TU) Introduction. Types of second order equations. Initial and initial boundary value problems. Fourier series. Quasi-linear, first order partial differential equations. Waves and reflections of waves. Separation of variables. Sturm-Liouville problems. Parabolic, elliptic and hyperbolic equations. Maximum principle. Diffusion and heat transport problems. Lectures (3 ECTS). II: (Wi3151TU) Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972.	
Assessment	Written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. The resit consists of a written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. H.M. Schuttelaars	
Co-Instructor	Dr.ir. W.T. van Horssen	

WI3151TU	Partial Differential Equations B	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	WI3150TU	
Course Contents	Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures and a feedback assignment.	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	The feedback assignment and a written exam. The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The resit consists of a written exam (100%). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Judgement	The feedback assignment is mandatory, and as long as the assignment is not completed no final grading will be given.	
Co-Instructor	Dr. H.M. Schuttelaars	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging programme CESE for HBO TI 2024	
Introduction 1	The bridging programme gives access to the masters programme in Embedded Systems and consists of courses in the field of calculus, mathematical modelling and computer engineering.
Introduction 2	Students will gain access to the Masters degree programme if they have a relevant HBO degree and if they completed the bridging programme.
Program Structure 1	

CSE1110	Software Quality and Testing	5
Responsible Instructor	F. Mulder	
Contact Hours / Week x/x/x/x	0/0/0/4 lecture; 0/0/0/4 shared lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course covers the most important testing techniques needed to build high quality software systems. The course covers both the testing techniques themselves, and the software design techniques needed to create testable systems. Specific topics covered are: * testing principles, test levels, and the testing pyramid * software testing automation and JUnit * specification-based testing * structural testing and the different coverage criteria * boundary analysis and testing * design by contract * property-based testing * test doubles and mock objects * design for testability * test-driven development (TDD) * larger tests (SQL and web testing) * test code quality and test smells	
Study Goals	By the end of the course, students will be able to: * Create unit, integration, and system tests using current existing tools (like JUnit, Mockito, and JaCoCo) that effectively test complex software systems. * Derive test cases that deal with exceptional, corner, and bad weather cases by performing different testing techniques as well as able to reflect about their limitations, when and when not to apply them in a given context. * Measure and reflect on the efficiency of the developed test suites by means of different test adequacy metrics (i.e., line, branch, condition, MC/DC coverage). * Design and implement testable software systems by means of architectural patterns such as dependency injection and ports and adapters. * Write maintainable test code by avoiding well-known test code smells. * Understand and apply state-of-the-art techniques for intelligent software testing, such as property-based testing.	
Education Method	This course follows a flipped classroom approach. Students read chapters of the book and work on (practical and theoretical) exercises during their study time; lectures are used for questions and broader discussions. We offer TA support during the labs.	
Books	Effective Software Testing, by Maurício Aniche: https://www.effective-software-testing.com.	
Assessment	The evaluation of the course is based on one exam. It happens on WebLab and is composed of coding questions and theoretical questions. Students can consult the documentation of certain testing libraries during the exam. There is also one possibility to do a resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. C.E. Brandt	

CSE1205 — Linear Algebra — 5

Responsible Instructor
Dr. E. Emsiz

Contact Hours / Week x/x/x/x
0/0/6/0 colstruction; 0/0/2/0 lecture

Education Period
3

Start Education
3

Exam Period
3
4

Course Language
English

Required for
Multivariate data Analysis; Computer Graphics; Image Processing

Course Contents
- Solving systems of linear equations (using Gaussian elimination)
- Vectors in R^n
- Linear dependence/independence
- Matrix algebra (sum, product, inverse)
- LU decomposition of a matrix
- Lineare transformations
- Subspaces in R^n (basis, dimension); column space and null space
- Determinants
- Eigenvalues and eigenvectors
- Diagonalization
- Inner products and orthogonal sets
- Orthogonal projections
- Method of least squares and linear models
- Symmetric matrices

Study Goals
An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample.

After successfully completing the course you will be able to:

- Solve systems of linear equations.
- Identify if a system of linear equations is solvable, and if the solution is unique.
- Show the equivalence between a vector equation, a matrix equation and a system of linear equations.
- Give a geometric description of solutions sets of vector equations.
- Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence.
- Identify if a transformation is linear and determine the standard matrix of a linear transformation.
- Determine the matrix of geometric linear transformations.
- Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties.
- Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics.
- Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace.
- Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem.
- Determine the LU-decomposition of a matrix.
- Determine the determinant of a matrix by applying the properties of a determinant.
- Apply the properties of determinants (product, transpose, inverse).
- Determine the eigenvalues and eigenvectors of a matrix.
- Determine the algebraic and geometric multiplicity of a matrix.
- Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix.
- Analyze the case of complex eigenvalues and eigenvectors.
- Determine the matrix of a linear transformation with respect to a basis.
- Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets.
- Calculate the orthogonal projection on a subspace.
- Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process.
- Explain the normal equation of the least-squares method and apply the least-squares method to linear models.
- Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.

Education Method
Lectures blended with exercise sessions, online homework exercises and instructions.

Disclaimer: information may change depending on the developments around the coronavirus.

Literature and Study Materials
Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E.
- Slides on Brightspace.
- Prelecture videos.
- Online homework assignments.
- Book exercises
- Old exams on Brightspace.

Disclaimer: information may change depending on the developments around the coronavirus.

Assessment
The final score F is calculated as follow:

IF the midterm score M is higher than the score E for the endterm AND E >= 5.0
THEN F = 0.25 M + 0.75 E
ELSE F = E

All grades are rounded to 1 decimal place.

Note that the midterm is not mandatory (and cannot be resited).

For the resit the midterm is not relevant anymore.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)

Permitted Materials during Tests
No material.

Enrolment / Application
Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations.
Registration for this course takes place via Osiris, separately from the exam registration. For more information, see:
https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/

Co-Instructor	Drs. A.T. Hensbergen

CSE1300	Reasoning and Logic	5
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. N. Yorke-Smith	
Course Coordinator	S. Hugtenburg	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Every course in which exact reasoning is required, e.g. to prove a theorem, or reasoning from the assumption that a theorem is true. Examples: * Calculus * Algorithms and Data Structures * Linear Algebra * Probability and Statistics * Machine Learning * Algorithm Design * Automata, Languages and Computability * Logic-Based Artificial Intelligence * Cryptography * Software Verification * Complexity Theory	
Expected prior knowledge	High school mathematics	
Course Contents	It is often useful or even essential to know if a certain statement is true, e.g. Pythagoras' famous a^2 + b^2 = c^2 theorem about right-angled triangles. Knowledge gathered from these statements or theorems can be broadly used to solve more complicated problems. This way of working, i.e., deriving more complex theorems from simpler ones, is useful in many fields, in particular also in computer science. An argument is a set of premises or assumptions, followed by a conclusion. To be certain of the truth of an argument, the conclusion has to be a logical consequence of the assumptions. To prove this, the conclusion is derived from the premisses. The derivation shows us that once all premises are true, the conclusion is true as well. The course Reasoning and Logic is about proving the logical validity of arguments. What is a valid argument? When is an argument logically valid and when is it not? How can we determine whether an argument is logically valid? How can we derive a logically valid conclusion from the premises? Or how can we prove that a conclusion is not a logical consequence of the premises? In this course we will first explain a number of basic proof techniques, such as proof by contradiction, proof by mathematical induction, proof by division into cases, and the use of invariants. The application of these techniques will be practiced by proving and rejecting simple mathematical theorems. These proof techniques can only lead to a valid argument when the formulation of the premises and the conclusion is sufficiently precise. To express statements precisely, multiple artificial languages exist, of which we will learn two: propositional calculus and predicate logic. For both languages we look at the syntax and semantics and study how to translate expressions from a natural language to the more exact languages. Furthermore we will look at how to establish the logical validity of an argument in both languages. Moreover, to be able to assign truth values to formulas in predicate logic and because of the importance of this subject in every exact science, in this course we will also pay attention to elementary set theory. Together, the course provides tools that have important applications across mathematics, computer science, philosophy, and beyond.	
Study Goals	After this course, the student is able to: 1. translate a logically precise claim to and from natural language. 2. describe the operation of logical connectors and quantifiers. 3. describe the notion of logical validity. 4. explain and apply basic set and graph operations. 5. define and perform computations with functions, relations and equivalence classes. 6. construct and interpret recursive definitions, including recursive data structures like trees. 7. construct an appropriate function or relation given a description (in natural language or formal notation). 8. construct a direct or indirect proof (by contradiction, division into cases, generalisation, or [structural] induction) or logical equivalence, or counterexample for (in)valid arguments, in propositional logic, predicate logic and set theory. 9. identify what type of proof is appropriate for a given claim. 10. solve simple SAT instances. 11. develop specifications for verification tools like SAT or SMT solvers. 12. interpret the response of verification tools like SAT or SMT solvers	
Education Method	This course uses a student-centred learning methodology. Students study provided material at home. Every week has two large-group sessions, during which questions about the material will be answered. There will also be small interactive exercises in some of these sessions to help the students to evaluate their understanding of the material and to apply it. In addition there is a 1 hour tutorial session in which students tackle a larger problem as a group under the supervision of lecturers and TAs. Optional homework assignments are provided. These should be done in groups of 4. Lab sessions are offered, in which answers to the homework questions can be discussed. A project is also included in the course. This must be done individually.	
Books	Delftse Foundations of Computation, Hugtenburg & Yorke-Smith, TU Delft Open 2019, textbooks.open.tudelft.nl, ISBN 978-94-6186-952-4	
Assessment	The final grade of the course is computed from several components, described below. A final exam (100%) in week 10 (F) Additionally there are two opportunities to achieve some bonus points: Midterm exam Week 5 (M) Project, deadline Week 9 (P) Between M and P, you can score a bonus that is computed in accordance with the rules of the BoE: B = min(10, max(2*FLOOR(M-5), 0) + max(2*FLOOR(P-5), 0)) (FLOOR is rounding down the grade to the largest integer value that is smaller or equal to the grade. This means one bonus opportunity can compensate for another, but an opportunity only gets you bonus if the grade is sufficient.)	

	Final grade The final grade X for the course is computed as follows (again in accordance with the rules for bonus points of the BoE): X = F + (B*(10-F)/100) (This means the bonus scales down linearly as your summative final grade increases.) The resit (R) covers all material, and your final grade for the resit opportunity is: Y = R + (B*(10-R)/100) As per TU delft regulations, the grade considered for the course is max(X,Y). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Permitted Materials during Tests	No materials are allowed for on-campus exams.
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/
Special Information	For any administrative questions about the course, please use rl-cs-ewi@tudelft.nl and not the instructors personal e-mail addresses.
Tags	Abstract Algorithmics Logics Mathematics Project Small groups
Co-Instructor	S. Dumani

CSE2310	Algorithm Design	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. E. Demirovi	
Contact Hours / Week x/x/x/x	0/4/0/0 lecture; 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: programming experience from CSE1100 or equivalent Strongly advised: experience with algorithms and data structures (priority queue, graph algorithms etc) such as from Algorithms and Datastructures (CSE1305); experience with constructing proofs, such as from Reasoning and Logic (CSE1300)	
Course Contents	Graph algorithms, greedy algorithms, divide and conquer, dynamic programming and network flow algorithms	
Study Goals	After completing this course, the student can apply, reproduce, and give relevant properties and definitions of 1. graph algorithms, 2. greedy algorithms 3. divide and conquer, 4. dynamic programming and 5. network flow algorithms. The student can also 6. prove the correctness of these algorithms, 7. determine the running time of algorithms both analytically as well as experimentally, 8. implement algorithms efficiently, and 9. provide well-argued advice about which problems can be solved best using which algorithms.	
Education Method	Lectures (possibly only online / pre-recorded) Each lecture includes a number of multiple choice questions and a selection of especially the difficult components of the material. Students are expected to study some of the topics at home. Tutorials (possibly only online) In these interactive sessions we take you through some example assignments (both implementation and written ones). Scheduled classroom lecture hours may be used for a combination of the above, as announced by the lecturer. Skill circuit assignments To train with the learning objectives, several assignments at exam level, but also more accessible ones are available on BrightSpace as elements in the skill circuit. For some of them feedback can be requested from teaching assistants or other students if handed in on time. Similarly, students can train for the digital test by making the implementation assignments in the skill circuit. Correctness of the code will be automatically checked.	
Literature and Study Materials	The study material consists of the book Algorithm Design by John Kleinberg and Eva Tardos (2005), Addison Wesley ISBN 13: 9780321372918 or 9780321295354 and some documents made available through BrightSpace (such as a manual of proving techniques and the master's method for solving simple recurrent relations).	
Assessment	The final grade of the course consists of the following components: Written exam part 1 (of 1.5h) in week 5 (W1) Written exam part 2 (of 1.5h) in week 10/11 (W2) Computer exam (3h) in week 10/11 (P) Final grade The final grade for the course is computed as follows: W = 0.4 W1 + 0.6 W2 If P < 5 or W < 5, the final grade is min(P,W). Else the final grade is 0.5 W + 0.5 P Both parts P and W can be retaken, in individual exam moments in the resit week. Written exam (W) Half-way and at the end of the quarter written examinations of 1.5 hours each are organized. The written exam primarily assesses proving skills and (application of) algorithms and algorithmic theory. Resitting W implies redoing both parts in a single three-hour exam. Computer exam (P) Algorithm design and implementation skills are assessed through a computer exam of three hours at the end of the course. The computer exam needs to be submitted through Weblab, but IntelliJ is available within the exam environment. These partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	course email address: ad-cs-ewi@tudelft.nl	
Co-Instructor	Dr. E. Demirovi	

CSE2315	Automata, Computability and Complexity	5
Responsible Instructor	Prof.dr. M.T.J. Spaan	
Responsible Instructor	I.D. van Kreveld	
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. J.W. Böhmer	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Proof techniques (such as by induction or contradiction), set theory, recursive definitions, trees and graphs. For example from CSE1300 or equivalent. Run-time complexity of algorithms. For example from CSE2310 or equivalent.	
Course Contents	This course introduces three areas of the theory of computation. First, automata are used to recognize words in (formal) languages, and we will discern several types of languages along with the types of automata that can recognize them. Second, once we have an intuition about languages, we discuss the topic of computability, where you will learn not only what kinds of problems a computer can solve but also how to prove this. Third, we will examine the class of computable problems and make a distinction between "easy" and "hard" problems and delve into the famous problem: P = NP?	
Study Goals	After this course, the student can: 1. parse and construct automata and languages. 2. prove a problem to be computable. 3. manipulate and construct grammars for context-free languages. 4. describe the differences between different types of automata and grammars and their limitations, specifically that there are problems that are algorithmically unsolvable. 5. prove languages to be non-regular using the pumping lemma. 6. describe the classes P, NP, and NPC and explain why P = NP? is an open problem. 7. verify a proof that shows that a problem is NPC.	
Education Method	Engaging interactive lectures for which the student studies the material at home. During the lecture, the students test their knowledge via multiple choice questions and the teacher explains difficult topics on demand. Lab course in the form of optional take-home assignments, 8 times during the quarter. Students solve these assignments individually and hand them in weekly according to predetermined deadlines. Students correct each others work followed by a round of feedback by teaching assistants.	
Literature and Study Materials	1. Textbook (see Books), chapters 0, 1, 2 (2.1-2.3), 3, 4, 5 (5.1), 7. 2. Supplementary material (lecture slides, video lectures).	
Books	Michael Sipser, Introduction to the Theory of Computation, Third, International Edition, Cengage Learning. ISBN 978-1-133-18781-3. Note: if you do not use the international version, then section numbers and exercises will be different!	
Assessment	A written, closed-book exam (see permitted materials during tests below). The final grade of the course is based on the final exam for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	You are allowed to bring a hand-written double sided A4 cheat sheet. No additional materials are permitted, including but not limited to: books, printouts of the slides, or devices.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Prof.dr. M.T.J. Spaan	

CSE2420	Digital Systems	5
Responsible Instructor	Dr.ing. M. Shahraki Moghaddam	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 8/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	No prior knowledge is expected from the student.	
Course Contents	This course introduces the following aspects in digital design: - combinational digital systems, - sequential systems, - complex combinatial and sequential building blocks - reconfigurable hardware and complex digital systems. All the important aspects of digital design will be dicussed: - number systems, - Boolean algebra, - combinational logic, - Two-level and multilevel networks and reduction techniques/design tools, - timing hazards, - Moore/Mealy machines, - Finite State Machines (FSM) and more. The lab starts with the realization of simple circuits and will end with complex systems realized on an FPGA.	
Study Goals	You (at the end of the course) have learned: Two fundamental notions of digital systems, namely combinational and sequential networks, and the basics of VHDL/Verilog. You will be able to: 1- To translate non-digital functions to a digital representation. 2- To establish the needed combinational and/or sequential networks that have to obey specific design requirements (e.g., area, timing, and functionality). 3- To model and/or understand simple circuits in VHDL/Verilog. Finally you have gained enough knowledge to follow more advanced course, such as, computer architecture, computer systems, and embedded systems.	
Education Method	Lectures and lab.	
Assessment	* Multiple-choice exam and successful completion of lab. * Weekly online tests (5x) that can lead to 10% of the final grade This rule does not apply for resits. There are two requirements for students to pass the course: --- - Obligatory Lab: Students must successfully complete the lab. Only successful completions are registered. - Written Exam: you must take a written exam comprising at least 20 multiple-choice questions and achieve a passing grade of 6 or higher (considering 90% for the final exam and 10% for the weekly quizzes). Resit & Repair -------------- - Each year, you have two opportunities to take the exam: the regular exam in the same quarter and a resit in the subsequent quarter. This follows the exam regulations, which also define the validity duration of the exam results. - If you do not complete the lab assignments on time, there will be an extra lab session to finish the assignments. Online Quizzes -------------- To motivate you to study consistently throughout the quarter, an online test system via Brightspace is set up with up to 5 quizzes, each containing 4 multiple-choice questions. The questions are made available online in sets of 4 over a period of 5 weeks, covering material from the preceding weeks. These weekly quizzes contribute 10% to the final grade. Note that this online test result will only count for the first upcoming exam in the same quarter and will not count for the resit or for exams in subsequent years. Study Groups ------------ Interested and motivated students have the opportunity to interact more with the lecturer through study groups, which meet 5 times over 5 weeks. Registration is required to participate, as the number of seats is limited. A task for students in these study groups is to create multiple-choice questions that may be used in the upcoming online quizzes. After discussion and vetting by the lecturer, each group will select up to 4 questions for the weekly quizzes. Each student in the group will receive a pass for the questions selected in the meeting. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Book and lecture slides, excluding old exams and tests.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, Minor Electronics for Robotics, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ing. A.M.J. Slats	

CSE2430	Operating Systems	5
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The lab component requires hardware, which will be made available to registered participants in the course. A deposit is required for the hardware, which will be returned when the hardware is handed in.	
Course Contents	This course gives an introduction to the design and implementation of Operating Systems, the software that manages hardware and software resources on modern computing machinery. Topics discussed in class include: - function and structure of Operating Systems, system calls, interrupts, and exceptions, dual-mode execution, kernel architecture, device drivers - processes, running multiple processes, and the various algorithms for CPU scheduling and their properties - threads, inter-process communication, programming for concurrent execution and the danger of race conditions - process synchronization, critical sections, mutual exclusion, and mechanisms for these, deadlocks and solutions - memory management techniques, virtual memory, paging and page table management - protection and security, access control, security policies and mechanisms, side-channel attacks - IO and storage, programming with IO, hard disk drives, disk scheduling, SSDs, RAID - file systems, block allocation, directory structures, file systems in practice; - virtualization, emulation, virtual machines. Concepts and principles discussed in in this course are generally valid for any operating system, specific implementations like Linux are used as illustrations.	
Study Goals	Students are able to describe the fundamental problems and solutions in Operating Systems (resource management, dealing with concurrent tasks, abstracting from hardware) and apply them to practical problems. Students are able to choose and defend solutions to new systems problems. Students are able to design and implement basic operating system algorithms such as process/thread scheduling, thread synchronization, page replacement, disk IO scheduling, etc. Students are able to conduct experiments with the goal of evaluating system performance, scalability, and other non-functional requirements.	
Education Method	Lectures and Lab assignments in Linux on the Raspberry Pi. Homework to practice exam-like questions are handed out during the semester. The homework are self-assessed with an option to discuss during lectures. Additional feedback is possible during lab sessions.	
Practical Guide	Lab manual, available on Brightspace	
Books	Operating System Concepts by Silberschatz, Galvin & Gagne, 10th edition, 2021.	
Assessment	The final grade of the course consists of the following components: - Mandatory Lab (pass/fail) consisting of: --- Multiple mandatory lab assignments with individual deadlines (pass/fail) --- Return of the borrowed hardware, if hardware is used (note that the lab is being redesigned not to necessitate hardware) (pass/fail) - Written exam (100%) Final grade calculation: - Mandatory Lab failed -> 1 - Mandatory Lab passed and Exam Grade < 5.8 -> Exam Grade - Mandatory Lab passed and Exam Grade >= 5.8 -> Exam Repair possibilities: - Mandatory Lab - Assignments: each assignment may be resubmitted and re-sign-offs may be attempted an unlimited amount of times before its corresponding deadline, as long as TA capacity permits. Based on previous years this effectively means that students who start early can do 2-4 sign-off attempts. Additionally, exactly one assignment (for which no passing result was obtained before its deadline) can be repaired. In this case, the deadline for the assignment being repaired and for all subsequent lab assignments moves by one week and normal policy applies (unlimited resubmissions and re-sign-offs under the conditions stated above). This moving of subsequent deadlines is to give the student breathing room to catch back up. - Written exam: a resit is provided in the next quarter - Mandatory Lab - Hardware Hand-in (if hardware is used, note that the lab is being redesigned not to need hardware): failure to hand in the borrowed hardware at the scheduled moment results in a failure for the mandatory lab. An individual appointment with the instructor can be made in the quarter after the course runs to still return the borrowed hardware. Returning the hardware at such an appointment is counted as a repair for this component. Transferring partial results to different academic years: - Lab results (pass/fail) from a previous year can be honored if the content has not changed. Prior approval from the instructors is required for lab results to be transferred from a previous year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	Email: os-cs-ewi@tudelft.nl. Please send any emails related to the course to this address.	
Co-Instructor	K. Atasu	

EE3D11	Computer Architecture and Organisation	5
Responsible Instructor	Prof.dr.ir. S. Hamdioui	
Instructor	Dr.ir. M. Taouil	
Instructor	Dr.ing. M. Shahraki Moghaddam	
Instructor	Ing. A.M.J. Slats	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	Digital systems or equivalent course on logic design Basic concepts of programming in C	
Course Contents	This course provides an overview of the architecture and organization of a computer hardware system and the important principles of computer organization. In addition, the course addresses also object-oriented programming techniques. The course demonstrates the interrelation between hardware and software, and illustrates how the computers operates and how they can be programmed, with the emphasis on processor design and implementation. Topics discussed are Computer system overview, Measuring and comparing performance, ISA: instruction set architecture (MIPS, x86, 8051, JVM), Computer arithmetic, processor implementation, fast processor implementation, Memory hierarchy and caching, Interfacing, Modern architectures and organizations (superscalar, VLIW/IA64/Itanium), etc.	
Study Goals	The main aims of the course are: 1. Understand the basics of computer abstraction and performance: what are the different computers? How to evaluate performance? What is the impact of technology scaling? Etc. 2. Understand and use the computer language: What is an ISA and how to write and extend it? How does it support procedures in hardware? Impact of language selection of hardware design? Etc. 3. Understand and perform computer arithmetic: how to perform addition, subtraction, multiplication and division for integers and floating point? What is the impact of the algorithm on hardware implementation? Etc. 4. Understand and use the principle and techniques used to design a processor: how does hardware execute instructions? What are the performance characteristics and limitations? How to accelerate the performance? Etc. 5. Understand the memory hierarchy and its impact on performance: What is the memory hierarchy? What are the characteristics and limitations of each hierarchy level? What are the different alternatives to improve the performance? Etc. 6. I/O systems and their interaction with the processor: what are the different I/O? How does the communication take place? How are the dependability, reliability and availability requirements for each I/O? etc. 7. Understand and describes the trends and the challenges in the field of computer architecture. 8. Use object-oriented programming techniques in C++ such as: classes and objects (definition and implementation) and inheritance (types, multiple inheritance, redefinition).	
Education Method	Lectures + lab	
Literature and Study Materials	Lecture Notes Computer Organization & Design MIPS Edition: The Hardware/Software Interface, by David A. Patterson &John L. Hennessy; 6th Edition; Morgan Kaufmann E-book "C++ Language Tutorial," Juan Soulie, www.cplusplus.com/doc/tutorial	
Assessment	The assessment consists of the following parts: 7 Lab assignments: Pass/Fail 6 homework: HW1, HW2, HW3, HW4, HW5, and HW6 (weekly). 1 written exam: WE Final grade of the first exam If [(Lab passed) AND(HW>=6)] then: 'Finale grade' = max((0.20*HW + 0.80*WE), WE) Where HW is the weighted average grade of all homework; HW = (HW1 + HW2 + HW3 + 2*HW4 + 2*HW5 +HW6)/ 8. WE is the grade of the written exam. Final grade of retake exam: If (Lab passed) then: 'Finale grade' = WE Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Judgement		
Co-Instructor	Dr.ir. M. Taouil	
Co-Instructor	Dr.ing. M. Shahraki Moghaddam	

IFEEMCS010500	Probability and Statistics	3
Responsible Instructor	Dr. R.J. Fokkink	
Responsible Instructor	Dr. C. Kraaikamp	
Contact Hours / Week x/x/x/x	x/x/4/x or x/x/x/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5	
Course Language	Dutch English	
Expected prior knowledge	Linear Algebra, Calculus for Engineering part 1 and part 2 or Calculus for Science part 1 and part 2.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test	
Study Goals	At the end of the course, you will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Use a computer program to perform elementary simulation and data analysis	
Education Method	Interactive lectures, pre-lecture videos, exercises	
Literature and Study Materials	Book: Mathematical introduction to Probability and Statistics, Dekking et al.	
Assessment	One midterm (week 5) on Part 1: Probability Theory and one endterm (week 10) on Part 2: Statistics. Final grade is the average of both tests. To pass the course, the grade for part 2 needs to be at least 4 or higher.	
Permitted Materials during Tests	1) Formula sheet to be used for exams of Probability and Statistics 2) Any kind of calculator	
Remarks	Please note that this course is available in Q3 and in Q4.	
Co-Instructor	Dr. C. Kraaikamp	
Co-Instructor	Dr. R.J. Fokkink	

IFEEMCS011100	Calculus for Science, part 1	3
Responsible Instructor	Dr. N.D. Verhulst	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Course Contents	The goal of the course is to lay a strong foundation in calculus of functions in one variable. We will discuss various techniques to differentiate, integrate, and solve differential equations. Applications in different fields will be discussed. Both a deep knowledge of the theoretical foundations and accuracy in computations will be important.	
Study Goals	At the end of this course, the student will be able to - Use differentiation for linearising a function and apply differentials. - Perform vector arithmetic (addition, scalar multiplication, dot product, cross product). - Work with functions and related concepts (domain, range, graphs of functions, injectivity,etc.). - Perform the basic operations of calculus (differentiation, product laws, chain rule, etc.). - Apply integration techniques for functions of one variable (substitution rule, integration by parts). - Solve simple first-order differential equations by analytical methods. - Draw and interpret slope fields - Do arithmetic with complex numbers, both in Cartesian and polar form (addition, multiplication, division, powers, etc.). - Use complex numbers, perform complex arithmetic in both cartesian and polar form.	
Education Method	Colstruction	
Assessment	One written exam at the end.	
Co-Instructor	Dr. C.O. Smet	

IFEEMCS011200	Calculus for Science, part 2	3
Responsible Instructor	Dr. N.D. Verhulst	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Science, part 1 or WI1708TH1 (EE/AE).	
Course Contents	The course starts with the introduction of a method to solve general second order linear differential equations. But the main focus will be on series. Convergence and absolute convergence of series takes pride of place. Taylor series will be introduced and used to solve differential equations and initial value problems. A start will be made with multivariate calculus. In particular, continuity, limits, partial derivatives, and linearizations of multivariate functions will be discussed.	
Study Goals	At the end of this course, the student is able to - Solve homogeneous and non-homogeneous second order differential equations. - Recall and apply various tests for (absolute) convergence of series. - Give precise definitions and relevant theorems about a.o. alternating series, power series, binomial series, etc. - Compute Taylor series and Taylor polynomials. - Solve differential equations using series. - Find limits and partial derivatives of multivariate functions. - Compute Tangent planes, linearizations, and differentials for multivariate functions.	
Education Method	Colstruction	
Assessment	One written exam at the end.	
Co-Instructor	Dr. C.O. Smet	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging programme CESE for HBO EE 2024

CSE1100	Introduction to Programming	5
Responsible Instructor	Prof.dr. A.E. Zaidman	
Responsible Instructor	Ir. T.A.R. Overklift Vaupel Klein	
Contact Hours / Week x/x/x/x	6/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introductory programming course focused on programming in Java. The course covers: - Primitive datatypes: value domains, operators, expression, declarations and assignments. - Control structures: conditional selection, repetition. - Composition of data: arrays, Strings, ArrayLists, Sets. - Classes: specification, implementation, equality & hashcodes. - Inheritance: modeling inheritance, type casts, polymorphism, static and dynamic binding. - Use of libraries (APIs). - Code quality: best practices, unit testing (with JUnit), debugging strategies. - Exception handling. - Generics. - Basic IO operations (user input / reading / writing). - Design & implementation of small programs. - Functional programming in Java (streams, map, filter).	
Study Goals	After completing this course, the student: - Can write methods and classes using the Java programming language. --- Can write methods and classes using correct indentation and coding style. --- Can write equals and hashCode methods for a class using the Java programming language. --- Can create and manipulate a data structure using methods and classes. - Can design, implement and debug a simple program that contains inheritance and interfaces using the Java programming language. - Can design, implement and debug a simple program that reads from files, and writes to files using the Java programming language. - Can write unit tests for a program in Java using the Junit testing framework. - Can effectively search for existing packages, classes and programming solutions, using the Java documentation as well as online resources, and apply this information in their own programs. - Can describe key characteristics of different programming paradigms and the Java programming language.	
Education Method	Lectures, tutorials, and practical sessions with programming exercises. The programming exercises are feedback assignments, i.e., they do not count towards your grade, but you can get feedback on them to improve your programming skills.	
Assessment	This course contains two summative exams: 1) Theoretical exam in Weblab (Multiple-Choice, Open & (Small) Programming questions; no auxiliary materials may be used). 2) Programming exam (Open book, IDE & additional resources/documentation allowed). This course contains two formative tests. These allow you to practice programming in a real exam environment: 1) A Midterm Theoretical Exam (in week 5). 2) A Mock Programming Exam (in week 8). Between the two formative tests you can score a bonus on your final grade. Bonus = MIN(10, Midterm Bonus + Mock Exam Bonus). Your final grade is composed of: 1) The Theoretical Exam (weight 0.5, minimal required grade 5/10) 2) The Programming Exam (weight 0.5, minimal required grade 5/10) 3) The bonus score you can obtain by participating (and scoring points) in the midterm exam and mock exam. Your final grade is calculated as follows: Exam = (Theoretical Exam Grade + Programming Exam Grade) /2 Final = Exam + (Bonus x (10 - Exam) / 100) (This formula linearly scales down the amount of bonus you get as your exam score increases, and follows the directions of the Board of Examiners for bonus points). If your score for the Theoretical Exam and Programming Exam are (1) both minimally a 5/10 and (2) the "Final" result of the grading formula is at least 5.8, you pass this course. Within the academic year you can resit the Theoretical Exam and the Programming Exam in quarter 2. Out of the two attempts for the Theoretical Exam and the Programming Exam respectively, the highest score will count towards your final grade. Partial grades for the Theoretical Exam or Programming Exam cannot be carried over to a next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	This course makes use of a course email: ip-cs-ewi@tudelft.nl	
Co-Instructor	Prof.dr. A.E. Zaidman	

CSE1405	Computer Networks	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/0/4 lecture; 0/0/0/4 shared lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Computer networks form the basis of all digital communication. They are typically structured in layers. Each layer has clearly defined functions and algorithms, which we will discuss in class. Furthermore, the course also discusses potential security issues and defences in computer networks, thereby also providing an introduction to cryptographic notions utilised by those defence mechanisms.	
Study Goals	1) The student can describe the principles of layered architectures, in particular the functions and purposes of the various network layers. 2) The student can describe the core algorithms per layer and provide arguments why these algorithms fulfil the function of the respective layer. 3) The student can apply the core algorithms to toy examples. 4) The student can identify potential network security issues and choose suitable mitigation measures. 5) The student can exemplify the concepts of ethical hacking and responsible disclosure.	
Education Method	Lectures (8 weeks x 2 lectures x 2 hours). During these lectures, new concepts are illustrated through the discussion of multiple examples from the book. Lab (5 weeks x 4 hours). During these labs the student gains experience with commonly used Internet software (e.g. Wireshark). Additionally the student applies given abstractions and network concepts.	
Literature and Study Materials	Computer Networks, Andrew S. Tanenbaum et al., 6th (Global) edition. We might provide additional material to further explain or practise with certain topics.	
Assessment	Written exam on paper, closed-book. There is a single resit opportunity of the exam. Exams are a combination of multiple choice questions and open questions. The final test is your final grade. The lab work is not graded. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	None	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme or an EEMCS bridging programme (CS, CESE) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris (a.k.a. MyTUDelft), separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ir. O.W. Visser	

EE2S2	Systems and Control	5
Responsible Instructor	Dr.ing. S. Grammatico	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Course Contents	In this course, we study properties of dynamic systems and how these properties can be influenced via automatic control. To this aim, we develop mathematical models for the dynamic behavior of systems, with focus on linear models. Then, we analyze the important system properties both in the time domain and in the frequency domain. Next, we introduce the concept of feedback control and discuss its effects on the system properties. Finally, we present some feedback control approaches, with an emphasis on proportional-integral-derivative controllers.	
Study Goals	After following this course you should be able to: 1. Describe simple physical systems (e.g. electrical, mechanical, electro-mechanicalsystems) via mathematical dynamic models. 2. Analyze the dynamic behavior of closed-loop systems described by state-space models or transfer functions in terms of step and impulse responses, zeros and poles. 3. Interpret Bode diagrams of 1st and 2nd order systems, and sketch Bode diagrams of systems given via a transfer function. 4. Design controllers for linear dynamical systems. 5. Apply the above approaches and methods via dedicated software.	
Education Method	Lectures; Instruction	
Literature and Study Materials	Required: G. F. Franklin, J. D. Powell, and A. Emami-Naeini, Feedback Control of Dynamic Systems 7th edition	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) - Lab assignments (graded pass/fail) Repair possibilities: - A resit - Resit for the course labs In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	G. Pantazis	

EE3D11	Computer Architecture and Organisation	5

Responsible Instructor	Prof.dr.ir. S. Hamdioui
Instructor	Dr.ir. M. Taouil
Instructor	Dr.ing. M. Shahraki Moghaddam
Instructor	Ing. A.M.J. Slats
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 prac
Education Period	3
Start Education	3
Exam Period	3 5
Course Language	English
Expected prior knowledge	Digital systems or equivalent course on logic design Basic concepts of programming in C
Course Contents	This course provides an overview of the architecture and organization of a computer hardware system and the important principles of computer organization. In addition, the course addresses also object-oriented programming techniques. The course demonstrates the interrelation between hardware and software, and illustrates how the computers operates and how they can be programmed, with the emphasis on processor design and implementation. Topics discussed are Computer system overview, Measuring and comparing performance, ISA: instruction set architecture (MIPS, x86, 8051, JVM), Computer arithmetic, processor implementation, fast processor implementation, Memory hierarchy and caching, Interfacing, Modern architectures and organizations (superscalar, VLIW/IA64/Itanium), etc.
Study Goals	The main aims of the course are: 1. Understand the basics of computer abstraction and performance: what are the different computers? How to evaluate performance? What is the impact of technology scaling? Etc. 2. Understand and use the computer language: What is an ISA and how to write and extend it? How does it support procedures in hardware? Impact of language selection of hardware design? Etc. 3. Understand and perform computer arithmetic: how to perform addition, subtraction, multiplication and division for integers and floating point? What is the impact of the algorithm on hardware implementation? Etc. 4. Understand and use the principle and techniques used to design a processor: how does hardware execute instructions? What are the performance characteristics and limitations? How to accelerate the performance? Etc. 5. Understand the memory hierarchy and its impact on performance: What is the memory hierarchy? What are the characteristics and limitations of each hierarchy level? What are the different alternatives to improve the performance? Etc. 6. I/O systems and their interaction with the processor: what are the different I/O? How does the communication take place? How are the dependability, reliability and availability requirements for each I/O? etc. 7. Understand and describes the trends and the challenges in the field of computer architecture. 8. Use object-oriented programming techniques in C++ such as: classes and objects (definition and implementation) and inheritance (types, multiple inheritance, redefinition).
Education Method	Lectures + lab
Literature and Study Materials	Lecture Notes Computer Organization & Design MIPS Edition: The Hardware/Software Interface, by David A. Patterson &John L. Hennessy; 6th Edition; Morgan Kaufmann E-book "C++ Language Tutorial," Juan Soulie, www.cplusplus.com/doc/tutorial
Assessment	The assessment consists of the following parts: 7 Lab assignments: Pass/Fail 6 homework: HW1, HW2, HW3, HW4, HW5, and HW6 (weekly). 1 written exam: WE Final grade of the first exam If [(Lab passed) AND(HW>=6)] then: 'Finale grade' = max((0.20*HW + 0.80*WE), WE) Where HW is the weighted average grade of all homework; HW = (HW1 + HW2 + HW3 + 2*HW4 + 2*HW5 +HW6)/ 8. WE is the grade of the written exam. Final grade of retake exam: If (Lab passed) then: 'Finale grade' = WE Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Judgement	
Co-Instructor	Dr.ir. M. Taouil
Co-Instructor	Dr.ing. M. Shahraki Moghaddam

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS010500	Probability and Statistics	3
Responsible Instructor	Dr. R.J. Fokkink	
Responsible Instructor	Dr. C. Kraaikamp	
Contact Hours / Week x/x/x/x	x/x/4/x or x/x/x/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5	
Course Language	Dutch English	
Expected prior knowledge	Linear Algebra, Calculus for Engineering part 1 and part 2 or Calculus for Science part 1 and part 2.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test	
Study Goals	At the end of the course, you will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Use a computer program to perform elementary simulation and data analysis	
Education Method	Interactive lectures, pre-lecture videos, exercises	
Literature and Study Materials	Book: Mathematical introduction to Probability and Statistics, Dekking et al.	
Assessment	One midterm (week 5) on Part 1: Probability Theory and one endterm (week 10) on Part 2: Statistics. Final grade is the average of both tests. To pass the course, the grade for part 2 needs to be at least 4 or higher.	
Permitted Materials during Tests	1) Formula sheet to be used for exams of Probability and Statistics 2) Any kind of calculator	
Remarks	Please note that this course is available in Q3 and in Q4.	
Co-Instructor	Dr. C. Kraaikamp	
Co-Instructor	Dr. R.J. Fokkink	

IFEEMCS011100	Calculus for Science, part 1	3
Responsible Instructor	Dr. N.D. Verhulst	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Course Contents	The goal of the course is to lay a strong foundation in calculus of functions in one variable. We will discuss various techniques to differentiate, integrate, and solve differential equations. Applications in different fields will be discussed. Both a deep knowledge of the theoretical foundations and accuracy in computations will be important.	
Study Goals	At the end of this course, the student will be able to - Use differentiation for linearising a function and apply differentials. - Perform vector arithmetic (addition, scalar multiplication, dot product, cross product). - Work with functions and related concepts (domain, range, graphs of functions, injectivity,etc.). - Perform the basic operations of calculus (differentiation, product laws, chain rule, etc.). - Apply integration techniques for functions of one variable (substitution rule, integration by parts). - Solve simple first-order differential equations by analytical methods. - Draw and interpret slope fields - Do arithmetic with complex numbers, both in Cartesian and polar form (addition, multiplication, division, powers, etc.). - Use complex numbers, perform complex arithmetic in both cartesian and polar form.	
Education Method	Colstruction	
Assessment	One written exam at the end.	
Co-Instructor	Dr. C.O. Smet	

IFEEMCS011200	Calculus for Science, part 2	3
Responsible Instructor	Dr. N.D. Verhulst	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Science, part 1 or WI1708TH1 (EE/AE).	
Course Contents	The course starts with the introduction of a method to solve general second order linear differential equations. But the main focus will be on series. Convergence and absolute convergence of series takes pride of place. Taylor series will be introduced and used to solve differential equations and initial value problems. A start will be made with multivariate calculus. In particular, continuity, limits, partial derivatives, and linearizations of multivariate functions will be discussed.	
Study Goals	At the end of this course, the student is able to - Solve homogeneous and non-homogeneous second order differential equations. - Recall and apply various tests for (absolute) convergence of series. - Give precise definitions and relevant theorems about a.o. alternating series, power series, binomial series, etc. - Compute Taylor series and Taylor polynomials. - Solve differential equations using series. - Find limits and partial derivatives of multivariate functions. - Compute Tangent planes, linearizations, and differentials for multivariate functions.	
Education Method	Colstruction	
Assessment	One written exam at the end.	
Co-Instructor	Dr. C.O. Smet	

TN2545	Systems and Signals	6
Responsible Instructor	Dr. F.M. Vos	
Instructor	Dr.ir. D.J. Verschuur	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	2A 2B	
Start Education	2A	
Exam Period	2A 2B 3A	
Course Language	English	
Expected prior knowledge	Complex numbers, elementary calculus, including derivatives & integrals, solving linear equations with few unknowns, geometric series, trigonometric & exponential functions. Create and analyze graphs of elementary functions	
Course Contents	This course covers the following topics: Systems and signals in continuous and discrete time; Basic deterministic signals; Convolution and properties of LTI systems; Fourier series and Fourier transforms with their properties; Filters in theory and practice: first- and second-order DE, Butterworth filters, Gaussian derivative filters; the estimation of spectra in a time window and the associated uncertainty relation; the DFT and FFT as practical tools; introduction to 2D Fourier transforms; modulation and demodulation techniques; the analytic signal and Hilbert transforms; sampling (Nyquist theorem), aliasing and reconstruction; laplace and Z-transforms with their ROC and properties; feedback systems and associated stability determination.	
Study Goals	The main learning goals of this course on Systems and Signals are to enable that a student can: 1. recognize and determine basic properties of systems including linearity, time invariance, impulse response, and determine their response based on convolution. 2. derive Fourier series coefficients and Fourier transform of elementary signals. 3. describe the effect of frequency-dependent filtering of discrete and continuous-time signals and relate this to linear difference/differential equations with constant coefficients. 4. analyze the effect of sampling of band-limited signals and reconstruct a continuous-time signal from the discrete samples. 5. perform amplitude-modulation to elementary signals and derive and analyze associated reconstruction methods. 6. assess properties of systems based on the Laplace and Z-transform.	
Education Method	Oral lectures by teachers, practical working classes with supervision	
Literature and Study Materials	A.V. Oppenheim and A.S.Willsky, with S.H.Nawab, Signals and Systems, Prentice-Hall, 2nd edition, Ch. 1 to 11 (obtainable in bookshop and VVTP). Reader with additional material en lecture notes/hand-outs.	
Assessment	There are 3 bonus tests during the teaching period before the exam takes place. They are optional and can give a student a bonus to the final grade (10% contribution per bonus test). If a bonus test grade (BG) is lower than the exam grade (EG) it will not contribute. Therefore, the final grade (FG) is computed as one of these options, depending on the bonus grades: FG = 0.7 EG + 0.1 (BG1+BG2+BG3) FG = 0.8 EG + 0.1 (BG1+BG2) FG = 0.9 EG + 0.1 BG1 FG = 1.0 EG Bonus test results are only valid in the academic year in which they are obtained.	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging programme CESE for TUD BSc ME/AE/AP 2024

CSE2420	Digital Systems	5
Responsible Instructor	Dr.ing. M. Shahraki Moghaddam	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 8/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	No prior knowledge is expected from the student.	
Course Contents	This course introduces the following aspects in digital design: - combinational digital systems, - sequential systems, - complex combinatial and sequential building blocks - reconfigurable hardware and complex digital systems. All the important aspects of digital design will be dicussed: - number systems, - Boolean algebra, - combinational logic, - Two-level and multilevel networks and reduction techniques/design tools, - timing hazards, - Moore/Mealy machines, - Finite State Machines (FSM) and more. The lab starts with the realization of simple circuits and will end with complex systems realized on an FPGA.	
Study Goals	You (at the end of the course) have learned: Two fundamental notions of digital systems, namely combinational and sequential networks, and the basics of VHDL/Verilog. You will be able to: 1- To translate non-digital functions to a digital representation. 2- To establish the needed combinational and/or sequential networks that have to obey specific design requirements (e.g., area, timing, and functionality). 3- To model and/or understand simple circuits in VHDL/Verilog. Finally you have gained enough knowledge to follow more advanced course, such as, computer architecture, computer systems, and embedded systems.	
Education Method	Lectures and lab.	
Assessment	* Multiple-choice exam and successful completion of lab. * Weekly online tests (5x) that can lead to 10% of the final grade This rule does not apply for resits. There are two requirements for students to pass the course: -- - Obligatory Lab: Students must successfully complete the lab. Only successful completions are registered. - Written Exam: you must take a written exam comprising at least 20 multiple-choice questions and achieve a passing grade of 6 or higher (considering 90% for the final exam and 10% for the weekly quizzes). Resit & Repair -------------- - Each year, you have two opportunities to take the exam: the regular exam in the same quarter and a resit in the subsequent quarter. This follows the exam regulations, which also define the validity duration of the exam results. - If you do not complete the lab assignments on time, there will be an extra lab session to finish the assignments. Online Quizzes -------------- To motivate you to study consistently throughout the quarter, an online test system via Brightspace is set up with up to 5 quizzes, each containing 4 multiple-choice questions. The questions are made available online in sets of 4 over a period of 5 weeks, covering material from the preceding weeks. These weekly quizzes contribute 10% to the final grade. Note that this online test result will only count for the first upcoming exam in the same quarter and will not count for the resit or for exams in subsequent years. Study Groups ------------ Interested and motivated students have the opportunity to interact more with the lecturer through study groups, which meet 5 times over 5 weeks. Registration is required to participate, as the number of seats is limited. A task for students in these study groups is to create multiple-choice questions that may be used in the upcoming online quizzes. After discussion and vetting by the lecturer, each group will select up to 4 questions for the weekly quizzes. Each student in the group will receive a pass for the questions selected in the meeting. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Book and lecture slides, excluding old exams and tests.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, Minor Electronics for Robotics, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ing. A.M.J. Slats	

CSE2425	Embedded Software	5
Responsible Instructor	Q. Wang	
Contact Hours / Week x/x/x/x	0/4/0/0 lecture; 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The Embedded Software course offers an introduction to real-time programming in an embedded systems context. The student acquires the principles of real-time programming in C (interrupts, multitasking) through a simple Real-Time Operating System (RTOS). The course looks at the principles of an RTOS, its operation, and how real-time applications may be programmed using an RTOS. The course also introduces the principles of running embedded machine learning models on embedded devices. The lecture will be accompanied by a practicum consisting of a tutorial on C programming and a hands-on lab in which students will program a home-brew robot to follow a line. The robot consists of an STM32F401 microcontroller board mounted on a robot base with two wheels, two infrared sensors, two ultrasonic distance sensors, among others.	
Study Goals	The essence of Embedded Software is how to perform real-time programming in an embedded system context. By the end of this course, you should be able to (LO: Learning Objective): LO1: Illustrate the principles of real-time programming on top of embedded real-time operating systems: LO1-a. Elaborate on the characteristics of embedded-systems programming. LO1-b. Formulate the principles of real-time programming. LO1-c. Explain the mechanisms behind embedded real-time operating systems. LO2: Employ the theoretical and experimental training on embedded systems to: LO2-a. Differentiate theoretically and experimentally the principles of good embedded systems designs. LO2-b. Quantify theoretically the estimated time on achieving an embedded-software project. LO3: Elaborate the principles of embedded machine learning systems. LO4: Program a car robot to follow a line by employing the training on embedded systems design, the necessary development environment, and toolchain.	
Education Method	Lectures (20 hrs) + programming lab (72 hrs) + self study (48 hrs)	
Literature and Study Materials	See course website	
Books	"An embedded systems primer", David E. Simon Addison Wesley, ISBN 0-201-61569-X	
Assessment	Multiple-choice and open questions exam + programming assignments. The end mark is a combination of the lab results (50%) and the written exam (50%); both parts (exam, lab) must be passed with a minimum mark of 5. A resit will be scheduled for the exam; individual lab assignments that do not meet the quality constraints may be retried once. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None (closed-book exam)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Prof.dr. K.G. Langendoen	

EE3115TU	Digital Communication Systems	4
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0 lecture (10 times) - 2/0/0/0 instruction (4 times)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Mathematics in general. Some basic knowledge on linear systems and signals, the Fourier transform and stochastic processes, will be an advantage.	
Summary	Transmission and distribution of information by means of telecommunication techniques form the backbone of our modern society. According to a binding international treaty among more than 180 countries, "telecommunication" covers every transfer, transmission or reception of signals, tokens, sounds, images or other data of any kind (including ideas), by means of electromagnetic systems, e.g. via radio-electric or optical means. The course Digital Communication Systems provides mathematical models and methods to describe and analyse communication systems for the transmission of signals that carry information in digital form.	
Course Contents	In this course, mathematical models and methods are given to describe and evaluate communication systems for the transmission of digital signals: - basics of signals and systems: * description of signals and decibel notation, * time-domain versus frequency-domain: LTI-systems, convolution, impulse response, Fourier-transformation, transfer function, filtering and definition of bandwidth, - propagation loss in guided (cable) and unguided (wireless) media, - noise and system noise calculations (link-budget), - signal sampling, analogue pulse modulation (PAM, PCM, Delta-modulation), - digital waveforms in baseband, power spectral density and spectral efficiency, - line codes, - bandpass signals and -systems: complex baseband description, modulation, demodulation, generic transmitter and receiver concepts: super heterodyne and direct conversion receiver, - digital modulation techniques (OOK, FSK, BPSK, QPSK, MSK, QAM), - matched filtering, - bit error probability after detection of digital signals.	
Study Goals	You have gained insight in the basic concepts of signal processing for telecommunications, as described under Course Contents, especially those related to the transmission of digital signals, and is able to apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures and working lectures. Recorded lectures will be made available via Collegerama. To support studying of the material of this course, sets of exercises are made available for each lecture and each working lecture via Brightspace. These exercises with answers will help you in mastering the course material.	
Literature and Study Materials	Couch, L.W., Digital and Analog Communication Systems, 8th edition, ISBN 978-0-273-77421-1, Prentice Hall, 2013.	
Assessment	The course is completed with a 3 hour written exam covering the full course content at the end of the 1st or 2nd quarter (resit). The final grade of Digital Communication Systems is based solely on the grade obtained for the exam, which is is rounded to the nearest half. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	The exam Digital Communication Systems is a closed-book exam. During the exam, you are allowed to use the following materials: - the overview Tables and Figures to be used at the exam Digital Communication Systems (EE3315TU), as made available on Brightspace, - an original own handwritten equation sheet: 1 page A4, written on both sides, (not type-written, no copy and it should not contain worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks		

EE3125TU	Advanced Electronics for Robotics	5
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Modern robots make extensive use of many sensors and actuators and rely for their power on high-performance batteries that need proper care for safety and longevity reasons. Often, especially for space systems and swarm robots sent to harsh environments, the robots will be equipped with solar panels. The quality of operation of these robots strongly depends on the quality of the sensor readout systems and the power efficiency of the electronics that drive the actuators and convert the solar energy to energy stored in the batteries. For an optimal design of these robots, every member of the multi-disciplinary design team must be part of design discussions, bringing in their expertise domain and understanding the essence of the design problems and opportunities in the other disciplines. This course intends to enable the students to function like this for Analog Circuit Design and Power Systems Design by introducing a systematic design methodology for Analog Electronic Circuits and Power Electronics. The principles of power conversion are taught considering solar-panel/battery charging systems and motor drivers. The systematic analog circuit design principles are applied to negative feedback amplifiers for sensor applications. Still, they apply to other electronics circuits, like oscillators, filters, and other feedback systems like power converters. The course intends to provide insight to the students into how these systems can be optimally designed by using the proper models at the right time, applying for the correct order in the design, and what design rules exist and how they are applied. It introduces a method to systematically separate a complex design problem into less complicated orthogonal sub-problems that are easier to solve, resulting in the optimal global solution. This insight will enable the students to understand circuit and system designs created by expert designers and critically evaluate the performance and correctness of complex structures. It will allow the students to specify circuit and system designs correctly and contribute effectively to interdisciplinary discussions on design decisions with other experts. The course treats: For analog electronics, the lectures focus on a systematic way to design a negative feedback amplifier with an operational amplifier used as controller. Attention is paid to noise, bandwidth and drive capability requirements for the operational amplifier and the feedback network. A synthesis model for negative feedback amplifiers will be discussed that includes the influence of the source and load impedance. A bandwidth analysis method will be discussed that can be used to find the performance requirements for the operational amplifier and how to implement frequency compensation using the Phantom Zero method to obtain the required frequency characteristics. For power systems, the principle of operation, analytical modelling and design of basic dc-dc power converters are presented. Also, the particularities of the solar-panels, such as the Maximum Power Point Tracking (MPPT), and battery charging principles and management systems and motor drivers. The influence in the selection of the control methods for the power electronics will be investigated. All this knowledge will be combined in order to derive a suited operational solution for the application at hand. The study will be verified using computational simulation tools. During the lectures many practical demonstrations will be given.	
Study Goals	After the lectures, the student: is be able to select a correct topology for a negative feedback amplifier matching with the sensor specifications is able to identify flaws in an amplifier based on its Transmission-1 matrix is able to use the nullor to ease amplifier synthesis is able to use the asymptotic gain model is be able to do a noise analysis for an amplifier is able to do a quick bandwidth prediction for an amplifier is able to evaluate the frequency performance of an amplifier is able to apply Phantom Zero frequency compensation is able to use the proper device and system models at every design phase and avoid unnecessary complexity knows the optimal design order is able to see the structure in a complex amplifier design and identify the potential flaws understands the particularities and requirements of renewable energy source based on photovoltaics, battery charging systems and motor drivers is be able to identify, analyse, control, and specify appropriate power electronics and their respective circuit components to understand power electronics circuit design trade-offs: Power Efficiency vs. Weight vs. Size vs. Cost. to verify the designs using computational aid tools suited for circuit analysis.	
Education Method	Lectures, practical demonstrations during the lectures, homework assignment, problem solving sessions and computational exercise	
Assessment	The assessment divided into two parts. The first part covers the topics of Power Electronics, the second part covers the topics of Analog Electronics. The weighted sum (60%/40%) of the results of both parts is used to compose the grade.	
Co-Instructor	A. Shekhar	

EE3130TU	Marsrover project	5
Responsible Instructor	Dr.ing. M. Shahraki Moghaddam	
Responsible Instructor	C. Galuzzi	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The Mars Rover is based on a simple vehicle that is driven by several electric stepper motors which are controlled by an onboard FPGA. In this project a group of 4 students have the task to develop a Mars Rover. The students have the task to feed the robot with solar energy and to enhance the robot with sensing and localization abilities such that the robot is capable to perform certain predefined tasks. In order to have control over the movement and the sensors the FPGA has to be programmed in VHDL. All the basic knowledge and skills needed for this project are provided by the courses in the minor.	
Study Goals	Applying the knowledge and skills obtained during the courses in the minor	
Education Method	Project in groups of 4 students	
Assessment	Delivery of a functional Mars Rover	
Co-Instructor	Dr.ing. M. Shahraki Moghaddam	
Co-Instructor	C. Galuzzi	

ET3033TU	Circuit Analysis	3
Responsible Instructor	Prof.dr.ir. L. Abelmann	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	The course requires familiarity with handling the complex numbers algebra.	
Course Contents	Electrical circuits fundamentals / basic concepts and components: current and charge, voltage and current sources, electrical circuits, circuit symbols, direct current, alternating current, resistors capacitors and inductors, Ohm's law, Kirchhoff's laws, power dissipation in resistors, series and parallel connections. Measurement of voltages and currents: common wave forms, measuring techniques and equipment. Resistance and DC Circuits: Thévenin and Norton theorems, superposition, nodal analysis, mesh analysis, (solving) circuit equations. Capacitance and electric fields: capacitors and capacitance, electric field strength and flux density, series and parallel connected capacitors, capacitor I/V relationship, sinusoidal voltage and current, energy stored in a capacitor. Inductance and Magnetic Fields: electromagnetism, reluctance, inductance, self-inductance, series and parallel connections, inductor I/V relationship, sinusoidal voltage and current, energy stored in an inductor, mutual inductance, transformers. Alternating voltages and currents: resistance vs. reactance, impedance, phasor diagrams, complex notation. Power in AC circuits: power dissipation in resistive circuits, power in capacitors and inductors, power in circuits with resistance and reactance, active and reactive power, power factor correction, three phase systems, power measurement. Frequency characteristics of AC circuits: two-port networks, the decibel, frequency response, filter networks, Bode diagrams, RLC circuits and resonance, stray capacitance and inductance. Transient behaviour: Charging of capacitors and energising of inductors, discharging and de-energising, first order systems, generalised response, second order systems, higher order systems.	
Study Goals	The course aims at building up insight and analysis instruments in the area of linear electrical circuits and components. It will also establish a solid foundation for understanding, analysing and synthesising electronic circuits using non-linear active components in both the analogue and digital domains.	
Education Method	Interactive lecturing as a preparation for more detailed self-study and engagement with the content. Lectures are not necessarily completely covering all topics.	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276. Material in the Brightspace learning environment.	
Assessment	Homeworks + written exam In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Permitted Materials during Tests	The exam is a closed book exam. However, a single-sided self-handwritten A4 as a cheat sheet is permitted. Only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines.	
Judgement	The final grade depends for 10% on the homework and for 90% on the written exam.	
maximum aantal deelnemers	100	

ET3604LR	Electronic Circuits	3
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	Major topics: Electronics in information processing (sensor readout) Design principles and trajectory Signal processing functions Electronic design analysis Negative feedback amplifiers Operational amplifiers Components and networks Semiconductor diodes Power supplies and voltage regulators Semiconductor transistors and IC technology (basic) Oscillators DC motor basics	
Course Contents	This course is intended to provide gain insight, in a structured way, in the function of electronics as information-processing technology. The way of modelling will be discussed in relation to the physical characteristics of electronic components, circuits and systems. The course will provide the students with the skills to interpret circuit diagrams and measurement results and formulate specifications for electronic circuits that enable expert electronics designers to take proper design decisions when requested to design a high-performance application specific circuit. The students will learn the terminology that enables them to understand the explanation for these decisions by the experts (and potentially critically review them with respect to system level decisions). During the course many lecture demonstrations will be given, where the students both can observe the practical behavior of the test object and see what type of measurement equipment is used and how it is operated.	
Study Goals	After this course, the student -Is able to interpret electronic circuit diagrams and draw them according to standards -Is able to recognize and select electronic components -Is able to interpret and create Bode plots that describe the frequency behavior of an electronic circuit -Is able to analyze and design basic frequency filters -Is able to explain the basic behavior of semiconductor devices, like bipolar transistors, field effect transistors, diodes and special devices like thyristors. -Is able to determine the proper readout conditions for a sensor -Is able to determine the proper drive conditions for an actuator -Is able to interpret basic diode circuits like rectifiers -Is able to explain the basics of negative feedback systems and able to calculate and interpret the essential parameters -Is able to design a negative feedback amplifier using an Operational Amplifier (OpAmp) as active circuit that optimally meets the requirements -Is able to explain the basics of power circuits -Is able to name the different oscillator classes and explain the working principle the oscillators in each class -Is able to determine the class of a given oscillator -Is able to name the specific good and bad properties of each oscillator class -Is able to communicate with experts in electronic design and translate system specifications to specifications for the electronics in such a way that the expert can create an optimal design	
Education Method	Lecture. There is a mandatory practical with its own course code: ET3604LRP	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276 (older versions can very well be used for this course)	
Assessment	The exam is multiple-choice. During the lecture period, there will be one bonus test on oscillators. A pass grade gives a bonus point for the exam. The bonus point remains valid for the re-sit exam. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging programme CS for HBO TI 2024	
Introduction 1	Students having obtained a relevant bachelors degree in Computer Science and Engineering from a Dutch institute of professional education (HBO), depending on the programme, can gain access to the Master's degree programme via a bridging programme. The bridging programmes consist of a minimum of 45 EC in the fields of computer science and mathematics.

CSE1110	Software Quality and Testing	5
Responsible Instructor	F. Mulder	
Contact Hours / Week x/x/x/x	0/0/0/4 lecture; 0/0/0/4 shared lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course covers the most important testing techniques needed to build high quality software systems. The course covers both the testing techniques themselves, and the software design techniques needed to create testable systems. Specific topics covered are: * testing principles, test levels, and the testing pyramid * software testing automation and JUnit * specification-based testing * structural testing and the different coverage criteria * boundary analysis and testing * design by contract * property-based testing * test doubles and mock objects * design for testability * test-driven development (TDD) * larger tests (SQL and web testing) * test code quality and test smells	
Study Goals	By the end of the course, students will be able to: * Create unit, integration, and system tests using current existing tools (like JUnit, Mockito, and JaCoCo) that effectively test complex software systems. * Derive test cases that deal with exceptional, corner, and bad weather cases by performing different testing techniques as well as able to reflect about their limitations, when and when not to apply them in a given context. * Measure and reflect on the efficiency of the developed test suites by means of different test adequacy metrics (i.e., line, branch, condition, MC/DC coverage). * Design and implement testable software systems by means of architectural patterns such as dependency injection and ports and adapters. * Write maintainable test code by avoiding well-known test code smells. * Understand and apply state-of-the-art techniques for intelligent software testing, such as property-based testing.	
Education Method	This course follows a flipped classroom approach. Students read chapters of the book and work on (practical and theoretical) exercises during their study time; lectures are used for questions and broader discussions. We offer TA support during the labs.	
Books	Effective Software Testing, by Maurício Aniche: https://www.effective-software-testing.com.	
Assessment	The evaluation of the course is based on one exam. It happens on WebLab and is composed of coding questions and theoretical questions. Students can consult the documentation of certain testing libraries during the exam. There is also one possibility to do a resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. C.E. Brandt	

CSE1200	Calculus	5
Responsible Instructor	N. de Kleijn	
Contact Hours / Week x/x/x/x	0/6/0/0 colstruction; 0/1/0/0 lecture	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Reasoning & Logic, High school mathematics.	
Course Contents	The main objective of this course is to provide a solid basis of mathematical concepts and skills that will be necessary in the rest of the Bachelor Computer Science & Engineering. The material treated in this course plays an important role in courses such as Algorithms and Data Structures, Linear Algebra, Computer Graphics, Probability and Statistics, Signal Processing and Image Processing.	
Study Goals	Here is a list of topics that will be treated in this course, and the associated learning objectives. Understanding of functions. The student can determine domain, range and inverse of a given function, and sketch its graph. The student can simplify compositions of functions, in particular in case of (inverse) trigonometric functions. Limits. The student knows and understands the concept of limits as well as the associated rules of calculation. The student can evaluate limits. The students knows the conditions under which 'Hospitals Rule can be applied, and can apply the rule in these cases. Differentiation. The student knows the rules of calculation for differentiation, in particular the Chain Rule. The student can apply those to determine derivatives (first order and higher) of a given function, if necessary by implicit differentiation. The student can determine linearisations and differentials of functions. Integration: The student knows and understands the definition of an integral in terms of Riemann sums, and can apply the Fundamental Theorem of Calculus to calculate integrals. The student knows the rules of calculation for integration, in particular the Substition Rule and Integration by parts, and can employ these to find anti-derivatives and evaluate integrals. Sequences and Series. The student can determine convergence and limits of a (possible recursively defined) sequence. The student masters techniques, in particular the Comparison test, Alternating Series Test and the Ratio Test, to determine convergence and absolute convergence of a given series. The student can determine the radius of convergence and interval of convergence of a given power series. The student is able to determine the limit of certain types of (power) series. Complex numbers. The student knows the rules of calculation for complex numbers and can apply those to solve algebraic equations. The student knows and can apply Euler's Identity and De Moivre's Rule. Multivariate functions. The student can sketch and interpret graphical representations of functions of several variables. The student can determine linearisations, tangent planes, directional derivatives, gradient and local extrema of multivariate functions. Integration of multivariate functions. The student can evaluate the integral of a function in two variables on a simple region and interpret the answer.	
Education Method	Every week, there will be 6 hours of lectures for this course.	
Books	For the course material and exercises we use the book "Calculus: Early Transcendentals', Interational Metric Edition, 9th edition, by James Stewart. ISBN: 978-1-305-27237-8 PLEASE NOTE: the use of older edition may cause problems, since the numbering of exercises and sections varies between editions.	
Assessment	The final grade of this course is determined by two tests; one in week 2.5 and one in week 2.10. There will be one overall retake. It is not possible to do a retake for separate tests. After the final test, the final grade for this course is calculated as follows: (grade test 1)/3 + 2*(grade test 2)/3 The course is passed if this grade is 5.75 or higher AND the grade of the second test is 5 or higher. If the grade for the resit is higher than the final grade, it will replace the final grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. C.O. Smet	

CSE1205	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	0/0/6/0 colstruction; 0/0/2/0 lecture	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Multivariate data Analysis; Computer Graphics; Image Processing	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Lineare transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Lectures blended with exercise sessions, online homework exercises and instructions. Disclaimer: information may change depending on the developments around the coronavirus.	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	The final score F is calculated as follow: IF the midterm score M is higher than the score E for the endterm AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	No material.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	

Co-Instructor	Drs. A.T. Hensbergen

CSE1210	**Probability Theory and Statistics**	**5**
Responsible Instructor	Dr. C.O. Smet	
Contact Hours / Week x/x/x/x	0/0/0/6 colstruction	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch English	
Expected prior knowledge	Calculus, in particular differentiating and integrating functions of one and two variables is indispensable.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, categorical variables, goodness of fit Confidence intervals for Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region, multiple testing, p-hacking Hypothesis tests on mean(s), proportion(s), variance, regression parameters	
Study Goals	At the end of the course, the students will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Knows how to avoid pitfalls in hypothesis testing - Use Python to perform elementary simulation and data analysis	
Education Method	Education will take place in a blended way: the student watches a prelecture video, makes some exercises related to this, participates in the lecture (which consist of lecturing by the teacher and making exercises) and then makes some more exercises at home. This cycle is repeated for every lecture.	
Literature and Study Materials	We use the "Blue Book": A Modern Introduction to Probability and Statistics Understanding Why and How Series: Springer Texts in Statistics Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover ISBN: 978-1-85233-896-1	
Assessment	The final grade of the course consists of the following components: - Midterm (50%) - Endterm (50%) Final grade calculation: (0.5 * Midterm + 0.5 * Endterm) Repair possibilities: - Resit in next quarter Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Formula sheet (one sheet, to be made by the student) and a simple, non-programmable, non-graphical calculator.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Tags	Mathematics Stochastics	
Co-Instructor	N. de Kleijn	

CSE1300	Reasoning and Logic	5
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. N. Yorke-Smith	
Course Coordinator	S. Hugtenburg	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Every course in which exact reasoning is required, e.g. to prove a theorem, or reasoning from the assumption that a theorem is true. Examples: * Calculus * Algorithms and Data Structures * Linear Algebra * Probability and Statistics * Machine Learning * Algorithm Design * Automata, Languages and Computability * Logic-Based Artificial Intelligence * Cryptography * Software Verification * Complexity Theory	
Expected prior knowledge	High school mathematics	
Course Contents	It is often useful or even essential to know if a certain statement is true, e.g. Pythagoras' famous a^2 + b^2 = c^2 theorem about right-angled triangles. Knowledge gathered from these statements or theorems can be broadly used to solve more complicated problems. This way of working, i.e., deriving more complex theorems from simpler ones, is useful in many fields, in particular also in computer science. An argument is a set of premises or assumptions, followed by a conclusion. To be certain of the truth of an argument, the conclusion has to be a logical consequence of the assumptions. To prove this, the conclusion is derived from the premises. The derivation shows us that once all premises are true, the conclusion is true as well. The course Reasoning and Logic is about proving the logical validity of arguments. What is a valid argument? When is an argument logically valid and when is it not? How can we determine whether an argument is logically valid? How can we derive a logically valid conclusion from the premises? Or how can we prove that a conclusion is not a logical consequence of the premises? In this course we will first explain a number of basic proof techniques, such as proof by contradiction, proof by mathematical induction, proof by division into cases, and the use of invariants. The application of these techniques will be practiced by proving and rejecting simple mathematical theorems. These proof techniques can only lead to a valid argument when the formulation of the premises and the conclusion is sufficiently precise. To express statements precisely, multiple artificial languages exist, of which we will learn two: propositional calculus and predicate logic. For both languages we look at the syntax and semantics and study how to translate expressions from a natural language to the more exact languages. Furthermore we will look at how to establish the logical validity of an argument in both languages. Moreover, to be able to assign truth values to formulas in predicate logic and because of the importance of this subject in every exact science, in this course we will also pay attention to elementary set theory. Together, the course provides tools that have important applications across mathematics, computer science, philosophy, and beyond.	
Study Goals	After this course, the student is able to: 1. translate a logically precise claim to and from natural language. 2. describe the operation of logical connectors and quantifiers. 3. describe the notion of logical validity. 4. explain and apply basic set and graph operations. 5. define and perform computations with functions, relations and equivalence classes. 6. construct and interpret recursive definitions, including recursive data structures like trees. 7. construct an appropriate function or relation given a description (in natural language or formal notation). 8. construct a direct or indirect proof (by contradiction, division into cases, generalisation, or [structural] induction) or logical equivalence, or counterexample for (in)valid arguments, in propositional logic, predicate logic and set theory. 9. identify what type of proof is appropriate for a given claim. 10. solve simple SAT instances. 11. develop specifications for verification tools like SAT or SMT solvers. 12. interpret the response of verification tools like SAT or SMT solvers	
Education Method	This course uses a student-centred learning methodology. Students study provided material at home. Every week has two large-group sessions, during which questions about the material will be answered. There will also be small interactive exercises in some of these sessions to help the students to evaluate their understanding of the material and to apply it. In addition there is a 1 hour tutorial session in which students tackle a larger problem as a group under the supervision of lecturers and TAs. Optional homework assignments are provided. These should be done in groups of 4. Lab sessions are offered, in which answers to the homework questions can be discussed. A project is also included in the course. This must be done individually.	
Books	Delftse Foundations of Computation, Hugtenburg & Yorke-Smith, TU Delft Open 2019, textbooks.open.tudelft.nl, ISBN 978-94-6186-952-4	
Assessment	The final grade of the course is computed from several components, described below. A final exam (100%) in week 10 (F) Additionally there are two opportunities to achieve some bonus points: Midterm exam Week 5 (M) Project, deadline Week 9 (P) Between M and P, you can score a bonus that is computed in accordance with the rules of the BoE: B = min(10, max(2*FLOOR(M-5), 0) + max(2*FLOOR(P-5), 0)) (FLOOR is rounding down the grade to the largest integer value that is smaller or equal to the grade. This means one bonus opportunity can compensate for another, but an opportunity only gets you bonus if the grade is sufficient.)	

	Final grade The final grade X for the course is computed as follows (again in accordance with the rules for bonus points of the BoE): X = F + (B*(10-F)/100) (This means the bonus scales down linearly as your summative final grade increases.) The resit (R) covers all material, and your final grade for the resit opportunity is: Y = R + (B*(10-R)/100) As per TU delft regulations, the grade considered for the course is max(X,Y). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Permitted Materials during Tests	No materials are allowed for on-campus exams.
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/
Special Information	For any administrative questions about the course, please use rl-cs-ewi@tudelft.nl and not the instructors personal e-mail addresses.
Tags	Abstract Algorithmics Logics Mathematics Project Small groups
Co-Instructor	S. Dumani

CSE1505	Information and Data Management	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	Dr. A. Katsifodimos	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic understanding relational query languages (SQL) and databases (as obtained in the Web and Database Technology Y1 Q2 course). This class also relies heavily on Algorithms and Data Structures.	
Course Contents	In this course you will learn how to model, manage, and query information in a Database Management System (DBMS). We will focus on different data modelling techniques, but also on aspects of query processing like for example query parsing or optimization. Also transaction management is a central topic of interest. The core focus is on classic relational OLTP databases, but we will also briefly touch alternative systems like the current NewSQL-trend and systems optimized for in-memory or OLAP applications. Additionally, NoSQL systems and their respective data models will be discussed. - Database normalization: modelling in an unambiguous and non-redundant manner in relational tables such that inconsistencies and anomalies are avoided. - File-organization and indexing: the data structures and their associated algorithms to store data in memory and on disk, such that it can be retrieved and processed efficiently when needed. - Query processing and optimization: the techniques that are used by a DBMS to execute certain SQL queries as efficiently as possible, taking into account how the data is stored and which indexes are present. This also covers Relational Algebra. - Transaction management and concurrency control: the techniques that a DBMS offers to allow different users and application to operate in parallel on the same data without interfering with each other or causing inconsistencies in the data. - NewSQL and alternative system: the systems beyond traditional relational database design space optimized for specialized requirements like extreme OLTP performance, OLAP queries, or in-memory storage. - NoSQL and non-relational data: in the recent years, systems not relying on the relational model became popular. We will also discuss how this affects data modelling, especially with respect to document models, key-value models, wide-columnar models, but also graph models.	
Study Goals	The course aims at teaching the basic competencies required to design and understand applications interacting with relational database management systems (RDBMS). Students will gain an insight into how RDBMS are implemented, learn about the available design choices, and the trade-offs imposed by different designs. This allows students to competently identify and solve problems in application design, and to improve and tune data models and queries. We will also deal with non-relational systems and discuss how they affect the data processing workflow. - Perform Data Modeling and be able to: -- Design and implement normalized relational schemas in a database system, by designing Entity Relationship diagrams and translating them to relational tables. -- Describe alternative data models such as the document model, key-value model, or graph models and analyze the trade-offs as in comparison to the relational model. -- Analyze the requirements of a given application scenario and be able to derive a suitable data model. - Get a grip of the internals of relational database systems by being able to: -- Describe and analyze the two main ways of storing and representing data in relational databases as well as analyze their performance trade-offs with respect to input/output (IO) costs. -- Describe different index structures and analyze their impact on query response time for range-, point- and join queries. -- Analyze the trade-offs between different physical operators for database operations such as joins, aggregates and filters. -- Describe and apply the query optimization process and design physical execution plan and compute the costs for their relational operations. -- Explain the need for transaction protocols and illustrate database update anomalies that can arise in the absence of such protocols. Be able to apply such protocols for given scenarios. -- Evaluate and apply different recovery algorithms such as UNDO/REDO logging, under different types of failures and their frequency.	
Education Method	Lectures and Homework Lab	
Computer Use	Students will require access to a personal computer. The labs will deal with larger data sets, so some HDD space (several GB) will be needed.	
Literature and Study Materials	The course and evaluation is based on the provided lecture slides and assignment material. For alternative explanation or more in -depth studies, the recommended book can be used.	
Books	Fundamentals of Database Systems by Ramez Elmasri and Shamkant Navathe, Pearson. Preferred 7th edition, but also the older ones are usable. Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, and Jennifer Widom (2nd edition preferred, older ones also OK). These books are NOT required, but might be helpful for more in-depth studies. Additional (optional) books/resources are mentioned in the lecture.	
Assessment	This year, there will only be a final exam which counts for 100% of the course grade. The lab assignments are not mandatory and will not contribute to the grade, but we STRONGLY recommend doing them anyway.	
Permitted Materials during Tests	1 handwritten sheet of A4 paper (any content chosen by the student, writing front and back permitted).	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme or an EEMCS bridging programme (CS, CESE) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ir. T.A.R. Overklift Vaupel Klein	
Co-Instructor	Dr. A.J. Power	

CSE2310	Algorithm Design	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. E. Demirovi	
Contact Hours / Week x/x/x/x	0/4/0/0 lecture; 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: programming experience from CSE1100 or equivalent Strongly advised: experience with algorithms and data structures (priority queue, graph algorithms etc) such as from Algorithms and Datastructures (CSE1305); experience with constructing proofs, such as from Reasoning and Logic (CSE1300)	
Course Contents	Graph algorithms, greedy algorithms, divide and conquer, dynamic programming and network flow algorithms	
Study Goals	After completing this course, the student can apply, reproduce, and give relevant properties and definitions of 1. graph algorithms, 2. greedy algorithms 3. divide and conquer, 4. dynamic programming and 5. network flow algorithms. The student can also 6. prove the correctness of these algorithms, 7. determine the running time of algorithms both analytically as well as experimentally, 8. implement algorithms efficiently, and 9. provide well-argued advice about which problems can be solved best using which algorithms.	
Education Method	Lectures (possibly only online / pre-recorded) Each lecture includes a number of multiple choice questions and a selection of especially the difficult components of the material. Students are expected to study some of the topics at home. Tutorials (possibly only online) In these interactive sessions we take you through some example assignments (both implementation and written ones). Scheduled classroom lecture hours may be used for a combination of the above, as announced by the lecturer. Skill circuit assignments To train with the learning objectives, several assignments at exam level, but also more accessible ones are available on BrightSpace as elements in the skill circuit. For some of them feedback can be requested from teaching assistants or other students if handed in on time. Similarly, students can train for the digital test by making the implementation assignments in the skill circuit. Correctness of the code will be automatically checked.	
Literature and Study Materials	The study material consists of the book Algorithm Design by John Kleinberg and Eva Tardos (2005), Addison Wesley ISBN 13: 9780321372918 or 9780321295354 and some documents made available through BrightSpace (such as a manual of proving techniques and the master's method for solving simple recurrent relations).	
Assessment	The final grade of the course consists of the following components: Written exam part 1 (of 1.5h) in week 5 (W1) Written exam part 2 (of 1.5h) in week 10/11 (W2) Computer exam (3h) in week 10/11 (P) Final grade The final grade for the course is computed as follows: W = 0.4 W1 + 0.6 W2 If P < 5 or W < 5, the final grade is min(P,W). Else the final grade is 0.5 W + 0.5 P Both parts P and W can be retaken, in individual exam moments in the resit week. Written exam (W) Half-way and at the end of the quarter written examinations of 1.5 hours each are organized. The written exam primarily assesses proving skills and (application of) algorithms and algorithmic theory. Resitting W implies redoing both parts in a single three-hour exam. Computer exam (P) Algorithm design and implementation skills are assessed through a computer exam of three hours at the end of the course. The computer exam needs to be submitted through Weblab, but IntelliJ is available within the exam environment. These partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	course email address: ad-cs-ewi@tudelft.nl	
Co-Instructor	Dr. E. Demirovi	

CSE2315	Automata, Computability and Complexity	5
Responsible Instructor	Prof.dr. M.T.J. Spaan	
Responsible Instructor	I.D. van Kreveld	
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. J.W. Böhmer	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Proof techniques (such as by induction or contradiction), set theory, recursive definitions, trees and graphs. For example from CSE1300 or equivalent. Run-time complexity of algorithms. For example from CSE2310 or equivalent.	
Course Contents	This course introduces three areas of the theory of computation. First, automata are used to recognize words in (formal) languages, and we will discern several types of languages along with the types of automata that can recognize them. Second, once we have an intuition about languages, we discuss the topic of computability, where you will learn not only what kinds of problems a computer can solve but also how to prove this. Third, we will examine the class of computable problems and make a distinction between "easy" and "hard" problems and delve into the famous problem: P = NP?	
Study Goals	After this course, the student can: 1. parse and construct automata and languages. 2. prove a problem to be computable. 3. manipulate and construct grammars for context-free languages. 4. describe the differences between different types of automata and grammars and their limitations, specifically that there are problems that are algorithmically unsolvable. 5. prove languages to be non-regular using the pumping lemma. 6. describe the classes P, NP, and NPC and explain why P = NP? is an open problem. 7. verify a proof that shows that a problem is NPC.	
Education Method	Engaging interactive lectures for which the student studies the material at home. During the lecture, the students test their knowledge via multiple choice questions and the teacher explains difficult topics on demand. Lab course in the form of optional take-home assignments, 8 times during the quarter. Students solve these assignments individually and hand them in weekly according to predetermined deadlines. Students correct each others work followed by a round of feedback by teaching assistants.	
Literature and Study Materials	1. Textbook (see Books), chapters 0, 1, 2 (2.1-2.3), 3, 4, 5 (5.1), 7. 2. Supplementary material (lecture slides, video lectures).	
Books	Michael Sipser, Introduction to the Theory of Computation, Third, International Edition, Cengage Learning. ISBN 978-1-133-18781-3. Note: if you do not use the international version, then section numbers and exercises will be different!	
Assessment	A written, closed-book exam (see permitted materials during tests below). The final grade of the course is based on the final exam for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	You are allowed to bring a hand-written double sided A4 cheat sheet. No additional materials are permitted, including but not limited to: books, printouts of the slides, or devices.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Prof.dr. M.T.J. Spaan	

CSE2510	Machine Learning	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Responsible Instructor	M.A. Migut	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	CSE1100 Introduction to Programming CSE1305 Algorithms and Data Structures CSE1200 Calculus CSE1205 Linear Algebra CSE1210 Probability Theory and Statistics	
Course Contents	The goal of the course is to acquaint students with the basic Machine Learning concepts and algorithms. Specifically, the course will cover parametric and non-parametric density estimation, linear and non-linear classification, unsupervised learning including clustering and dimensionality reduction, performance evaluation of predictive algorithms and ethical issues in machine learning.	
Study Goals	After succesfully completing this course, the student is able to: - explain the basic concepts and algorithms of machine learning and their underlying statistical concepts. - implement, apply and evaluate basic ML algorithms in Python. - explain the concept of and identify (implicit) bias in data and ML algorithms.	
Education Method	The course comprises two lectures and one (four-hour) lab session per week.	
Assessment	Computer exam: 100% of the final grade. Reparation: There is one resit per year for the digital exam. The passing grade is 5.8. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. D.M.J. Tax	
Co-Instructor	I.D. van Kreveld	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging programme CS for WO TUD BSc TW and EE 2024

CSE1100	Introduction to Programming	5
Responsible Instructor	Prof.dr. A.E. Zaidman	
Responsible Instructor	Ir. T.A.R. Overklift Vaupel Klein	
Contact Hours / Week x/x/x/x	6/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introductory programming course focused on programming in Java. The course covers: - Primitive datatypes: value domains, operators, expression, declarations and assignments. - Control structures: conditional selection, repetition. - Composition of data: arrays, Strings, ArrayLists, Sets. - Classes: specification, implementation, equality & hashcodes. - Inheritance: modeling inheritance, type casts, polymorphism, static and dynamic binding. - Use of libraries (APIs). - Code quality: best practices, unit testing (with JUnit), debugging strategies. - Exception handling. - Generics. - Basic IO operations (user input / reading / writing). - Design & implementation of small programs. - Functional programming in Java (streams, map, filter).	
Study Goals	After completing this course, the student: - Can write methods and classes using the Java programming language. --- Can write methods and classes using correct indentation and coding style. --- Can write equals and hashCode methods for a class using the Java programming language. --- Can create and manipulate a data structure using methods and classes. - Can design, implement and debug a simple program that contains inheritance and interfaces using the Java programming language. - Can design, implement and debug a simple program that reads from files, and writes to files using the Java programming language. - Can write unit tests for a program in Java using the Junit testing framework. - Can effectively search for existing packages, classes and programming solutions, using the Java documentation as well as online resources, and apply this information in their own programs. - Can describe key characteristics of different programming paradigms and the Java programming language.	
Education Method	Lectures, tutorials, and practical sessions with programming exercises. The programming exercises are feedback assignments, i.e., they do not count towards your grade, but you can get feedback on them to improve your programming skills.	
Assessment	This course contains two summative exams: 1) Theoretical exam in Weblab (Multiple-Choice, Open & (Small) Programming questions; no auxiliary materials may be used). 2) Programming exam (Open book, IDE & additional resources/documentation allowed). This course contains two formative tests. These allow you to practice programming in a real exam environment: 1) A Midterm Theoretical Exam (in week 5). 2) A Mock Programming Exam (in week 8). Between the two formative tests you can score a bonus on your final grade. Bonus = MIN(10, Midterm Bonus + Mock Exam Bonus). Your final grade is composed of: 1) The Theoretical Exam (weight 0.5, minimal required grade 5/10) 2) The Programming Exam (weight 0.5, minimal required grade 5/10) 3) The bonus score you can obtain by participating (and scoring points) in the midterm exam and mock exam. Your final grade is calculated as follows: Exam = (Theoretical Exam Grade + Programming Exam Grade) /2 Final = Exam + (Bonus x (10 - Exam) / 100) (This formula linearly scales down the amount of bonus you get as your exam score increases, and follows the directions of the Board of Examiners for bonus points). If your score for the Theoretical Exam and Programming Exam are (1) both minimally a 5/10 and (2) the "Final" result of the grading formula is at least 5.8, you pass this course. Within the academic year you can resit the Theoretical Exam and the Programming Exam in quarter 2. Out of the two attempts for the Theoretical Exam and the Programming Exam respectively, the highest score will count towards your final grade. Partial grades for the Theoretical Exam or Programming Exam cannot be carried over to a next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	This course makes use of a course email: ip-cs-ewi@tudelft.nl	
Co-Instructor	Prof.dr. A.E. Zaidman	

CSE1105	Collaborative Software Engineering Project	5
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/2/0/0 lecture; 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Modern software development is not just about knowing the right programming language or the right technology stack. It is as important to know how to work together in a team to get a serious programming project done. In this course, you will learn how to collaboratively create a software system. The contents of this course include learning about teamwork, interpreting requirements given by a client and breaking them down into actionable tasks, using popular collaboration tools like GitLab to plan and track project progress, using collaborative tools like Git, and learning popular frameworks and libraries to bring the project to success. The first weeks of the course will be structured through lectures to get you and your group started, afterwards, the course will transition into the implementation phase, in which you will work on the project. Every group will be closely supervised by a tutor. *Important:* The course has submission deadlines from week 1 onwards. Plan your travel to Delft accordingly.	
Study Goals	After completing this course, you will be able to ... 1) Act professionally in a team of software developers ... Make agreements within your team about your collaboration ... Chair a meeting and take minutes ... Give and receive feedback 2) Manage a complex software project in a group ... Break down a formal product backlog into actionable tasks ... Plan and manage the project tasks in GitLab ... Organize and distribute tasks between team members 3) Follow a collaborative software engineering process ... Use Git to version and share source code contributions ... Integrate testing in your process ... Contribute source code to a group project through GitLab merge requests ... Evaluate code contributions of other team members in GitLab merge requests 4) Create a client/server application from a provided set of project requirements ... Build a server application with Spring Boot ... Build a graphical client with Java FX ... Exchange data through REST using automated object mapping ... Manage the lifecycle of an application through dependency injection 5) Apply usability principles in a graphical user interface	
Education Method	Following this course will require you to... - Follow interactive lectures and pre-recorded material for self-study - Participate in a collaborative programming project in a small team - Assess the work of yourself and others through merge requests (code) and peer reviews (process) - Actively participate in TA sessions	
Assessment	This course has the following knock-out criteria which you need to meet: - Weekly contribution criteria. If you fail to meet these in two subsequent weeks, you fail the course. - End of course minimum contribution criteria. If you do not reach these by the end of the course you fail the course. - There are weekly mandatory TA meetings starting in week 2. Missing 2 without providing a valid reason beforehand results in failing the course. The course has several pass/fail assignments, which must all be passed. Assignments marked as (repairable) will have an opportunity for submitting an improved version, if an initial submission was insufficient. - Conducting an effective meeting - (repairable) Code of conduct - (repairable) Providing constructive feedback with the AID model - (repairable) Git assignment The following assignments determine your final grade. Graded items marked as repairable can be repaired by resubmitting an improved version if the grade of the initial submission was >=4.0 and <6.0. In the case of a resubmission the grade will be capped at a 6.0. - Code contribution and code reviews (25%) - Tasks and Planning (25%) - (repairable) Implemented Features (25%) - (repairable) Technology (15%) - (repairable) Usability (10%) Each partial grade has to be at least a 5.0 and the final grade has to be a 6. Partial grades are rounded to .1 and the final grade to .5. --- Please Note --- - The full specification of the minimum criteria, deadlines and more information about the assessment will be posted on Brightspace. - There is NO resit opportunity for this course. - Partial grades are not carried over to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	In addition to your regular enrollment of the course, you must also register in Brightspace, AT THE LATEST during the first lecture, so you can be assigned to a group. Late registrations after the group creation cannot be considered anymore.	

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Co-Instructor	I.C. de Bruin

CSE1110	**Software Quality and Testing**	**5**
Responsible Instructor	F. Mulder	
Contact Hours / Week x/x/x/x	0/0/0/4 lecture; 0/0/0/4 shared lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course covers the most important testing techniques needed to build high quality software systems. The course covers both the testing techniques themselves, and the software design techniques needed to create testable systems. Specific topics covered are: * testing principles, test levels, and the testing pyramid * software testing automation and JUnit * specification-based testing * structural testing and the different coverage criteria * boundary analysis and testing * design by contract * property-based testing * test doubles and mock objects * design for testability * test-driven development (TDD) * larger tests (SQL and web testing) * test code quality and test smells	
Study Goals	By the end of the course, students will be able to: * Create unit, integration, and system tests using current existing tools (like JUnit, Mockito, and JaCoCo) that effectively test complex software systems. * Derive test cases that deal with exceptional, corner, and bad weather cases by performing different testing techniques as well as able to reflect about their limitations, when and when not to apply them in a given context. * Measure and reflect on the efficiency of the developed test suites by means of different test adequacy metrics (i.e., line, branch, condition, MC/DC coverage). * Design and implement testable software systems by means of architectural patterns such as dependency injection and ports and adapters. * Write maintainable test code by avoiding well-known test code smells. * Understand and apply state-of-the-art techniques for intelligent software testing, such as property-based testing.	
Education Method	This course follows a flipped classroom approach. Students read chapters of the book and work on (practical and theoretical) exercises during their study time; lectures are used for questions and broader discussions. We offer TA support during the labs.	
Books	Effective Software Testing, by Maurício Aniche: https://www.effective-software-testing.com.	
Assessment	The evaluation of the course is based on one exam. It happens on WebLab and is composed of coding questions and theoretical questions. Students can consult the documentation of certain testing libraries during the exam. There is also one possibility to do a resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. C.E. Brandt	

CSE1305	Algorithms and Data Structures	5
Responsible Instructor	I.D. van Kreveld	
Responsible Instructor	M. Khosla	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Java programming, constructing proofs, trees and graphs	

Course Contents

Algorithms and data structures are fundamental notions in computer science . Understanding how they can be exploited in combination for better programming implementations in terms of time and space complexity is vital for writing efficient code.This course enables the student to:

- Explain and compare the structure and properties of standard algorithms and data structures.
- Execute and visualize standard algorithms and data structures on given inputs.
- Use mathematical methods to analyze the time and space complexity of algorithms and data structures.
- Implement algorithms and data structures using the Java programming language.
- Solve programming tasks using standard algorithms and data structures.

Study Goals

1. Data Structures

- data containers
-- vector
-- list
-- tree
-- set
- ordered data structures
-- stack
-- queue
-- priority queue
-- heap
-- map
- operations on data structures
-- iterative implementations
-- recursive implementations

The student can explain essential data structures, explain and analyze the core properties of these data structures, and explain, develop, and adopt iterative and recursive implementations of operations on these data structures, and analyze and compare space and time complexity of these operations.

2. Sorting

- selection sort
- insertion sort
- heap sort
- merge sort
- quick sort
- bucket sort
- radix sort

The student can explain and implement various sorting algorithms, analyze and compare space and time complexity of these algorithms, and choose sorting algorithms for different problem scenarios.

3. Searching

- search structures
-- search trees
-- AVL trees
-- (2,4) trees
- backtracking

The student can explain data structures which support efficient search and their invariants, explain, develop, and adopt implementations of operations on these data structures, analyze and compare space and time complexity of these operations, choose search structures for different problem scenarios, and explain, implement, and analyze backtracking algorithms.

4. Graphs and Graph Algorithms

- graph data structures
-- directed graphs
-- undirected graphs
-- weights
-- representations
- graph algorithms
-- graph traversals
-- path finding
-- cycle finding
-- connectivity
-- topological ordering
-- shortest path
-- minimum spanning tree

The student can explain various kinds of graphs, explain, analyze and compare different data structures for graphs, explain, develop and adopt implementations of operations on these data structures, analyze and compare space and time complexity of these operations, choose graph data structures for different problem scenarios, explain, implement, apply and analyze generic graph traversal strategies, and explain, develop and adopt implementations of various graph algorithms.

Education Method

Every week has two lecture sessions during which the key concepts will be explained, questions about the material will be answered, and small exercises will be completed with the students. Students will study the relevant material before every session through videos, slides, and the course book.

	Homework assignments are provided. Lab sessions are offered, in which answers to the homework can be discussed.
Books	Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser Data Structures And Algorithms In Java 6th edition ISBN 978-1-118-80857-3
Assessment	The course comprises the following assessment moments: Week 5: a formative analysis midterm exam (MA) Week 5: a formative implementation midterm exam (MI) Week 10: a summative analysis final exam (FA) Week 10: a summative implementation final exam (FI) The final course grade GF will be as follows: If FA >= 5 and FI >= 5, then GF = (FA+FI)/2 Otherwise, GF = NV For successful completion of the course: - both grades FA and FI need to be at least 5 (before rounding), - the final course grade GF needs to be at least 5.8. The standard TU Delft rounding rules apply. Note: the midterms do not contribute to the final course grade, their purpose is only to give the student an idea of the progress halfway through the course. There are separate resits for the analysis (RA) and the implementation (RI) components. The course grade GR after the resit will be determined as follows: If MAX(FA,RA) >= 5 and MAX(FI,RI) >= 5, then GR = (MAX(FA,RA)+MAX(FI,RI))/2 Otherwise, GR = NV For successful completion of the course after the resit: - FA or RA needs to be at least 5 and also FI or RI needs to be at least 5 (all before rounding), - the course grade after resit GR needs to be at least 5.8. The standard TU Delft rounding rules apply. Partial grades cannot be carried over to future editions of the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Permitted Materials during Tests	Paper-based analysis exams: you are allowed to bring a hand-written double sided A4 cheat sheet. No additional materials are permitted, including but not limited to: books, printouts of the slides, or devices. Computer-based implementation exams: no materials are allowed during the computer-based implementation exams.
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/
Co-Instructor	I.D. van Kreveld

CSE1405	Computer Networks	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/0/4 lecture; 0/0/0/4 shared lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Computer networks form the basis of all digital communication. They are typically structured in layers. Each layer has clearly defined functions and algorithms, which we will discuss in class. Furthermore, the course also discusses potential security issues and defences in computer networks, thereby also providing an introduction to cryptographic notions utilised by those defence mechanisms.	
Study Goals	1) The student can describe the principles of layered architectures, in particular the functions and purposes of the various network layers. 2) The student can describe the core algorithms per layer and provide arguments why these algorithms fulfil the function of the respective layer. 3) The student can apply the core algorithms to toy examples. 4) The student can identify potential network security issues and choose suitable mitigation measures. 5) The student can exemplify the concepts of ethical hacking and responsible disclosure.	
Education Method	Lectures (8 weeks x 2 lectures x 2 hours). During these lectures, new concepts are illustrated through the discussion of multiple examples from the book. Lab (5 weeks x 4 hours). During these labs the student gains experience with commonly used Internet software (e.g. Wireshark). Additionally the student applies given abstractions and network concepts.	
Literature and Study Materials	Computer Networks, Andrew S. Tanenbaum et al., 6th (Global) edition. We might provide additional material to further explain or practise with certain topics.	
Assessment	Written exam on paper, closed-book. There is a single resit opportunity of the exam. Exams are a combination of multiple choice questions and open questions. The final test is your final grade. The lab work is not graded. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	None	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme or an EEMCS bridging programme (CS, CESE) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris (a.k.a. MyTUDelft), separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ir. O.W. Visser	

CSE1500	Web and Database Technology	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	Dr. rer. nat. U.K. Gadiraju	
Contact Hours / Week x/x/x/x	0/4/0/0 lecture; 0/4/0/0 shared lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language.	
Course Contents	Web & Database Technologies is a first-year course that provides an introduction to web and database technologies and programming. Topics related to web technology and programming include: = Introduction to the web and HTTP; = Introduction to web development and app design; = Front-end development: HTML, CSS, JavaScript; = Back-end development: Node.js. Topics related to database technologies include: = Introduction to databases; = Overview of database Languages and architectures; = The basic relational model; database design theory; = SQL: data definition, constraints, updates, queries, views, triggers; = Relational databases and object persistence; = Introduction to non-relational database systems: key-value stores, document stores, graph databases.	
Study Goals	Upon completion of this course, participants will have gained knowledge of web and database system concepts and the ability to: 1. Explain the basic architecture of the Internet and the web; 2. Analyse the requirements for a web application given a description of an application idea; 3. Create an application front-end with HTML and CSS based on a given static design; 4. Create an interactive web application with client-side JavaScript; 5. Develop server-side code for authorization, cookies, sessions and database access; 6. Compare the strengths and weaknesses of different data management techniques, describing their most suited applications trough meaningful examples; 7. Enumerate and define the main elements of the relational model; 8. Describe and design SQL queries for the creation, altering, and manipulation of tables, indexes, and views; 9. Develop logical database schema, with principled design that enforce data integrity; 10. Prototype and deploy database applications using open-source database systems.	
Education Method	Lectures, and practical work (lab assignments).	
Literature and Study Materials	In addition to the official course book(s), students will be provided with: = The slides presented in class, cross-referenced to the content of the book(s); = Additional study materials, selected to provide insights on the topics covered during the lectures.	
Practical Guide	All necessary information about the practical setup (schedule, grading, exams, resits, etc.) of the course is available in the course syllabus (accessible via Brightspace).	
Books	WEB There is no required book for the web technologies part of the course. DATABASES The book is not required but helpful: Fundamentals of Database Systems, 7th Edition, Augustus 2016 (Global Edition). Authors: Ramez Elmasri, Shamkant Navathe. ISBN10: 1292097612. ISBN13: 9781292097619 URL: http://catalogue.pearsoned.co.uk/educator/product/Fundamentals-of-Database-Systems-Global-Edition/9781292097619.page	
Assessment	There are two exams: Midterm in W2.5 and Final in W2.10. The exams will be executed in a combination of written and weblab-based manner, with multiple choice, short answer questions, and open questions.	
Permitted Materials during Tests	No books or other materials are allowed during exams.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Tags	Databases Software	
Co-Instructor	F. Mulder	
Co-Instructor	Dr. R. Hai	
Co-Instructor	Dr. A.J. Power	

CSE1505	Information and Data Management	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	Dr. A. Katsifodimos	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic understanding relational query languages (SQL) and databases (as obtained in the Web and Database Technology Y1 Q2 course). This class also relies heavily on Algorithms and Data Structures.	
Course Contents	In this course you will learn how to model, manage, and query information in a Database Management System (DBMS). We will focus on different data modelling techniques, but also on aspects of query processing like for example query parsing or optimization. Also transaction management is a central topic of interest. The core focus is on classic relational OLTP databases, but we will also briefly touch alternative systems like the current NewSQL-trend and systems optimized for in-memory or OLAP applications. Additionally, NoSQL systems and their respective data models will be discussed. - Database normalization: modelling in an unambiguous and non-redundant manner in relational tables such that inconsistencies and anomalies are avoided. - File-organization and indexing: the data structures and their associated algorithms to store data in memory and on disk, such that it can be retrieved and processed efficiently when needed. - Query processing and optimization: the techniques that are used by a DBMS to execute certain SQL queries as efficiently as possible, taking into account how the data is stored and which indexes are present. This also covers Relational Algebra. - Transaction management and concurrency control: the techniques that a DBMS offers to allow different users and application to operate in parallel on the same data without interfering with each other or causing inconsistencies in the data. - NewSQL and alternative system: the systems beyond traditional relational database design space optimized for specialized requirements like extreme OLTP performance, OLAP queries, or in-memory storage. - NoSQL and non-relational data: in the recent years, systems not relying on the relational model became popular. We will also discuss how this affects data modelling, especially with respect to document models, key-value models, wide-columnar models, but also graph models.	
Study Goals	The course aims at teaching the basic competencies required to design and understand applications interacting with relational database management systems (RDBMS). Students will gain an insight into how RDBMS are implemented, learn about the available design choices, and the trade-offs imposed by different designs. This allows students to competently identify and solve problems in application design, and to improve and tune data models and queries. We will also deal with non-relational systems and discuss how they affect the data processing workflow. - Perform Data Modeling and be able to: -- Design and implement normalized relational schemas in a database system, by designing Entity Relationship diagrams and translating them to relational tables. -- Describe alternative data models such as the document model, key-value model, or graph models and analyze the trade-offs as in comparison to the relational model. -- Analyze the requirements of a given application scenario and be able to derive a suitable data model. - Get a grip of the internals of relational database systems by being able to: -- Describe and analyze the two main ways of storing and representing data in relational databases as well as analyze their performance trade-offs with respect to input/output (IO) costs. -- Describe different index structures and analyze their impact on query response time for range-, point- and join queries. -- Analyze the trade-offs between different physical operators for database operations such as joins, aggregates and filters. -- Describe and apply the query optimization process and design physical execution plan and compute the costs for their relational operations. -- Explain the need for transaction protocols and illustrate database update anomalies that can arise in the absence of such protocols. Be able to apply such protocols for given scenarios. -- Evaluate and apply different recovery algorithms such as UNDO/REDO logging, under different types of failures and their frequency.	
Education Method	Lectures and Homework Lab	
Computer Use	Students will require access to a personal computer. The labs will deal with larger data sets, so some HDD space (several GB) will be needed.	
Literature and Study Materials	The course and evaluation is based on the provided lecture slides and assignment material. For alternative explanation or more in-depth studies, the recommended book can be used.	
Books	Fundamentals of Database Systems by Ramez Elmasri and Shamkant Navathe, Pearson. Preferred 7th edition, but also the older ones are usable. Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, and Jennifer Widom (2nd edition preferred, older ones also OK). These books are NOT required, but might be helpful for more in-depth studies. Additional (optional) books/resources are mentioned in the lecture.	
Assessment	This year, there will only be a final exam which counts for 100% of the course grade. The lab assignments are not mandatory and will not contribute to the grade, but we STRONGLY recommend doing them anyway.	
Permitted Materials during Tests	1 handwritten sheet of A4 paper (any content chosen by the student, writing front and back permitted).	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme or an EEMCS bridging programme (CS, CESE) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ir. T.A.R. Overklift Vaupel Klein	
Co-Instructor	Dr. A.J. Power	

CSE2115	Software Engineering Methods	5

Responsible Instructor	Prof.dr. A. van Deursen
Responsible Instructor	T.V. Aerts
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	English
Course Contents	The course covers the most important software engineering practices needed to build high-quality software systems. The overall goal is to give the students the instruments to become Software Professionals. The course covers both high-level software project management, and low-level software design techniques needed to create evolvable and maintainable systems. Specific topics covered include design patterns, requirements engineering, UML modeling, software architecture, and software analytics. In the lab work, students will put theory into practice using various real-world software systems as case studies.
Study Goals	(1) Students can select appropriate software engineering methods, which are designed and used to build maintainable and evolvable software systems. (2) Students can judge the benefits/drawbacks of the presented software engineering methods that are commonly used in current software development. (3) Students can use presented software visualization techniques and software analytics to obtain an overview of an existing software system and to detect anomalies in the structure of the code. (4) Students can judge the real benefits/drawbacks of using certain design patterns and software validation techniques in a given software project. (5) Students can derive the valid requirements for a new software project, given natural-language descriptions and interaction with stakeholders. (6) Students can apply the most appropriate software engineering practices in a project, given the context of development.
Education Method	Labwork + Lectures
Literature and Study Materials	Materials will be provided via Brightspace
Assessment	-- Final grade composition -- The final grade of this course is calculated using the following weights; * Digital exam: 100% * Oral Presentations: Pass/Fail * Attendance to weekly meetings (Oral Presentations): Pass/Fail -- General remarks -- * Each grade must be sufficient to pass the course. * Passing the course is conditioned by passing the Oral Presentation module. * Partial grades are not carried over to the next edition of this course. * In case you have completed Oral Presentations before then you may be eligible for an exemption for the attendance to the weekly lectures. Check Brightspace for more information. -- Resits -- * The re-sit will replace the grade for the exam only. * For the Oral Presentations part we offer a single resit possibility. -- Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/
Special Information	Contact with the course staff via: sem-cs-ewi@tudelft.nl (use your TU Delft email address).
Co-Instructor	Prof.dr. A. van Deursen

CSE2310	Algorithm Design	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. E. Demirovi	
Contact Hours / Week x/x/x/x	0/4/0/0 lecture; 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: programming experience from CSE1100 or equivalent Strongly advised: experience with algorithms and data structures (priority queue, graph algorithms etc) such as from Algorithms and Datastructures (CSE1305); experience with constructing proofs, such as from Reasoning and Logic (CSE1300)	
Course Contents	Graph algorithms, greedy algorithms, divide and conquer, dynamic programming and network flow algorithms	
Study Goals	After completing this course, the student can apply, reproduce, and give relevant properties and definitions of 1. graph algorithms, 2. greedy algorithms 3. divide and conquer, 4. dynamic programming and 5. network flow algorithms. The student can also 6. prove the correctness of these algorithms, 7. determine the running time of algorithms both analytically as well as experimentally, 8. implement algorithms efficiently, and 9. provide well-argued advice about which problems can be solved best using which algorithms.	
Education Method	Lectures (possibly only online / pre-recorded) Each lecture includes a number of multiple choice questions and a selection of especially the difficult components of the material. Students are expected to study some of the topics at home. Tutorials (possibly only online) In these interactive sessions we take you through some example assignments (both implementation and written ones). Scheduled classroom lecture hours may be used for a combination of the above, as announced by the lecturer. Skill circuit assignments To train with the learning objectives, several assignments at exam level, but also more accessible ones are available on BrightSpace as elements in the skill circuit. For some of them feedback can be requested from teaching assistants or other students if handed in on time. Similarly, students can train for the digital test by making the implementation assignments in the skill circuit. Correctness of the code will be automatically checked.	
Literature and Study Materials	The study material consists of the book Algorithm Design by John Kleinberg and Eva Tardos (2005), Addison Wesley ISBN 13: 9780321372918 or 9780321295354 and some documents made available through BrightSpace (such as a manual of proving techniques and the master's method for solving simple recurrent relations).	
Assessment	The final grade of the course consists of the following components: Written exam part 1 (of 1.5h) in week 5 (W1) Written exam part 2 (of 1.5h) in week 10/11 (W2) Computer exam (3h) in week 10/11 (P) Final grade The final grade for the course is computed as follows: W = 0.4 W1 + 0.6 W2 If P < 5 or W < 5, the final grade is min(P,W). Else the final grade is 0.5 W + 0.5 P Both parts P and W can be retaken, in individual exam moments in the resit week. Written exam (W) Half-way and at the end of the quarter written examinations of 1.5 hours each are organized. The written exam primarily assesses proving skills and (application of) algorithms and algorithmic theory. Resitting W implies redoing both parts in a single three-hour exam. Computer exam (P) Algorithm design and implementation skills are assessed through a computer exam of three hours at the end of the course. The computer exam needs to be submitted through Weblab, but IntelliJ is available within the exam environment. These partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	course email address: ad-cs-ewi@tudelft.nl	
Co-Instructor	Dr. E. Demirovi	

CSE2315	Automata, Computability and Complexity	5
Responsible Instructor	Prof.dr. M.T.J. Spaan	
Responsible Instructor	I.D. van Kreveld	
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. J.W. Böhmer	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Proof techniques (such as by induction or contradiction), set theory, recursive definitions, trees and graphs. For example from CSE1300 or equivalent. Run-time complexity of algorithms. For example from CSE2310 or equivalent.	
Course Contents	This course introduces three areas of the theory of computation. First, automata are used to recognize words in (formal) languages, and we will discern several types of languages along with the types of automata that can recognize them. Second, once we have an intuition about languages, we discuss the topic of computability, where you will learn not only what kinds of problems a computer can solve but also how to prove this. Third, we will examine the class of computable problems and make a distinction between "easy" and "hard" problems and delve into the famous problem: P = NP?	
Study Goals	After this course, the student can: 1. parse and construct automata and languages. 2. prove a problem to be computable. 3. manipulate and construct grammars for context-free languages. 4. describe the differences between different types of automata and grammars and their limitations, specifically that there are problems that are algorithmically unsolvable. 5. prove languages to be non-regular using the pumping lemma. 6. describe the classes P, NP, and NPC and explain why P = NP? is an open problem. 7. verify a proof that shows that a problem is NPC.	
Education Method	Engaging interactive lectures for which the student studies the material at home. During the lecture, the students test their knowledge via multiple choice questions and the teacher explains difficult topics on demand. Lab course in the form of optional take-home assignments, 8 times during the quarter. Students solve these assignments individually and hand them in weekly according to predetermined deadlines. Students correct each others work followed by a round of feedback by teaching assistants.	
Literature and Study Materials	1. Textbook (see Books), chapters 0, 1, 2 (2.1-2.3), 3, 4, 5 (5.1), 7. 2. Supplementary material (lecture slides, video lectures).	
Books	Michael Sipser, Introduction to the Theory of Computation, Third, International Edition, Cengage Learning. ISBN 978-1-133-18781-3. Note: if you do not use the international version, then section numbers and exercises will be different!	
Assessment	A written, closed-book exam (see permitted materials during tests below). The final grade of the course is based on the final exam for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	You are allowed to bring a hand-written double sided A4 cheat sheet. No additional materials are permitted, including but not limited to: books, printouts of the slides, or devices.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Prof.dr. M.T.J. Spaan	

CSE2510	Machine Learning	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Responsible Instructor	M.A. Migut	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	CSE1100 Introduction to Programming CSE1305 Algorithms and Data Structures CSE1200 Calculus CSE1205 Linear Algebra CSE1210 Probability Theory and Statistics	
Course Contents	The goal of the course is to acquaint students with the basic Machine Learning concepts and algorithms. Specifically, the course will cover parametric and non-parametric density estimation, linear and non-linear classification, unsupervised learning including clustering and dimensionality reduction, performance evaluation of predictive algorithms and ethical issues in machine learning.	
Study Goals	After succesfully completing this course, the student is able to: - explain the basic concepts and algorithms of machine learning and their underlying statistical concepts. - implement, apply and evaluate basic ML algorithms in Python. - explain the concept of and identify (implicit) bias in data and ML algorithms.	
Education Method	The course comprises two lectures and one (four-hour) lab session per week.	
Assessment	Computer exam: 100% of the final grade. Reparation: There is one resit per year for the digital exam. The passing grade is 5.8. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. D.M.J. Tax	
Co-Instructor	I.D. van Kreveld	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging Programme DSAIT for WO MSc 2024

CSE1100	Introduction to Programming	5

Responsible Instructor	Prof.dr. A.E. Zaidman
Responsible Instructor	Ir. T.A.R. Overklift Vaupel Klein
Contact Hours / Week x/x/x/x	6/0/0/0 lecture; 4/0/0/0 lab
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	English
Course Contents	Introductory programming course focused on programming in Java. The course covers: - Primitive datatypes: value domains, operators, expression, declarations and assignments. - Control structures: conditional selection, repetition. - Composition of data: arrays, Strings, ArrayLists, Sets. - Classes: specification, implementation, equality & hashcodes. - Inheritance: modeling inheritance, type casts, polymorphism, static and dynamic binding. - Use of libraries (APIs). - Code quality: best practices, unit testing (with JUnit), debugging strategies. - Exception handling. - Generics. - Basic IO operations (user input / reading / writing). - Design & implementation of small programs. - Functional programming in Java (streams, map, filter).
Study Goals	After completing this course, the student: - Can write methods and classes using the Java programming language. --- Can write methods and classes using correct indentation and coding style. --- Can write equals and hashCode methods for a class using the Java programming language. --- Can create and manipulate a data structure using methods and classes. - Can design, implement and debug a simple program that contains inheritance and interfaces using the Java programming language. - Can design, implement and debug a simple program that reads from files, and writes to files using the Java programming language. - Can write unit tests for a program in Java using the Junit testing framework. - Can effectively search for existing packages, classes and programming solutions, using the Java documentation as well as online resources, and apply this information in their own programs. - Can describe key characteristics of different programming paradigms and the Java programming language.
Education Method	Lectures, tutorials, and practical sessions with programming exercises. The programming exercises are feedback assignments, i.e., they do not count towards your grade, but you can get feedback on them to improve your programming skills.
Assessment	This course contains two summative exams: 1) Theoretical exam in Weblab (Multiple-Choice, Open & (Small) Programming questions; no auxiliary materials may be used). 2) Programming exam (Open book, IDE & additional resources/documentation allowed). This course contains two formative tests. These allow you to practice programming in a real exam environment: 1) A Midterm Theoretical Exam (in week 5). 2) A Mock Programming Exam (in week 8). Between the two formative tests you can score a bonus on your final grade. Bonus = MIN(10, Midterm Bonus + Mock Exam Bonus). Your final grade is composed of: 1) The Theoretical Exam (weight 0.5, minimal required grade 5/10) 2) The Programming Exam (weight 0.5, minimal required grade 5/10) 3) The bonus score you can obtain by participating (and scoring points) in the midterm exam and mock exam. Your final grade is calculated as follows: Exam = (Theoretical Exam Grade + Programming Exam Grade) /2 Final = Exam + (Bonus x (10 - Exam) / 100) (This formula linearly scales down the amount of bonus you get as your exam score increases, and follows the directions of the Board of Examiners for bonus points). If your score for the Theoretical Exam and Programming Exam are (1) both minimally a 5/10 and (2) the "Final" result of the grading formula is at least 5.8, you pass this course. Within the academic year you can resit the Theoretical Exam and the Programming Exam in quarter 2. Out of the two attempts for the Theoretical Exam and the Programming Exam respectively, the highest score will count towards your final grade. Partial grades for the Theoretical Exam or Programming Exam cannot be carried over to a next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/
Special Information	This course makes use of a course email: ip-cs-ewi@tudelft.nl
Co-Instructor	Prof.dr. A.E. Zaidman

CSE1105	Collaborative Software Engineering Project	5
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/2/0/0 lecture; 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Modern software development is not just about knowing the right programming language or the right technology stack. It is as important to know how to work together in a team to get a serious programming project done. In this course, you will learn how to collaboratively create a software system. The contents of this course include learning about teamwork, interpreting requirements given by a client and breaking them down into actionable tasks, using popular collaboration tools like GitLab to plan and track project progress, using collaborative tools like Git, and learning popular frameworks and libraries to bring the project to success. The first weeks of the course will be structured through lectures to get you and your group started, afterwards, the course will transition into the implementation phase, in which you will work on the project. Every group will be closely supervised by a tutor. *Important:* The course has submission deadlines from week 1 onwards. Plan your travel to Delft accordingly.	
Study Goals	After completing this course, you will be able to ... 1) Act professionally in a team of software developers ... Make agreements within your team about your collaboration ... Chair a meeting and take minutes ... Give and receive feedback 2) Manage a complex software project in a group ... Break down a formal product backlog into actionable tasks ... Plan and manage the project tasks in GitLab ... Organize and distribute tasks between team members 3) Follow a collaborative software engineering process ... Use Git to version and share source code contributions ... Integrate testing in your process ... Contribute source code to a group project through GitLab merge requests ... Evaluate code contributions of other team members in GitLab merge requests 4) Create a client/server application from a provided set of project requirements ... Build a server application with Spring Boot ... Build a graphical client with Java FX ... Exchange data through REST using automated object mapping ... Manage the lifecycle of an application through dependency injection 5) Apply usability principles in a graphical user interface	
Education Method	Following this course will require you to... - Follow interactive lectures and pre-recorded material for self-study - Participate in a collaborative programming project in a small team - Assess the work of yourself and others through merge requests (code) and peer reviews (process) - Actively participate in TA sessions	
Assessment	This course has the following knock-out criteria which you need to meet: - Weekly contribution criteria. If you fail to meet these in two subsequent weeks, you fail the course. - End of course minimum contribution criteria. If you do not reach these by the end of the course you fail the course. - There are weekly mandatory TA meetings starting in week 2. Missing 2 without providing a valid reason beforehand results in failing the course. The course has several pass/fail assignments, which must all be passed. Assignments marked as (repairable) will have an opportunity for submitting an improved version, if an initial submission was insufficient. - Conducting an effective meeting - (repairable) Code of conduct - (repairable) Providing constructive feedback with the AID model - (repairable) Git assignment The following assignments determine your final grade. Graded items marked as repairable can be repaired by resubmitting an improved version if the grade of the initial submission was >=4.0 and <6.0. In the case of a resubmission the grade will be capped at a 6.0. - Code contribution and code reviews (25%) - Tasks and Planning (25%) - (repairable) Implemented Features (25%) - (repairable) Technology (15%) - (repairable) Usability (10%) Each partial grade has to be at least a 5.0 and the final grade has to be a 6. Partial grades are rounded to .1 and the final grade to .5. --- Please Note --- - The full specification of the minimum criteria, deadlines and more information about the assessment will be posted on Brightspace. - There is NO resit opportunity for this course. - Partial grades are not carried over to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	In addition to your regular enrollment of the course, you must also register in Brightspace, AT THE LATEST during the first lecture, so you can be assigned to a group. Late registrations after the group creation cannot be considered anymore.	

	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/
Co-Instructor	I.C. de Bruin

CSE1110	**Software Quality and Testing**	**5**
Responsible Instructor	F. Mulder	
Contact Hours / Week x/x/x/x	0/0/0/4 lecture; 0/0/0/4 shared lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course covers the most important testing techniques needed to build high quality software systems. The course covers both the testing techniques themselves, and the software design techniques needed to create testable systems. Specific topics covered are: * testing principles, test levels, and the testing pyramid * software testing automation and JUnit * specification-based testing * structural testing and the different coverage criteria * boundary analysis and testing * design by contract * property-based testing * test doubles and mock objects * design for testability * test-driven development (TDD) * larger tests (SQL and web testing) * test code quality and test smells	
Study Goals	By the end of the course, students will be able to: * Create unit, integration, and system tests using current existing tools (like JUnit, Mockito, and JaCoCo) that effectively test complex software systems. * Derive test cases that deal with exceptional, corner, and bad weather cases by performing different testing techniques as well as able to reflect about their limitations, when and when not to apply them in a given context. * Measure and reflect on the efficiency of the developed test suites by means of different test adequacy metrics (i.e., line, branch, condition, MC/DC coverage). * Design and implement testable software systems by means of architectural patterns such as dependency injection and ports and adapters. * Write maintainable test code by avoiding well-known test code smells. * Understand and apply state-of-the-art techniques for intelligent software testing, such as property-based testing.	
Education Method	This course follows a flipped classroom approach. Students read chapters of the book and work on (practical and theoretical) exercises during their study time; lectures are used for questions and broader discussions. We offer TA support during the labs.	
Books	Effective Software Testing, by Maurício Aniche: https://www.effective-software-testing.com.	
Assessment	The evaluation of the course is based on one exam. It happens on WebLab and is composed of coding questions and theoretical questions. Students can consult the documentation of certain testing libraries during the exam. There is also one possibility to do a resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. C.E. Brandt	

CSE1400	Computer Organisation	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	Ir. O.W. Visser	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 1/0/0/0 tutorial; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	High-school-level math.	
Course Contents	Rationale Computers are everywhere. Topics include: the architecture, the structure, the operation and the interconnection of computer components into computer systems, including modern architectures, data representation, assembler programming, virtual machines, the structure of translators, compiling and loading, basic operating systems concepts (I/O, interrupt handling, process).	
Study Goals	Students can: 1. Explain the basic concepts, historical objectives, and modern functions of digital computers. 2. Evaluate and recommend number representation systems and digital circuits when given a set of realistic system constraints. 3. Derive conversion functions for Boolean algebra. 4. Use arithmetic and conversion functions on well-known number representation formats. 5. Describe the architecture and operation of digital computers. 6. Perform operations using each of the main components of a digital computer: the basic processing unit, the I/O system, the memory system, the interconnection system. 7. Explain system mechanisms for building faster processors, such as pipelining, memory organization, caching, and parallelism. 8. Analyze the trade-offs inherent in the design of digital computers, concerning among others performance, cost, scalability, availability, and energy consumption. 9. Demonstrate proficiency with basic assembly programming by implementing realistic basic programs used in digital computers.	
Education Method	Lectures, tutorial, and practical work (Lab). Self-study with peer reviewing. This course uses gamification.	
Literature and Study Materials	Course and Lab guides are also provided via Brightspace.	
Books	C. Hamacher e.a., Computer Organization, 6th edition, McGraw-Hill, 2012	
Assessment	(Mandatory) 3 basic lab assignments. Does not award any points (checked by TAs). If you do not pass, you can repair by doing one of the bonuses (NB: this bonus done as repair does not award any points!). (Mandatory) exam, written. Mid-term in week 5; second part in week 10. There is a complete resit of both parts in the next quarter. Minimum total exam score: 4000 points. (Voluntary) in-class exercises, oral and written. (Voluntary) self-study of relevant questions with peer feedback. (Voluntary) advanced bonus lab assignments (checked by TAs). All partial results (including the lab and the exam) are only valid during this academic year. Your end grade is the total points scored divided by 1000. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Pen/pencil. Everything else is not allowed: no books, notes, calculator, etc.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. A.J. Power	

CSE1405	Computer Networks	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/0/4 lecture; 0/0/0/4 shared lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Computer networks form the basis of all digital communication. They are typically structured in layers. Each layer has clearly defined functions and algorithms, which we will discuss in class. Furthermore, the course also discusses potential security issues and defences in computer networks, thereby also providing an introduction to cryptographic notions utilised by those defence mechanisms.	
Study Goals	1) The student can describe the principles of layered architectures, in particular the functions and purposes of the various network layers. 2) The student can describe the core algorithms per layer and provide arguments why these algorithms fulfil the function of the respective layer. 3) The student can apply the core algorithms to toy examples. 4) The student can identify potential network security issues and choose suitable mitigation measures. 5) The student can exemplify the concepts of ethical hacking and responsible disclosure.	
Education Method	Lectures (8 weeks x 2 lectures x 2 hours). During these lectures, new concepts are illustrated through the discussion of multiple examples from the book. Lab (5 weeks x 4 hours). During these labs the student gains experience with commonly used Internet software (e.g. Wireshark). Additionally the student applies given abstractions and network concepts.	
Literature and Study Materials	Computer Networks, Andrew S. Tanenbaum et al., 6th (Global) edition. We might provide additional material to further explain or practise with certain topics.	
Assessment	Written exam on paper, closed-book. There is a single resit opportunity of the exam. Exams are a combination of multiple choice questions and open questions. The final test is your final grade. The lab work is not graded. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	None	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme or an EEMCS bridging programme (CS, CESE) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris (a.k.a. MyTUDelft), separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ir. O.W. Visser	

CSE1505	Information and Data Management	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	Dr. A. Katsifodimos	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic understanding relational query languages (SQL) and databases (as obtained in the Web and Database Technology Y1 Q2 course). This class also relies heavily on Algorithms and Data Structures.	
Course Contents	In this course you will learn how to model, manage, and query information in a Database Management System (DBMS). We will focus on different data modelling techniques, but also on aspects of query processing like for example query parsing or optimization. Also transaction management is a central topic of interest. The core focus is on classic relational OLTP databases, but we will also briefly touch alternative systems like the current NewSQL-trend and systems optimized for in-memory or OLAP applications. Additionally, NoSQL systems and their respective data models will be discussed. - Database normalization: modelling in an unambiguous and non-redundant manner in relational tables such that inconsistencies and anomalies are avoided. - File-organization and indexing: the data structures and their associated algorithms to store data in memory and on disk, such that it can be retrieved and processed efficiently when needed. - Query processing and optimization: the techniques that are used by a DBMS to execute certain SQL queries as efficiently as possible, taking into account how the data is stored and which indexes are present. This also covers Relational Algebra. - Transaction management and concurrency control: the techniques that a DBMS offers to allow different users and application to operate in parallel on the same data without interfering with each other or causing inconsistencies in the data. - NewSQL and alternative system: the systems beyond traditional relational database design space optimized for specialized requirements like extreme OLTP performance, OLAP queries, or in-memory storage. - NoSQL and non-relational data: in the recent years, systems not relying on the relational model became popular. We will also discuss how this affects data modelling, especially with respect to document models, key-value models, wide-columnar models, but also graph models.	
Study Goals	The course aims at teaching the basic competencies required to design and understand applications interacting with relational database management systems (RDBMS). Students will gain an insight into how RDBMS are implemented, learn about the available design choices, and the trade-offs imposed by different designs. This allows students to competently identify and solve problems in application design, and to improve and tune data models and queries. We will also deal with non-relational systems and discuss how they affect the data processing workflow. - Perform Data Modeling and be able to: -- Design and implement normalized relational schemas in a database system, by designing Entity Relationship diagrams and translating them to relational tables. -- Describe alternative data models such as the document model, key-value model, or graph models and analyze the trade-offs as in comparison to the relational model. -- Analyze the requirements of a given application scenario and be able to derive a suitable data model. - Get a grip of the internals of relational database systems by being able to: -- Describe and analyze the two main ways of storing and representing data in relational databases as well as analyze their performance trade-offs with respect to input/output (IO) costs. -- Describe different index structures and analyze their impact on query response time for range-, point- and join queries. -- Analyze the trade-offs between different physical operators for database operations such as joins, aggregates and filters. -- Describe and apply the query optimization process and design physical execution plan and compute the costs for their relational operations. -- Explain the need for transaction protocols and illustrate database update anomalies that can arise in the absence of such protocols. Be able to apply such protocols for given scenarios. -- Evaluate and apply different recovery algorithms such as UNDO/REDO logging, under different types of failures and their frequency.	
Education Method	Lectures and Homework Lab	
Computer Use	Students will require access to a personal computer. The labs will deal with larger data sets, so some HDD space (several GB) will be needed.	
Literature and Study Materials	The course and evaluation is based on the provided lecture slides and assignment material. For alternative explanation or more in -depth studies, the recommended book can be used.	
Books	Fundamentals of Database Systems by Ramez Elmasri and Shamkant Navathe, Pearson. Preferred 7th edition, but also the older ones are usable. Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, and Jennifer Widom (2nd edition preferred, older ones also OK). These books are NOT required, but might be helpful for more in-depth studies. Additional (optional) books/resources are mentioned in the lecture.	
Assessment	This year, there will only be a final exam which counts for 100% of the course grade. The lab assignments are not mandatory and will not contribute to the grade, but we STRONGLY recommend doing them anyway.	
Permitted Materials during Tests	1 handwritten sheet of A4 paper (any content chosen by the student, writing front and back permitted).	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme or an EEMCS bridging programme (CS, CESE) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ir. T.A.R. Overklift Vaupel Klein	
Co-Instructor	Dr. A.J. Power	

CSE2115	Software Engineering Methods	5
Responsible Instructor	Prof.dr. A. van Deursen	
Responsible Instructor	T.V. Aerts	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course covers the most important software engineering practices needed to build high-quality software systems. The overall goal is to give the students the instruments to become Software Professionals. The course covers both high-level software project management, and low-level software design techniques needed to create evolvable and maintainable systems. Specific topics covered include design patterns, requirements engineering, UML modeling, software architecture, and software analytics. In the lab work, students will put theory into practice using various real-world software systems as case studies.	
Study Goals	(1) Students can select appropriate software engineering methods, which are designed and used to build maintainable and evolvable software systems. (2) Students can judge the benefits/drawbacks of the presented software engineering methods that are commonly used in current software development. (3) Students can use presented software visualization techniques and software analytics to obtain an overview of an existing software system and to detect anomalies in the structure of the code. (4) Students can judge the real benefits/drawbacks of using certain design patterns and software validation techniques in a given software project. (5) Students can derive the valid requirements for a new software project, given natural-language descriptions and interaction with stakeholders. (6) Students can apply the most appropriate software engineering practices in a project, given the context of development.	
Education Method	Labwork + Lectures	
Literature and Study Materials	Materials will be provided via Brightspace	
Assessment	-- Final grade composition -- The final grade of this course is calculated using the following weights; * Digital exam: 100% * Oral Presentations: Pass/Fail * Attendance to weekly meetings (Oral Presentations): Pass/Fail -- General remarks -- * Each grade must be sufficient to pass the course. * Passing the course is conditioned by passing the Oral Presentation module. * Partial grades are not carried over to the next edition of this course. * In case you have completed Oral Presentations before then you may be eligible for an exemption for the attendance to the weekly lectures. Check Brightspace for more information. -- Resits -- * The re-sit will replace the grade for the exam only. * For the Oral Presentations part we offer a single resit possibility. -- Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	Contact with the course staff via: sem-cs-ewi@tudelft.nl (use your TU Delft email address).	
Co-Instructor	Prof.dr. A. van Deursen	

CSE2310	Algorithm Design	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. E. Demirovi	
Contact Hours / Week x/x/x/x	0/4/0/0 lecture; 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: programming experience from CSE1100 or equivalent Strongly advised: experience with algorithms and data structures (priority queue, graph algorithms etc) such as from Algorithms and Datastructures (CSE1305); experience with constructing proofs, such as from Reasoning and Logic (CSE1300)	
Course Contents	Graph algorithms, greedy algorithms, divide and conquer, dynamic programming and network flow algorithms	
Study Goals	After completing this course, the student can apply, reproduce, and give relevant properties and definitions of 1. graph algorithms, 2. greedy algorithms 3. divide and conquer, 4. dynamic programming and 5. network flow algorithms. The student can also 6. prove the correctness of these algorithms, 7. determine the running time of algorithms both analytically as well as experimentally, 8. implement algorithms efficiently, and 9. provide well-argued advice about which problems can be solved best using which algorithms.	
Education Method	Lectures (possibly only online / pre-recorded) Each lecture includes a number of multiple choice questions and a selection of especially the difficult components of the material. Students are expected to study some of the topics at home. Tutorials (possibly only online) In these interactive sessions we take you through some example assignments (both implementation and written ones). Scheduled classroom lecture hours may be used for a combination of the above, as announced by the lecturer. Skill circuit assignments To train with the learning objectives, several assignments at exam level, but also more accessible ones are available on BrightSpace as elements in the skill circuit. For some of them feedback can be requested from teaching assistants or other students if handed in on time. Similarly, students can train for the digital test by making the implementation assignments in the skill circuit. Correctness of the code will be automatically checked.	
Literature and Study Materials	The study material consists of the book Algorithm Design by John Kleinberg and Eva Tardos (2005), Addison Wesley ISBN 13: 9780321372918 or 9780321295354 and some documents made available through BrightSpace (such as a manual of proving techniques and the master's method for solving simple recurrent relations).	
Assessment	The final grade of the course consists of the following components: Written exam part 1 (of 1.5h) in week 5 (W1) Written exam part 2 (of 1.5h) in week 10/11 (W2) Computer exam (3h) in week 10/11 (P) Final grade The final grade for the course is computed as follows: W = 0.4 W1 + 0.6 W2 If P < 5 or W < 5, the final grade is min(P,W). Else the final grade is 0.5 W + 0.5 P Both parts P and W can be retaken, in individual exam moments in the resit week. Written exam (W) Half-way and at the end of the quarter written examinations of 1.5 hours each are organized. The written exam primarily assesses proving skills and (application of) algorithms and algorithmic theory. Resitting W implies redoing both parts in a single three-hour exam. Computer exam (P) Algorithm design and implementation skills are assessed through a computer exam of three hours at the end of the course. The computer exam needs to be submitted through Weblab, but IntelliJ is available within the exam environment. These partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	course email address: ad-cs-ewi@tudelft.nl	
Co-Instructor	Dr. E. Demirovi	

CSE2315	Automata, Computability and Complexity 5
Responsible Instructor	Prof.dr. M.T.J. Spaan
Responsible Instructor	I.D. van Kreveld
Responsible Instructor	S. Hugtenburg
Responsible Instructor	Dr. J.W. Böhmer
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab
Education Period	3
Start Education	3
Exam Period	3 4
Course Language	English
Expected prior knowledge	Proof techniques (such as by induction or contradiction), set theory, recursive definitions, trees and graphs. For example from CSE1300 or equivalent. Run-time complexity of algorithms. For example from CSE2310 or equivalent.
Course Contents	This course introduces three areas of the theory of computation. First, automata are used to recognize words in (formal) languages, and we will discern several types of languages along with the types of automata that can recognize them. Second, once we have an intuition about languages, we discuss the topic of computability, where you will learn not only what kinds of problems a computer can solve but also how to prove this. Third, we will examine the class of computable problems and make a distinction between "easy" and "hard" problems and delve into the famous problem: P = NP?
Study Goals	After this course, the student can: 1. parse and construct automata and languages. 2. prove a problem to be computable. 3. manipulate and construct grammars for context-free languages. 4. describe the differences between different types of automata and grammars and their limitations, specifically that there are problems that are algorithmically unsolvable. 5. prove languages to be non-regular using the pumping lemma. 6. describe the classes P, NP, and NPC and explain why P = NP? is an open problem. 7. verify a proof that shows that a problem is NPC.
Education Method	Engaging interactive lectures for which the student studies the material at home. During the lecture, the students test their knowledge via multiple choice questions and the teacher explains difficult topics on demand. Lab course in the form of optional take-home assignments, 8 times during the quarter. Students solve these assignments individually and hand them in weekly according to predetermined deadlines. Students correct each others work followed by a round of feedback by teaching assistants.
Literature and Study Materials	1. Textbook (see Books), chapters 0, 1, 2 (2.1-2.3), 3, 4, 5 (5.1), 7. 2. Supplementary material (lecture slides, video lectures).
Books	Michael Sipser, Introduction to the Theory of Computation, Third, International Edition, Cengage Learning. ISBN 978-1-133-18781-3. Note: if you do not use the international version, then section numbers and exercises will be different!
Assessment	A written, closed-book exam (see permitted materials during tests below). The final grade of the course is based on the final exam for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Permitted Materials during Tests	You are allowed to bring a hand-written double sided A4 cheat sheet. No additional materials are permitted, including but not limited to: books, printouts of the slides, or devices.
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/
Co-Instructor	Prof.dr. M.T.J. Spaan

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging Programme DSAIT for WO BSc TW and EE 2024

CSE1100	Introduction to Programming	5
Responsible Instructor	Prof.dr. A.E. Zaidman	
Responsible Instructor	Ir. T.A.R. Overklift Vaupel Klein	
Contact Hours / Week x/x/x/x	6/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introductory programming course focused on programming in Java. The course covers: - Primitive datatypes: value domains, operators, expression, declarations and assignments. - Control structures: conditional selection, repetition. - Composition of data: arrays, Strings, ArrayLists, Sets. - Classes: specification, implementation, equality & hashcodes. - Inheritance: modeling inheritance, type casts, polymorphism, static and dynamic binding. - Use of libraries (APIs). - Code quality: best practices, unit testing (with JUnit), debugging strategies. - Exception handling. - Generics. - Basic IO operations (user input / reading / writing). - Design & implementation of small programs. - Functional programming in Java (streams, map, filter).	
Study Goals	After completing this course, the student: - Can write methods and classes using the Java programming language. --- Can write methods and classes using correct indentation and coding style. --- Can write equals and hashCode methods for a class using the Java programming language. --- Can create and manipulate a data structure using methods and classes. - Can design, implement and debug a simple program that contains inheritance and interfaces using the Java programming language. - Can design, implement and debug a simple program that reads from files, and writes to files using the Java programming language. - Can write unit tests for a program in Java using the Junit testing framework. - Can effectively search for existing packages, classes and programming solutions, using the Java documentation as well as online resources, and apply this information in their own programs. - Can describe key characteristics of different programming paradigms and the Java programming language.	
Education Method	Lectures, tutorials, and practical sessions with programming exercises. The programming exercises are feedback assignments, i.e., they do not count towards your grade, but you can get feedback on them to improve your programming skills.	
Assessment	This course contains two summative exams: 1) Theoretical exam in Weblab (Multiple-Choice, Open & (Small) Programming questions; no auxiliary materials may be used). 2) Programming exam (Open book, IDE & additional resources/documentation allowed). This course contains two formative tests. These allow you to practice programming in a real exam environment: 1) A Midterm Theoretical Exam (in week 5). 2) A Mock Programming Exam (in week 8). Between the two formative tests you can score a bonus on your final grade. Bonus = MIN(10, Midterm Bonus + Mock Exam Bonus). Your final grade is composed of: 1) The Theoretical Exam (weight 0.5, minimal required grade 5/10) 2) The Programming Exam (weight 0.5, minimal required grade 5/10) 3) The bonus score you can obtain by participating (and scoring points) in the midterm exam and mock exam. Your final grade is calculated as follows: Exam = (Theoretical Exam Grade + Programming Exam Grade) /2 Final = Exam + (Bonus x (10 - Exam) / 100) (This formula linearly scales down the amount of bonus you get as your exam score increases, and follows the directions of the Board of Examiners for bonus points). If your score for the Theoretical Exam and Programming Exam are (1) both minimally a 5/10 and (2) the "Final" result of the grading formula is at least 5.8, you pass this course. Within the academic year you can resit the Theoretical Exam and the Programming Exam in quarter 2. Out of the two attempts for the Theoretical Exam and the Programming Exam respectively, the highest score will count towards your final grade. Partial grades for the Theoretical Exam or Programming Exam cannot be carried over to a next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	This course makes use of a course email: ip-cs-ewi@tudelft.nl	
Co-Instructor	Prof.dr. A.E. Zaidman	

CSE1105	Collaborative Software Engineering Project	5
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/2/0/0 lecture; 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Modern software development is not just about knowing the right programming language or the right technology stack. It is as important to know how to work together in a team to get a serious programming project done. In this course, you will learn how to collaboratively create a software system. The contents of this course include learning about teamwork, interpreting requirements given by a client and breaking them down into actionable tasks, using popular collaboration tools like GitLab to plan and track project progress, using collaborative tools like Git, and learning popular frameworks and libraries to bring the project to success. The first weeks of the course will be structured through lectures to get you and your group started, afterwards, the course will transition into the implementation phase, in which you will work on the project. Every group will be closely supervised by a tutor. *Important:* The course has submission deadlines from week 1 onwards. Plan your travel to Delft accordingly.	
Study Goals	After completing this course, you will be able to ... 1) Act professionally in a team of software developers ... Make agreements within your team about your collaboration ... Chair a meeting and take minutes ... Give and receive feedback 2) Manage a complex software project in a group ... Break down a formal product backlog into actionable tasks ... Plan and manage the project tasks in GitLab ... Organize and distribute tasks between team members 3) Follow a collaborative software engineering process ... Use Git to version and share source code contributions ... Integrate testing in your process ... Contribute source code to a group project through GitLab merge requests ... Evaluate code contributions of other team members in GitLab merge requests 4) Create a client/server application from a provided set of project requirements ... Build a server application with Spring Boot ... Build a graphical client with Java FX ... Exchange data through REST using automated object mapping ... Manage the lifecycle of an application through dependency injection 5) Apply usability principles in a graphical user interface	
Education Method	Following this course will require you to... - Follow interactive lectures and pre-recorded material for self-study - Participate in a collaborative programming project in a small team - Assess the work of yourself and others through merge requests (code) and peer reviews (process) - Actively participate in TA sessions	
Assessment	This course has the following knock-out criteria which you need to meet: - Weekly contribution criteria. If you fail to meet these in two subsequent weeks, you fail the course. - End of course minimum contribution criteria. If you do not reach these by the end of the course you fail the course. - There are weekly mandatory TA meetings starting in week 2. Missing 2 without providing a valid reason beforehand results in failing the course. The course has several pass/fail assignments, which must all be passed. Assignments marked as (repairable) will have an opportunity for submitting an improved version, if an initial submission was insufficient. - Conducting an effective meeting - (repairable) Code of conduct - (repairable) Providing constructive feedback with the AID model - (repairable) Git assignment The following assignments determine your final grade. Graded items marked as repairable can be repaired by resubmitting an improved version if the grade of the initial submission was >=4.0 and <6.0. In the case of a resubmission the grade will be capped at a 6.0. - Code contribution and code reviews (25%) - Tasks and Planning (25%) - (repairable) Implemented Features (25%) - (repairable) Technology (15%) - (repairable) Usability (10%) Each partial grade has to be at least a 5.0 and the final grade has to be a 6. Partial grades are rounded to .1 and the final grade to .5. --- Please Note --- - The full specification of the minimum criteria, deadlines and more information about the assessment will be posted on Brightspace. - There is NO resit opportunity for this course. - Partial grades are not carried over to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	In addition to your regular enrollment of the course, you must also register in Brightspace, AT THE LATEST during the first lecture, so you can be assigned to a group. Late registrations after the group creation cannot be considered anymore.	

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Co-Instructor	I.C. de Bruin

CSE1110	**Software Quality and Testing**	**5**
Responsible Instructor	F. Mulder	
Contact Hours / Week x/x/x/x	0/0/0/4 lecture; 0/0/0/4 shared lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course covers the most important testing techniques needed to build high quality software systems. The course covers both the testing techniques themselves, and the software design techniques needed to create testable systems. Specific topics covered are: * testing principles, test levels, and the testing pyramid * software testing automation and JUnit * specification-based testing * structural testing and the different coverage criteria * boundary analysis and testing * design by contract * property-based testing * test doubles and mock objects * design for testability * test-driven development (TDD) * larger tests (SQL and web testing) * test code quality and test smells	
Study Goals	By the end of the course, students will be able to: * Create unit, integration, and system tests using current existing tools (like JUnit, Mockito, and JaCoCo) that effectively test complex software systems. * Derive test cases that deal with exceptional, corner, and bad weather cases by performing different testing techniques as well as able to reflect about their limitations, when and when not to apply them in a given context. * Measure and reflect on the efficiency of the developed test suites by means of different test adequacy metrics (i.e., line, branch, condition, MC/DC coverage). * Design and implement testable software systems by means of architectural patterns such as dependency injection and ports and adapters. * Write maintainable test code by avoiding well-known test code smells. * Understand and apply state-of-the-art techniques for intelligent software testing, such as property-based testing.	
Education Method	This course follows a flipped classroom approach. Students read chapters of the book and work on (practical and theoretical) exercises during their study time; lectures are used for questions and broader discussions. We offer TA support during the labs.	
Books	Effective Software Testing, by Maurício Aniche: https://www.effective-software-testing.com.	
Assessment	The evaluation of the course is based on one exam. It happens on WebLab and is composed of coding questions and theoretical questions. Students can consult the documentation of certain testing libraries during the exam. There is also one possibility to do a resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. C.E. Brandt	

CSE1305	Algorithms and Data Structures	5

Responsible Instructor	I.D. van Kreveld
Responsible Instructor	M. Khosla
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab
Education Period	3
Start Education	3
Exam Period	3 4
Course Language	English
Expected prior knowledge	Java programming, constructing proofs, trees and graphs

Course Contents

Algorithms and data structures are fundamental notions in computer science . Understanding how they can be exploited in combination for better programming implementations in terms of time and space complexity is vital for writing efficient code.This course enables the student to:

- Explain and compare the structure and properties of standard algorithms and data structures.
- Execute and visualize standard algorithms and data structures on given inputs.
- Use mathematical methods to analyze the time and space complexity of algorithms and data structures.
- Implement algorithms and data structures using the Java programming language.
- Solve programming tasks using standard algorithms and data structures.

Study Goals

1. Data Structures

- data containers
-- vector
-- list
-- tree
-- set
- ordered data structures
-- stack
-- queue
-- priority queue
-- heap
-- map
- operations on data structures
-- iterative implementations
-- recursive implementations

The student can explain essential data structures, explain and analyze the core properties of these data structures, and explain, develop, and adopt iterative and recursive implementations of operations on these data structures, and analyze and compare space and time complexity of these operations.

2. Sorting

- selection sort
- insertion sort
- heap sort
- merge sort
- quick sort
- bucket sort
- radix sort

The student can explain and implement various sorting algorithms, analyze and compare space and time complexity of these algorithms, and choose sorting algorithms for different problem scenarios.

3. Searching

- search structures
-- search trees
-- AVL trees
-- (2,4) trees
- backtracking

The student can explain data structures which support efficient search and their invariants, explain, develop, and adopt implementations of operations on these data structures, analyze and compare space and time complexity of these operations, choose search structures for different problem scenarios, and explain, implement, and analyze backtracking algorithms.

4. Graphs and Graph Algorithms

- graph data structures
-- directed graphs
-- undirected graphs
-- weights
-- representations
- graph algorithms
-- graph traversals
-- path finding
-- cycle finding
-- connectivity
-- topological ordering
-- shortest path
-- minimum spanning tree

The student can explain various kinds of graphs, explain, analyze and compare different data structures for graphs, explain, develop and adopt implementations of operations on these data structures, analyze and compare space and time complexity of these operations, choose graph data structures for different problem scenarios, explain, implement, apply and analyze generic graph traversal strategies, and explain, develop and adopt implementations of various graph algorithms.

Education Method

Every week has two lecture sessions during which the key concepts will be explained, questions about the material will be answered, and small exercises will be completed with the students. Students will study the relevant material before every session through videos, slides, and the course book.

	Homework assignments are provided. Lab sessions are offered, in which answers to the homework can be discussed.
Books	Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser Data Structures And Algorithms In Java 6th edition ISBN 978-1-118-80857-3
Assessment	The course comprises the following assessment moments: Week 5: a formative analysis midterm exam (MA) Week 5: a formative implementation midterm exam (MI) Week 10: a summative analysis final exam (FA) Week 10: a summative implementation final exam (FI) The final course grade GF will be as follows: If FA >= 5 and FI >= 5, then GF = (FA+FI)/2 Otherwise, GF = NV For successful completion of the course: - both grades FA and FI need to be at least 5 (before rounding), - the final course grade GF needs to be at least 5.8. The standard TU Delft rounding rules apply. Note: the midterms do not contribute to the final course grade, their purpose is only to give the student an idea of the progress halfway through the course. There are separate resits for the analysis (RA) and the implementation (RI) components. The course grade GR after the resit will be determined as follows: If MAX(FA,RA) >= 5 and MAX(FI,RI) >= 5, then GR = (MAX(FA,RA)+MAX(FI,RI))/2 Otherwise, GR = NV For successful completion of the course after the resit: - FA or RA needs to be at least 5 and also FI or RI needs to be at least 5 (all before rounding), - the course grade after resit GR needs to be at least 5.8. The standard TU Delft rounding rules apply. Partial grades cannot be carried over to future editions of the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Permitted Materials during Tests	Paper-based analysis exams: you are allowed to bring a hand-written double sided A4 cheat sheet. No additional materials are permitted, including but not limited to: books, printouts of the slides, or devices. Computer-based implementation exams: no materials are allowed during the computer-based implementation exams.
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/
Co-Instructor	I.D. van Kreveld

CSE1405	Computer Networks	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/0/4 lecture; 0/0/0/4 shared lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Computer networks form the basis of all digital communication. They are typically structured in layers. Each layer has clearly defined functions and algorithms, which we will discuss in class. Furthermore, the course also discusses potential security issues and defences in computer networks, thereby also providing an introduction to cryptographic notions utilised by those defence mechanisms.	
Study Goals	1) The student can describe the principles of layered architectures, in particular the functions and purposes of the various network layers. 2) The student can describe the core algorithms per layer and provide arguments why these algorithms fulfil the function of the respective layer. 3) The student can apply the core algorithms to toy examples. 4) The student can identify potential network security issues and choose suitable mitigation measures. 5) The student can exemplify the concepts of ethical hacking and responsible disclosure.	
Education Method	Lectures (8 weeks x 2 lectures x 2 hours). During these lectures, new concepts are illustrated through the discussion of multiple examples from the book. Lab (5 weeks x 4 hours). During these labs the student gains experience with commonly used Internet software (e.g. Wireshark). Additionally the student applies given abstractions and network concepts.	
Literature and Study Materials	Computer Networks, Andrew S. Tanenbaum et al., 6th (Global) edition. We might provide additional material to further explain or practise with certain topics.	
Assessment	Written exam on paper, closed-book. There is a single resit opportunity of the exam. Exams are a combination of multiple choice questions and open questions. The final test is your final grade. The lab work is not graded. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	None	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme or an EEMCS bridging programme (CS, CESE) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris (a.k.a. MyTUDelft), separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ir. O.W. Visser	

CSE1500	Web and Database Technology	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	Dr. rer. nat. U.K. Gadiraju	
Contact Hours / Week x/x/x/x	0/4/0/0 lecture; 0/4/0/0 shared lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language.	
Course Contents	Web & Database Technologies is a first-year course that provides an introduction to web and database technologies and programming. Topics related to web technology and programming include: = Introduction to the web and HTTP; = Introduction to web development and app design; = Front-end development: HTML, CSS, JavaScript; = Back-end development: Node.js. Topics related to database technologies include: = Introduction to databases; = Overview of database Languages and architectures; = The basic relational model; database design theory; = SQL: data definition, constraints, updates, queries, views, triggers; = Relational databases and object persistence; = Introduction to non-relational database systems: key-value stores, document stores, graph databases.	
Study Goals	Upon completion of this course, participants will have gained knowledge of web and database system concepts and the ability to: 1. Explain the basic architecture of the Internet and the web; 2. Analyse the requirements for a web application given a description of an application idea; 3. Create an application front-end with HTML and CSS based on a given static design; 4. Create an interactive web application with client-side JavaScript; 5. Develop server-side code for authorization, cookies, sessions and database access; 6. Compare the strengths and weaknesses of different data management techniques, describing their most suited applications trough meaningful examples; 7. Enumerate and define the main elements of the relational model; 8. Describe and design SQL queries for the creation, altering, and manipulation of tables, indexes, and views; 9. Develop logical database schema, with principled design that enforce data integrity; 10. Prototype and deploy database applications using open-source database systems.	
Education Method	Lectures, and practical work (lab assignments).	
Literature and Study Materials	In addition to the official course book(s), students will be provided with: = The slides presented in class, cross-referenced to the content of the book(s); = Additional study materials, selected to provide insights on the topics covered during the lectures.	
Practical Guide	All necessary information about the practical setup (schedule, grading, exams, resits, etc.) of the course is available in the course syllabus (accessible via Brightspace).	
Books	WEB There is no required book for the web technologies part of the course. DATABASES The book is not required but helpful: Fundamentals of Database Systems, 7th Edition, Augustus 2016 (Global Edition). Authors: Ramez Elmasri, Shamkant Navathe. ISBN10: 1292097612. ISBN13: 9781292097619 URL: http://catalogue.pearsoned.co.uk/educator/product/Fundamentals-of-Database-Systems-Global-Edition/9781292097619.page	
Assessment	There are two exams: Midterm in W2.5 and Final in W2.10. The exams will be executed in a combination of written and weblab-based manner, with multiple choice, short answer questions, and open questions.	
Permitted Materials during Tests	No books or other materials are allowed during exams.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Tags	Databases Software	
Co-Instructor	F. Mulder	
Co-Instructor	Dr. R. Hai	
Co-Instructor	Dr. A.J. Power	

CSE1505	Information and Data Management	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	Dr. A. Katsifodimos	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic understanding relational query languages (SQL) and databases (as obtained in the Web and Database Technology Y1 Q2 course). This class also relies heavily on Algorithms and Data Structures.	
Course Contents	In this course you will learn how to model, manage, and query information in a Database Management System (DBMS). We will focus on different data modelling techniques, but also on aspects of query processing like for example query parsing or optimization. Also transaction management is a central topic of interest. The core focus is on classic relational OLTP databases, but we will also briefly touch alternative systems like the current NewSQL-trend and systems optimized for in-memory or OLAP applications. Additionally, NoSQL systems and their respective data models will be discussed. - Database normalization: modelling in an unambiguous and non-redundant manner in relational tables such that inconsistencies and anomalies are avoided. - File-organization and indexing: the data structures and their associated algorithms to store data in memory and on disk, such that it can be retrieved and processed efficiently when needed. - Query processing and optimization: the techniques that are used by a DBMS to execute certain SQL queries as efficiently as possible, taking into account how the data is stored and which indexes are present. This also covers Relational Algebra. - Transaction management and concurrency control: the techniques that a DBMS offers to allow different users and application to operate in parallel on the same data without interfering with each other or causing inconsistencies in the data. - NewSQL and alternative system: the systems beyond traditional relational database design space optimized for specialized requirements like extreme OLTP performance, OLAP queries, or in-memory storage. - NoSQL and non-relational data: in the recent years, systems not relying on the relational model became popular. We will also discuss how this affects data modelling, especially with respect to document models, key-value models, wide-columnar models, but also graph models.	
Study Goals	The course aims at teaching the basic competencies required to design and understand applications interacting with relational database management systems (RDBMS). Students will gain an insight into how RDBMS are implemented, learn about the available design choices, and the trade-offs imposed by different designs. This allows students to competently identify and solve problems in application design, and to improve and tune data models and queries. We will also deal with non-relational systems and discuss how they affect the data processing workflow. - Perform Data Modeling and be able to: -- Design and implement normalized relational schemas in a database system, by designing Entity Relationship diagrams and translating them to relational tables. -- Describe alternative data models such as the document model, key-value model, or graph models and analyze the trade-offs as in comparison to the relational model. -- Analyze the requirements of a given application scenario and be able to derive a suitable data model. - Get a grip of the internals of relational database systems by being able to: -- Describe and analyze the two main ways of storing and representing data in relational databases as well as analyze their performance trade-offs with respect to input/output (IO) costs. -- Describe different index structures and analyze their impact on query response time for range-, point- and join queries. -- Analyze the trade-offs between different physical operators for database operations such as joins, aggregates and filters. -- Describe and apply the query optimization process and design physical execution plan and compute the costs for their relational operations. -- Explain the need for transaction protocols and illustrate database update anomalies that can arise in the absence of such protocols. Be able to apply such protocols for given scenarios. -- Evaluate and apply different recovery algorithms such as UNDO/REDO logging, under different types of failures and their frequency.	
Education Method	Lectures and Homework Lab	
Computer Use	Students will require access to a personal computer. The labs will deal with larger data sets, so some HDD space (several GB) will be needed.	
Literature and Study Materials	The course and evaluation is based on the provided lecture slides and assignment material. For alternative explanation or more in-depth studies, the recommended book can be used.	
Books	Fundamentals of Database Systems by Ramez Elmasri and Shamkant Navathe, Pearson. Preferred 7th edition, but also the older ones are usable. Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, and Jennifer Widom (2nd edition preferred, older ones also OK). These books are NOT required, but might be helpful for more in-depth studies. Additional (optional) books/resources are mentioned in the lecture.	
Assessment	This year, there will only be a final exam which counts for 100% of the course grade. The lab assignments are not mandatory and will not contribute to the grade, but we STRONGLY recommend doing them anyway.	
Permitted Materials during Tests	1 handwritten sheet of A4 paper (any content chosen by the student, writing front and back permitted).	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme or an EEMCS bridging programme (CS, CESE) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via Osiris, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ir. T.A.R. Overklift Vaupel Klein	
Co-Instructor	Dr. A.J. Power	

CSE2115	Software Engineering Methods	5
Responsible Instructor	Prof.dr. A. van Deursen	
Responsible Instructor	T.V. Aerts	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course covers the most important software engineering practices needed to build high-quality software systems. The overall goal is to give the students the instruments to become Software Professionals. The course covers both high-level software project management, and low-level software design techniques needed to create evolvable and maintainable systems. Specific topics covered include design patterns, requirements engineering, UML modeling, software architecture, and software analytics. In the lab work, students will put theory into practice using various real-world software systems as case studies.	
Study Goals	(1) Students can select appropriate software engineering methods, which are designed and used to build maintainable and evolvable software systems. (2) Students can judge the benefits/drawbacks of the presented software engineering methods that are commonly used in current software development. (3) Students can use presented software visualization techniques and software analytics to obtain an overview of an existing software system and to detect anomalies in the structure of the code. (4) Students can judge the real benefits/drawbacks of using certain design patterns and software validation techniques in a given software project. (5) Students can derive the valid requirements for a new software project, given natural-language descriptions and interaction with stakeholders. (6) Students can apply the most appropriate software engineering practices in a project, given the context of development.	
Education Method	Labwork + Lectures	
Literature and Study Materials	Materials will be provided via Brightspace	
Assessment	-- Final grade composition -- The final grade of this course is calculated using the following weights; * Digital exam: 100% * Oral Presentations: Pass/Fail * Attendance to weekly meetings (Oral Presentations): Pass/Fail -- General remarks -- * Each grade must be sufficient to pass the course. * Passing the course is conditioned by passing the Oral Presentation module. * Partial grades are not carried over to the next edition of this course. * In case you have completed Oral Presentations before then you may be eligible for an exemption for the attendance to the weekly lectures. Check Brightspace for more information. -- Resits -- * The re-sit will replace the grade for the exam only. * For the Oral Presentations part we offer a single resit possibility. -- Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	Contact with the course staff via: sem-cs-ewi@tudelft.nl (use your TU Delft email address).	
Co-Instructor	Prof.dr. A. van Deursen	

CSE2310	Algorithm Design	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. E. Demirovi	
Contact Hours / Week x/x/x/x	0/4/0/0 lecture; 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: programming experience from CSE1100 or equivalent Strongly advised: experience with algorithms and data structures (priority queue, graph algorithms etc) such as from Algorithms and Datastructures (CSE1305); experience with constructing proofs, such as from Reasoning and Logic (CSE1300)	
Course Contents	Graph algorithms, greedy algorithms, divide and conquer, dynamic programming and network flow algorithms	
Study Goals	After completing this course, the student can apply, reproduce, and give relevant properties and definitions of 1. graph algorithms, 2. greedy algorithms 3. divide and conquer, 4. dynamic programming and 5. network flow algorithms. The student can also 6. prove the correctness of these algorithms, 7. determine the running time of algorithms both analytically as well as experimentally, 8. implement algorithms efficiently, and 9. provide well-argued advice about which problems can be solved best using which algorithms.	
Education Method	Lectures (possibly only online / pre-recorded) Each lecture includes a number of multiple choice questions and a selection of especially the difficult components of the material. Students are expected to study some of the topics at home. Tutorials (possibly only online) In these interactive sessions we take you through some example assignments (both implementation and written ones). Scheduled classroom lecture hours may be used for a combination of the above, as announced by the lecturer. Skill circuit assignments To train with the learning objectives, several assignments at exam level, but also more accessible ones are available on BrightSpace as elements in the skill circuit. For some of them feedback can be requested from teaching assistants or other students if handed in on time. Similarly, students can train for the digital test by making the implementation assignments in the skill circuit. Correctness of the code will be automatically checked.	
Literature and Study Materials	The study material consists of the book Algorithm Design by John Kleinberg and Eva Tardos (2005), Addison Wesley ISBN 13: 9780321372918 or 9780321295354 and some documents made available through BrightSpace (such as a manual of proving techniques and the master's method for solving simple recurrent relations).	
Assessment	The final grade of the course consists of the following components: Written exam part 1 (of 1.5h) in week 5 (W1) Written exam part 2 (of 1.5h) in week 10/11 (W2) Computer exam (3h) in week 10/11 (P) Final grade The final grade for the course is computed as follows: W = 0.4 W1 + 0.6 W2 If P < 5 or W < 5, the final grade is min(P,W). Else the final grade is 0.5 W + 0.5 P Both parts P and W can be retaken, in individual exam moments in the resit week. Written exam (W) Half-way and at the end of the quarter written examinations of 1.5 hours each are organized. The written exam primarily assesses proving skills and (application of) algorithms and algorithmic theory. Resitting W implies redoing both parts in a single three-hour exam. Computer exam (P) Algorithm design and implementation skills are assessed through a computer exam of three hours at the end of the course. The computer exam needs to be submitted through Weblab, but IntelliJ is available within the exam environment. These partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Special Information	course email address: ad-cs-ewi@tudelft.nl	
Co-Instructor	Dr. E. Demirovi	

CSE2315	Automata, Computability and Complexity	5
Responsible Instructor	Prof.dr. M.T.J. Spaan	
Responsible Instructor	I.D. van Kreveld	
Responsible Instructor	S. Hugtenburg	
Responsible Instructor	Dr. J.W. Böhmer	
Contact Hours / Week x/x/x/x	0/0/4/0 lecture; 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Proof techniques (such as by induction or contradiction), set theory, recursive definitions, trees and graphs. For example from CSE1300 or equivalent. Run-time complexity of algorithms. For example from CSE2310 or equivalent.	
Course Contents	This course introduces three areas of the theory of computation. First, automata are used to recognize words in (formal) languages, and we will discern several types of languages along with the types of automata that can recognize them. Second, once we have an intuition about languages, we discuss the topic of computability, where you will learn not only what kinds of problems a computer can solve but also how to prove this. Third, we will examine the class of computable problems and make a distinction between "easy" and "hard" problems and delve into the famous problem: P = NP?	
Study Goals	After this course, the student can: 1. parse and construct automata and languages. 2. prove a problem to be computable. 3. manipulate and construct grammars for context-free languages. 4. describe the differences between different types of automata and grammars and their limitations, specifically that there are problems that are algorithmically unsolvable. 5. prove languages to be non-regular using the pumping lemma. 6. describe the classes P, NP, and NPC and explain why P = NP? is an open problem. 7. verify a proof that shows that a problem is NPC.	
Education Method	Engaging interactive lectures for which the student studies the material at home. During the lecture, the students test their knowledge via multiple choice questions and the teacher explains difficult topics on demand. Lab course in the form of optional take-home assignments, 8 times during the quarter. Students solve these assignments individually and hand them in weekly according to predetermined deadlines. Students correct each others work followed by a round of feedback by teaching assistants.	
Literature and Study Materials	1. Textbook (see Books), chapters 0, 1, 2 (2.1-2.3), 3, 4, 5 (5.1), 7. 2. Supplementary material (lecture slides, video lectures).	
Books	Michael Sipser, Introduction to the Theory of Computation, Third, International Edition, Cengage Learning. ISBN 978-1-133-18781-3. Note: if you do not use the international version, then section numbers and exercises will be different!	
Assessment	A written, closed-book exam (see permitted materials during tests below). The final grade of the course is based on the final exam for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	You are allowed to bring a hand-written double sided A4 cheat sheet. No additional materials are permitted, including but not limited to: books, printouts of the slides, or devices.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Prof.dr. M.T.J. Spaan	

CSE2510	Machine Learning	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Responsible Instructor	M.A. Migut	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	CSE1100 Introduction to Programming CSE1305 Algorithms and Data Structures CSE1200 Calculus CSE1205 Linear Algebra CSE1210 Probability Theory and Statistics	
Course Contents	The goal of the course is to acquaint students with the basic Machine Learning concepts and algorithms. Specifically, the course will cover parametric and non-parametric density estimation, linear and non-linear classification, unsupervised learning including clustering and dimensionality reduction, performance evaluation of predictive algorithms and ethical issues in machine learning.	
Study Goals	After succesfully completing this course, the student is able to: - explain the basic concepts and algorithms of machine learning and their underlying statistical concepts. - implement, apply and evaluate basic ML algorithms in Python. - explain the concept of and identify (implicit) bias in data and ML algorithms.	
Education Method	The course comprises two lectures and one (four-hour) lab session per week.	
Assessment	Computer exam: 100% of the final grade. Reparation: There is one resit per year for the digital exam. The passing grade is 5.8. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Dr. D.M.J. Tax	
Co-Instructor	I.D. van Kreveld	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging programme EE for HBO EE 2024

EE1E1	Electrical Energy Fundamentals	5
Responsible Instructor	Dr. M. Cvetkovic	
Responsible Instructor	Dr. J. Dong	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Strongly advised: EE1P1 Electricity and magnetism	
Course Contents	Starting with the energy conversion principles, this course will introduce the electrical engineering students to the critical energy conversion components and fundamentals of power systems. After the course, the students will be able to analyse and model energy components and power systems. This is essential knowledge for future engineers in the energy transition.	
Study Goals	By the end of this course, you should be able to: Electrical energy conversion components part: - Explain the power system based on AC and DC connections; - Tell how electromagnetic principles are applied in energy conversion; - Describe the functioning of momentum of inertia, pulley & belt and gearboxes. - Calculate the steady state operation of system containing mechanical components. - Explain the construction and operation principle of a DC machine; - Explain the nameplate of a given machine; - Apply steady state performance calculation of a DC machine - Explain the common operation principle of AC machines - Explain the construction and operation principle of an induction machine; - Apply steady state performance calculation of an induction machine; - Explain the construction and operation principle of a synchronous machine; - Apply steady state performance calculation of a synchronous machine; - Tell the conditions to connect a synchronous machine to the AC power grid; - Tell the roles of AC machines in modern power grid. - Explain how to use power semiconductors and passive components to convert power; - Tell main DC-DC converter topology (buck, boost) - Analyze DC-DC converters at steady state; - Describe the operation principles of H-brdige inverter and rectifier Power system part: - Describe a power system structure - Model a power system - Describe operation principles during normal conditions - Describe operation principles during disturbances - Apply methods to analyze a power system	
Education Method	32 hrs lectures 2 hrs practical self-grade assignments each week	
Literature and Study Materials	Pieter Schavemaker, Lou van der Sluis, "Electrical Power System Essentials", John Wiley and sons; James L. Kirtley, "Electric Power Principles", John Wiley and sons. Handouts (copies of the lecture slides). Supplementary reading material offered via Brightspace.	
Assessment	1 final exam; 1 lab report, pass/no-pass; A resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	Final exam: 3 hrs Resit: 3 hrs	
Permitted Materials during Tests	During the exam, students can use a simple non-programmable and non-graphical calculator (like the TI-30 or Casio FX-82). The exam is NOT an open book exam. During the exam, students are allowed to ask questions to clarify the wording.	
Co-Instructor	Dr. M. Cvetkovic	
Co-Instructor	Dr. J. Dong	

EE1M1	Calculus	5
Responsible Instructor	Dr. W.M. Schouten-Straatman	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Calculus and linear algebra Electricity and magnetism	
Expected prior knowledge	Strongly advised: Introdution to Electrical Engineering	
Course Contents	"In this course the basic concepts from calculus are treated. The course is roughly divided into three parts. The first part of the course focusses on single variable calculus. This parts main subjects are general understanding of functions of a single variable and integration of these functions. General understanding of functions contains a variety of topics, including inverse (trigonometric) functions, limits, derivatives, implicit differentiation, Taylor polynomials and lHospitals rule. Subsequently, the subject of integration mainly focusses on the common integration techniques the substitution rule, integration by parts and integration by partial fraction definition and ends with a treatise of improper integrals. The second part of the course focusses on multivariate calculus. After starting with some general geometrical concepts such as the dot and cross product, as well as graphs and contour plots for functions of several variables, one of the main subjects of this part is partial derivatives and various related topics, including tangent planes, linear approximations, the chain rule, directional derivatives, the gradient and extreme values. This part finishes with a treatise of double and triple integrals, including polar, cylindrical and spherical coordinates, as well as general changes of coordinates. The final part of the course focusses on vector calculus. In this part, vector fields are introduced. In addition, line integrals are covered both for scalar functions and for vector fields and connected to double integrals by means of Greens theorem. "	
Study Goals	"At the end of this course, you should be able to 1. Apply the definitions, theorems and formulas related to the topics mentioned in the Course Contents in concrete mathematical problems. 2. Relate applications within the Electrical Engineering program to these definitions, theorems and formulas. 3. Decompose mathematical problems in relevant subproblems. "	
Education Method	Colstruction (lectures combined with instruction) Typical routine: - Preparation: Students do homework, study a pre-lecture video (or do pre-lecture exercises) - Participation: Students come to the lecture which is a block of two hours. In the lecture a new topic will be introduced by the teacher. During the lecture there will be time to do exercises. Sometimes a practical application of the theory is discussed. - Post-lecture work: Students do homework assignments. These contain book exercises as well as exercises in a digital environment.	
Literature and Study Materials	"Calculus: Early Transcendentals', Interational Metric Edition, 9th edition, by James Stewart, Daniel Clegg and Saleem Watson. ISBN: 9780357113516 Open Linear Algebra (distributed via BrightSpace)"	
Assessment	Examination method: this course will be assessed using examination software with a written component. The exams will contain a mix of short-answer exercises, where only the final answer is graded, and open questions, where the full reasoning and calculation are considered. The examination software (Grasple) will be used; therefore, the exams will take place in a computer room at the TU Delft. The written component will be on paper. Course grade The final grade of the course consists of the following components: - Partial exam 1 This exam covers the material of the first half of the course. This exam lasts 2 hours. - Partial exam 2 - This exam mainly covers the material of the second half of the course. However, some techniques from the first half of the course might be used as well. This exam lasts 2 hours. Exam grade calculation: (0.5 * Partial exam 1 + 0.5 * Partial exam 2) Bonus calculation: At the end of each lecture, a set of practice exercises will be posted on the Brightspace page using the online platform Grasple. Completing each exercise set results in a red, yellow or green check mark, depending on the score. You are allowed to take multiple attempts at each set to improve your score, but the questions may vary slightly per attempt. Participation is voluntary. The bonus grade is based on the Grasple exercises in the following way: each red checkmark counts as a 3, each yellow checkmark as a 6 and each green checkmark as a 10, while a set not made results in a grade equal to 0. For this, the results on the day of the final exam are the ones that are taken into account. The bonus grade is the average over the best 75% of all these grades. Final grade calculation: (exam grade) + ((bonus grade)*(10 - exam grade)/100) Repair possibilities: - Endterm: The resit is taken in one part (i.e., contains all course topics). The result of the sub-exams expire before the resit, and are not transferable to the next academic year. This covers the entire course contents. This exam lasts 3 hours. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Drs. A.T. Hensbergen	

EE1M2	Calculus and Linear Algebra	5
Responsible Instructor	Dr. W.M. Schouten-Straatman	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	Linear algebra and differential equations	
Expected prior knowledge	Required: Calculus	
Course Contents	"The goal of this course is to lay a strong foundation for the various and numerous applications of calculus and linear algebra, both directly within the field of electrical engineering as well as in general engineering mathematics. This course consists of two halves: calculus and linear algebra. The first half of this course starts directly where the previous calculus course ended, by introducing parametric surfaces and surface integrals, as well as the curl and divergence of a vector field. Then, all topics in multivariate and vector calculus are connected by means of Stokes theorem and the divergence theorem. Afterwards, the focussed is shifted towards series. These are introduced and analysed by means of various convergence tests, such as the integral test, the alternating series test, comparison tests and the ratio test. Finally, power series are introduced, which results in a treatise of Taylor series. The second half of this course starts from scratch with methods from Linear Algebra to solve and analyse systems of linear equations. Here concepts like vectors, linear combination, span and linear independence play a central role. In addition, matrices and matrix algebra are key concepts that will be discussed, with a major focus on invertibility of matrices. Finally, a lot of time will be devoted towards the more geometrical aspects of linear algebra, such as linear transformations, LU factorisation, subspaces, coordinates, dimension, orthogonality, orthogonal projections and least-squares problems. "	
Study Goals	"At the end of this course, you should be able to 1. Apply the definitions, theorems and formulas related to the topics in calculus mentioned in the Course Contents in concrete mathematical problems. 2. Perform calculations in the field of linear algebra. 3. Determine whether a given statement in linear algebra is true or false. 4. Connect various properties of matrices. 5. Relate applications within the Electrical Engineering program to the definitions, theorems and formulas as formulated in the Course Contents. 6. Decompose mathematical problems in relevant subproblems "	
Education Method	Colstruction (lectures combined with instruction) Typical routine: - Preparation: Students do homework, study a pre-lecture video (or do pre-lecture exercises) - Participation: Students come to the lecture which is a block of two hours. In the lecture a new topic will be introduced by the teacher. During the lecture there will be time to do exercises. Sometimes a practical application of the theory is discussed. - Post-lecture work: Students do homework assignments. These contain book exercises as well as exercises in a digital environment.	
Literature and Study Materials	Calculus: Early Transcendentals', Interational Metric Edition, 9th edition, by James Stewart, Daniel Clegg and Saleem Watson. ISBN: 9780357113516 Open Linear Algebra (distributed via BrightSpace)	
Assessment	Examination method: this course will be assessed using examination software with a written component. The exams will contain a mix of short-answer exercises, where only the final answer is graded, and open questions, where the full reasoning and calculation are considered. The examination software (Grasple) will be used; therefore, the exams will take place in a computer room at the TU Delft. The written component will be on paper. Course grade The final grade of the course consists of the following components: - Partial exam 1 This exam covers the material of the first half of the course. This exam lasts 2 hours. - Partial exam 2 - This exam covers the material of the second half of the course. This exam lasts 2 hours. Exam grade calculation: (0.5 * Partial exam 1 + 0.5 * Partial exam 2) Bonus calculation: At the end of each lecture, a set of practice exercises will be posted on the Brightspace page using the online platform Grasple. Completing each exercise set results in a red, yellow or green check mark, depending on the score. You are allowed to take multiple attempts at each set to improve your score, but the questions may vary slightly per attempt. Participation is voluntary. The bonus grade is based on the Grasple exercises in the following way: each red checkmark counts as a 3, each yellow checkmark as a 6 and each green checkmark as a 10, while a set not made results in a grade equal to 0. For this, the results on the day of the final exam are the ones that are taken into account. The bonus grade is the average over the 75% best grades of the Calculus part and the 75% best grades of the Linear Algebra part. Final grade calculation: (exam grade) + ((bonus grade)*(10 - exam grade)/100) Repair possibilities: - Endterm: The resit is taken in one part (i.e., contains all course topics). The result of the sub-exams expire before the resit, and are not transferable to the next academic year. This covers the entire course contents. This exam lasts 3 hours. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Drs. A.T. Hensbergen	

EE1M3	Linear Algebra and Differential Equations	5
Responsible Instructor	Dr. B.J. Meulenbroek	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	EE1M2 Calculus and linear Algebra	
Summary	Part 1: Linear Algebra: -- Determinants; -- Eigenvalues/Eigenvectors; Diagonalizability of matrices; -- Quadratic Forms; -- Singular Value Decomposition Differential Equations: -- Homogeneous Linear Systems Part 2: Differential Equations: -- Linear Systems: --- Phase Plane --- Matrix Exponential --- Inhomogenous Systems -- Second order linear DEs; Variation of Parameters -- Non-linear Autonomous Systems: --- Phase Plane; --- Trajectories; --- Stability; --- Linearization; Classification of critical points -- Partial Differential Equations: --- Separation of Variables --- Fourier Series --- Heat Equation --- Wave Equation --- Laplace Equation	
Course Contents	Linear Algebra: -Determinants -Eigenvalues and eigenvectors -Diagonalizability -Symmetric matrices -Singular value decomposition Differential equations -First and second order linear differential equations -Systems of linear differential equations -Systems of nonlinear differential equations -Boundary value problems -Fourier series -Partial differential equations, including heat equation, wave equation, Laplace's equation, Maxwell's equations	
Study Goals	At the end of this course, you should be able to: 1. Apply the definitions, theorems and formulas related to Linear Algebra (see course description) in concrete mathematical problems. 2. Relate applications within the Electrical Engineering program to these definitions, theorems and formulas. 3. Decompose mathematical problems in relevant subproblems.	
Study Goals continuation	See Summary	
Education Method	Colstruction (lectures combined with instruction)	
Literature and Study Materials	Required: - Open linear algebra (linked via Brightspace) - Elementary Differential Equations and Boundary Value Problems, by William E. Boyce (any edition from the 9th is suitable)	
Assessment	The final grade of the course consists of the following components: - Written Partial Exam 1 (50%) - Written Partial Exam 2 (50%) Final grade calculation: Final grade = 0.5 * Written Partial Exam 1 + 0.5 * Written Partial Exam 2 Duration of partial exams: 2 hours Repair possibility: Written Final Exam (=Resit) Duration of resit: 3 hours	
Permitted Materials during Tests	None	
Test and result scale	???	
Pop-up tekst bij intekenen	???	
Co-Instructor	Dr. W.M. Schouten-Straatman	

EE1P1	Electricity and Magnetism	5
Responsible Instructor	Dr.ing. I.E. Lager	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	N.A.	
Expected prior knowledge	N.A.	
Course Contents	This course examines the basics of the electromagnetic field in its static and stationary (slowly-varying) states. The course starts by summarising some fundamental prerequisites by insisting on their specific role and impact in the description of electromagnetic field phenomena. The electrostatic interaction is defined by starting from the electric charge (fundamental electric property of matter), the Coulomb force and electrostatic field. It then introduces the ensemble field description via Gauss's law. The energetic description of the electrostatic field is firstly given in terms of the electric potential, then the electrostatic energy (density) is defined and, finally, the capacitors (as energy storage devices) are examined. The description of the electric state is rounded off by introducing the electric current, in its classical interpretation. The second half of the course concerns the magnetic interaction that is firstly evidenced via the pertaining forces and the manners in which the field is produced. The effect of the presence of matter is briefly touched upon. The fundamental laws governing the magnetic interaction, namely Gausss law, Ampères law and Faradays law are discussed at length, together with some pertaining practical applications. The course ends by highlighting the need for an integrated electromagnetic field description in terms of Maxwell's equations.	
Study Goals	After following this course the student should be able to: Estimate, examine and evaluate the basic electric/magnetic field sources: electric charge and electric current. Estimate, examine and evaluate the basic electric/magnetic field quantities: electric field and magnetic field; discriminate their intrinsically VECTOR substance. Estimate, examine and evaluate derived quantities, namely electrostatic/electric potential and the electric and magnetic fluxes; discriminate their intrinsically SCALAR substance. Identify, interpret and examine relatively simple electrostatic configurations; identify sources and calculate forces and potentials by applying superposition. Express Gausss law, and apply it for calculating the electrostatic field in configurations with high degree of symmetry. Examine and quantify the electrostatic energy stored in relatively simple electrostatic configurations; in particular, apply this to capacitors having canonical shapes (primarily, but not restricted to parallel-plate and cylindrical capacitors). Estimate, examine and evaluate the constant electric currents in wire-like and massive conductors; express and evaluate electric currents with variable current densities and/or in nonhomogeneous, massive conductors. Interpret, examine and evaluate magnetic core guided magnetic field systems by organising them into lumped parameter elements making use of magnetic circuit analogies. Identify, interpret and examine the relation between electric field and the current density, and the relation between the current density and the electric charge + drift velocity; discriminate the current densitys intrinsically VECTOR substance. Identify, interpret and examine the magnetic field produced by relatively simple current-carrying, wire-like conductors. Express Ampères law and Faradays law, and apply them for calculating the magnetic field and the induced electric field in configurations with high degree of symmetry. Understand the significance of Gausss law for the magnetic field and infer the inexistence of magnetic monopoles.	
Education Method	32 hrs lectures 16 hrs seminars 6 graded homework assignments (GHAs) 2 partial exams	
Literature and Study Materials	Richard Wolfson: Essential University Physics, Pearson, 4th edition, 2021. Handouts (copies of the lecture slides) are made available after the lectures the lectures are made available. Supplementary reading offered via Brightspace.	
Assessment	This course has: graded homework assignments (GHAs), handled via a TU Delft platform → the GHAs yield a sliding-weight bonus for the exam (details will be communicated at the start of the course) 2 examinations, (weeks 3.5 and 3.10) A resit. The result of the GHAs is expressed as B, and ranges from 0 to 10. R is calculated as the average of the results of all but one of the GHAs, in which any non-made GHA is graded with a 0; either the least-scoring GHA (when all GHAs were done) or a non-made one are omitted from the average. The results of both partial exams are averaged, this gives the exam grade E. The final course grade C is calculated as follows: C = E + B * (10-E) / 100. To pass the course, a final grade C of at least 5.76 points in scale 0-10 is needed. The final grade will be rounded to half points. The GHAs result is valid for the partial exams and the resit, but only in the year the assignments have been made. The resit includes the contents of both partial exams. The result of the partial exams expires for the resit and is not transferable to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	Partial exams: 2 hrs Resit: 3 hrs	
Permitted Materials during Tests	During the exam, students can use a simple non-programmable and non-graphical calculator (like the TI-30 or Casio FX-82). Students are provided a complete list of formulas. The exam is NOT an open book exam. During the exam, students are allowed to ask questions to clarify the wording.	
Co-Instructor	Dr. J. Dong	

EE2C1	Transistor Circuits	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course extends the analysis of electronic circuits to circuits containing non-linear components as key building blocks for realizing essential functions such a switching and amplification. Noise is introduced as a fundamental limitation to signal quality, motivating the need for signal amplification. The physics of diodes and MOSFETs is explained at an intuitive level, and their characteristic equations are introduced. Large-signal operation is presented as approach to realize digital logic, with the CMOS inverter as basic building block. Biasing and small-signal operation are presented as approaches to obtain linearised behavior from non-linear circuits, and are applied to single-MOSFETs amplifiers (common-source, common-gate, common-drain stages). Amplifiers with feedback are introduced as a means of realizing ideal transfer functions. Finally, opamp-based amplifiers and single-MOSFET amplifiers are shown to be practical approximation of these ideal amplifiers.	
Study Goals	After following this course, you should be able to: 1. Calculate noise levels and SNR based on the thermal noise of resistors. 2. Describe the behaviour of diodes and MOSFETs based on their characteristic equations. 3. Analyse the static large-signal behaviour of simple non-linear circuits, including the CMOS inverter. 4. Analyse non-linear circuits using small-signal analysis. 5. Calculate the small-signal parameters of single-transistor amplifiers based on MOSFETs. 6. Synthesize ideal amplifier transfer functions using passive feedback around active amplifiers. 7. Synthesize CMOS logic circuits to implement Boolean operations	
Education Method	Lectures, instructions, practical (course lab), Self-graded homework	
Literature and Study Materials	"Microelectronic Circuits" by Sedra, Smith, Carusone and Gaudet. 8th international edition, ISBN 9780190853501	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) - Lab assignments (graded pass/fail) Repair possibilities: - A resit - Resit for the course labs In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	Final: 3 hours Resit: 3 hours	
Co-Instructor	Dr.ir. M.A.P. Pertijs	
Co-Instructor	Dr. S. Du	

EE2M1	Probability and statistics	5
Responsible Instructor	Dr. C.O. Smet	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, in particular differentiating and integrating functions of one and two variables is indispensable.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test Parametric bootstrap	
Study Goals	At the end of the course, the students will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - perform a bootstrap analysis - Use Python to perform elementary simulation and data analysis	
Education Method	Education will take place in a blended way: the student watches a prelecture video, makes some exercises related to this, participates in the lecture (which consist of lecturing by the teacher and making exercises) and then makes some more exercises at home. This cycle is repeated for every lecture.	
Literature and Study Materials	We use the "Blue Book": A Modern Introduction to Probability and Statistics Understanding Why and How Series: Springer Texts in Statistics Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover ISBN: 978-1-85233-896-1	
Assessment	The final grade of the course consists of the following components: - Midterm on Probability Theory (50%) - Endterm on Statistics (50%) Final grade calculation: (0.5 * Midterm + 0.5 * Endterm) Repair possibilities: - Resit in next quarter Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Permitted Materials during Tests	Formula sheet (one sheet, to be made by the student) and a simple, non-programmable, non-graphical calculator.	
Co-Instructor	N. de Kleijn	

EE2T1	Telecommunication and Sensing	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Analysis (in particular integration, differentiation and complex calculation). Signals and Systems: LTI-systems, convolution, impulse response, Fourier-transformation and filtering. Probability and Statistics: probability model, conditional probability, independence, probability density function (PDF), expectation and variance, auto-correlation and power spectral density (PSD) function.	
Course Contents	The aim of this course is for you to gain basic insight and understanding of modern telecommunication and sensing systems with a unique combination of physics and signals & systems theory. The course starts with the description of fundamental concepts in telecommunication and sensing including channel capacity, received signal power and noise, link budget, signal modulation and detection, and range-doppler profile. Baseband techniques and bandpass signals with analog and digital modulation methods are discussed in detail. The last part of this course provides an introduction to data communication networking aspects. The lectures are supported by practical numerical problems throughout the course as weekly homework exercises and in the instruction lectures, in order to make you skilled in solving system related problems supported by calculations. The list of topics and distribution over the lectures is as follows: L1. Introduction: a general introduction on modern communication and sensing systems, OSI model, ideal communication and ranging systems, channel capacity (Shannon). L2-3. Wireless and guided links and noise: propagation of e.m. waves, antenna radiation pattern, polarization, received signal power in communication and sensing systems, thermal noise sources, antenna noise, system noise calculation and link budget evaluation L4. Properties of signals: review ideal/practical signals, dB relations, power spectral density, distortionless transmission and bandwidth definitions. L5. Signal sampling and PCM: Signal sampling, natural sampling, flat-top Pule Amplitude Modulation, Pulse Code Modulation: digital encoding, Signal-to-Noise ratio for PCM. L6. Baseband signaling: description of digital information carrying waveforms, line codes and eye pattern, spectral efficiency, inter-symbol interference (ISI). L7. Bandpass signals and systems, distortionless transmission and group delay, bandpass sampling, superheterodyne and homodyne receivers. Analog modulation techniques: Amplitude Modulation (AM), Double Sideband - Suppressed Carrier (DSB -SC) modulation. L8. Analog modulation techniques (continued). Phase Modulation (PM) and Frequency Modulation (FM). Signal-to-Noise Ratio (SNR) after detection for analog modulation techniques. Envelope and product detector. L9. Digital modulation techniques: Binary techniques: OOK, BPSK, DPSK, FSK, Multilevel modulation: QPSK, MSK, QAM. L10. Optimum signal detection for communication and sensing. Optimal receiver, matched filter receiver, waveform ambiguity function. L11. Bit Error Probability framework, BER in baseband systems: NRZ unipolar, polar, bi-polar using a lowpass filter and matched filter. L12. BER for digital bandpass signals using a for coherent (matched-filter) demodulation and non-coherent detection: OOK, BPSK, FSK, DPSK, QPSK, MSK. L13. Fundamentals of sensing: ranging, doppler, range-doppler plane. L14-15 An introduction to the OSI (Open Systems Interconnect) layered model and discuss the OSI layers and their protocols.	
Study Goals	After following this course you are able to: 1. Explain the basic working principles of communication and radar systems, as well as the key similarities and differences between the two applications. 2. Apply the fundamentals of information, signal and EM theory in the performance analysis of simple communication and radar scenarios to estimate quantitatively, (1) signal-to-noise ratio and bit error rate in communication systems, (2) range and velocity of objects in sensing systems. 3. Solve communication and radar system related problems supported by calculations.	
Education Method	Lectures and working lectures. Recorded lectures will be made available via Collegerama. To support studying of the material of this course, exercises are made available for each lecture and each working lecture via Brightspace. These exercises with answers will help you in mastering the course material.	
Literature and Study Materials	Couch, L.W., Digital and Analog Communication Systems, and handout or copies of same chapters of other books, e.g. T.Rappaport, Wireless Communications: Principle and Practice, G. Stimson, Introduction to Airborne Radar, P. Van Mieghem, Data Communications Networking, Chapters 1-7, 2011, ISBN: 978-0-273-77421-1.	
Assessment	The course Telecommunications and Sensing is completed with a 3-hour closed-book written exam covering the full course content at the end of the 2nd or 3rd quarter (resit). The final grade of the course Telecommunications and Sensing is based solely on the grade obtained for the exam, which is is rounded to the nearest half. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the 3-hour closed-book exam you are allowed to use the following materials: - a non-programmable electronic calculator, - the overview Tables and Figures to be used at the exam Telecommunications and Sensing (EE2T1), as made available on Brightspace, - an original self-handwritten equation form: 1 page A4, written on both sides, (not type-written, no copy and it should not contain worked out exercises or examples).	
Co-Instructor	Prof.dr. O. Yarovyi	
Co-Instructor	M.A. Kitsak	

WB2235	Signal Analysis	6
Responsible Instructor	Dr.ing. R. van de Plas	
Instructor	Dr. N.J. Myers	
Instructor	G. de Albuquerque Gleizer	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	WI2030WBMT Wiskunde 3 (Differentiaalvergelijkingen) WI2031WBMT Wiskunde 4 (Kansrekening en Statistiek) should be heard in parallel to the course	
Course Contents	This course treats the analysis, filtering and detection of signals influenced by linear time-invariant systems as they occur in typical mechanical and mechatronic applications. Signals will be considered in both deterministic and stochastic formulations. Special emphasis is put on discrete-time techniques that can be easily implemented in practice. The main topics of the course are: 1. Elementary properties of signals and systems (even, odd, linear, stable, causal, linear-phase, group-delay, all-pass, minimum-phase) 2. Fourier and Hilbert transform, one and two-sided Laplace and Z-transforms 3. Bode plots and Bodes phase-gain relationship 4. Continuous-time low-pass filters 5. Modulation and demodulation 6. Analog-to-digital and digital-to-analog conversion 7. Design of discrete-time filters using windowing and impulse invariance 8. Elementary properties of random processes (i.i.d., w.s.s., joint w.s.s., independence); key indicators such as mean, auto and cross-correlation/covariance, and spectral densities 9. Identification of linear time-invariant discrete-time systems in the frequency domain 10. Discrete-time Wiener filtering (finite impulse response and non-causal) 11. Detection of discrete-time signals in Gaussian noise	
Study Goals	Students can 1. Evaluate if a signal or system possesses an elementary property; utilize these in simple problems; give examples that exhibit desired elementary properties 2. Compute Fourier, Laplace, z and Hilbert transforms, both directly and using transform pairs and properties 3. Sketch Bode plots for rational transfer functions 4. Determine the specifications of continuous-time low-pass filters from their frequency response; design low-pass filters that fulfill given specifications 5. Analyze typical modulation and demodulation schemes; add missing components 6. Evaluate the impact of A/D and D/A conversion schemes; choose sampling intervals 7. Design discrete-time filters using impulse invariance, windowing, and Wiener techniques 8. Evaluate if a random process possesses an elementary property; utilize them in simple problems; determine key indicators of random processes 9. Identify the frequency response of a linear time-invariant discrete-time system from actual measurements; specify the bias and the variance 10. Derive optimal detection rules for a known signal in Gaussian noise	
Education Method	Lectures, problem sets and practice sessions (werkcolleges).	
Literature and Study Materials	- A.V. Oppenheim, A.S. Willsky and S. Hamid, Signals and Systems, Pearson New International Edition, 2nd ed, 2013. - A.V. Oppenheim and G. Verghese, Signals, Systems and Inference, Class Notes for 6.011: Introduction to Communication, Control and Signal Processing, MIT, Spring 2010. Available online: http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-011-introduction-to-communication-control-and-signal-processing-spring-2010/readings/ - For some parts of the lecture, external materials will be used. These materials will either be available online from within the network of TU Delft, or be provided via Brightspace.	
Assessment	Written exam. Students will only be allowed to bring a non-programmable and non-graphing calculator. A formula sheet will be provided.	
Department	ME Department Delft Center for Systems and Control	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bridging Programme EWI**

Bridging programme EE for TUD BSc ME/AE/AP 2024

CSE2420	Digital Systems	5
Responsible Instructor	Dr.ing. M. Shahraki Moghaddam	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 8/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	No prior knowledge is expected from the student.	
Course Contents	This course introduces the following aspects in digital design: - combinational digital systems, - sequential systems, - complex combinatial and sequential building blocks - reconfigurable hardware and complex digital systems. All the important aspects of digital design will be dicussed: - number systems, - Boolean algebra, - combinational logic, - Two-level and multilevel networks and reduction techniques/design tools, - timing hazards, - Moore/Mealy machines, - Finite State Machines (FSM) and more. The lab starts with the realization of simple circuits and will end with complex systems realized on an FPGA.	
Study Goals	You (at the end of the course) have learned: Two fundamental notions of digital systems, namely combinational and sequential networks, and the basics of VHDL/Verilog. You will be able to: 1- To translate non-digital functions to a digital representation. 2- To establish the needed combinational and/or sequential networks that have to obey specific design requirements (e.g., area, timing, and functionality). 3- To model and/or understand simple circuits in VHDL/Verilog. Finally you have gained enough knowledge to follow more advanced course, such as, computer architecture, computer systems, and embedded systems.	
Education Method	Lectures and lab.	
Assessment	* Multiple-choice exam and successful completion of lab. * Weekly online tests (5x) that can lead to 10% of the final grade This rule does not apply for resits. There are two requirements for students to pass the course: --- - Obligatory Lab: Students must successfully complete the lab. Only successful completions are registered. - Written Exam: you must take a written exam comprising at least 20 multiple-choice questions and achieve a passing grade of 6 or higher (considering 90% for the final exam and 10% for the weekly quizzes). Resit & Repair -------------- - Each year, you have two opportunities to take the exam: the regular exam in the same quarter and a resit in the subsequent quarter. This follows the exam regulations, which also define the validity duration of the exam results. - If you do not complete the lab assignments on time, there will be an extra lab session to finish the assignments. Online Quizzes -------------- To motivate you to study consistently throughout the quarter, an online test system via Brightspace is set up with up to 5 quizzes, each containing 4 multiple-choice questions. The questions are made available online in sets of 4 over a period of 5 weeks, covering material from the preceding weeks. These weekly quizzes contribute 10% to the final grade. Note that this online test result will only count for the first upcoming exam in the same quarter and will not count for the resit or for exams in subsequent years. Study Groups ------------ Interested and motivated students have the opportunity to interact more with the lecturer through study groups, which meet 5 times over 5 weeks. Registration is required to participate, as the number of seats is limited. A task for students in these study groups is to create multiple-choice questions that may be used in the upcoming online quizzes. After discussion and vetting by the lecturer, each group will select up to 4 questions for the weekly quizzes. Each student in the group will receive a pass for the questions selected in the meeting. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Book and lecture slides, excluding old exams and tests.	
Enrolment / Application	Courses of the bachelor CSE are only open to students with an active and valid registration for the B-CSE programme, Minor Electronics for Robotics, EEMCS Exchange or an EEMCS bridging programme (CS, CESE, DSAIT) in accordance with article 13 of the Implementation Regulation in the Onderwijs en Examenregeling / Teaching and Exam Regulations. Registration for this course takes place via MyTUDelft, separately from the exam registration. For more information, see: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/registration-courses-and-examinations/	
Co-Instructor	Ing. A.M.J. Slats	

EE3115TU	Digital Communication Systems	4
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0 lecture (10 times) - 2/0/0/0 instruction (4 times)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Mathematics in general. Some basic knowledge on linear systems and signals, the Fourier transform and stochastic processes, will be an advantage.	
Summary	Transmission and distribution of information by means of telecommunication techniques form the backbone of our modern society. According to a binding international treaty among more than 180 countries, "telecommunication" covers every transfer, transmission or reception of signals, tokens, sounds, images or other data of any kind (including ideas), by means of electromagnetic systems, e.g. via radio-electric or optical means. The course Digital Communication Systems provides mathematical models and methods to describe and analyse communication systems for the transmission of signals that carry information in digital form.	
Course Contents	In this course, mathematical models and methods are given to describe and evaluate communication systems for the transmission of digital signals: - basics of signals and systems: * description of signals and decibel notation, * time-domain versus frequency-domain: LTI-systems, convolution, impulse response, Fourier-transformation, transfer function, filtering and definition of bandwidth, - propagation loss in guided (cable) and unguided (wireless) media, - noise and system noise calculations (link-budget), - signal sampling, analogue pulse modulation (PAM, PCM, Delta-modulation), - digital waveforms in baseband, power spectral density and spectral efficiency, - line codes, - bandpass signals and -systems: complex baseband description, modulation, demodulation, generic transmitter and receiver concepts: super heterodyne and direct conversion receiver, - digital modulation techniques (OOK, FSK, BPSK, QPSK, MSK, QAM), - matched filtering, - bit error probability after detection of digital signals.	
Study Goals	You have gained insight in the basic concepts of signal processing for telecommunications, as described under Course Contents, especially those related to the transmission of digital signals, and is able to apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures and working lectures. Recorded lectures will be made available via Collegerama. To support studying of the material of this course, sets of exercises are made available for each lecture and each working lecture via Brightspace. These exercises with answers will help you in mastering the course material.	
Literature and Study Materials	Couch, L.W., Digital and Analog Communication Systems, 8th edition, ISBN 978-0-273-77421-1, Prentice Hall, 2013.	
Assessment	The course is completed with a 3 hour written exam covering the full course content at the end of the 1st or 2nd quarter (resit). The final grade of Digital Communication Systems is based solely on the grade obtained for the exam, which is is rounded to the nearest half. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	The exam Digital Communication Systems is a closed-book exam. During the exam, you are allowed to use the following materials: - the overview Tables and Figures to be used at the exam Digital Communication Systems (EE3315TU), as made available on Brightspace, - an original own handwritten equation sheet: 1 page A4, written on both sides, (not type-written, no copy and it should not contain worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks		

EE3120TU	Guided and Wireless EM Transfer	5
Responsible Instructor	Dr. S.N. Haider	
Contact Hours / Week x/x/x/x	0/5/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Complex calculus, vector algebra, integration and differentiation.	
Course Contents	Course Contents - Basic electromagnetic concepts Traveling waves Electrostatics Magnetostatics Time-varying electromagnetic fields Plane waves and Polarization - Transmission lines Theory of transmission lines Characteristic parameters of transmission lines Generator and load mismatch Examples - Antennas Antenna characteristic parameters Short and half-wave dipoles Antenna links Antenna arrays Examples	
Study Goals	By the end of this course, students will be able to: - explain the basic concepts of electromagnetism and describe the properties of electromagnetic waves - list and define the antenna and transmission lines characteristic parameters - solve transmission lines problems and design matching networks - solve antenna links and power budget problems, derive and analyze the expression of the array factor for antenna arrays	
Education Method	- Lectures through PowerPoint slide projection - Practicing sessions with exercises - Solution of practical examples with brightspace and slides	
Literature and Study Materials	- PowerPoint slide in Brightspace - Example exercises with solutions and old exams with solutions in Brightspace	
Assessment	To successfully complete the course, you will need to score at least 5.8/10 (rounded to 6) as overall grade, based on the three tests below, and at least 5/10in each of the three tests. 1. Assignments on transmission lines counts 25% of final grade 2. Assignments on antennas counts 25% of final grade 3. Final written exam on theory counts 50% of final grade All tests are mandatory. Assessment criteria for the tests are detailed in the course. Permitted Materials during Tests - An equation sheet will be provided during the tests - Calculator (handling complex number) is needed If a students cannot attend or fails one or two tests, at the resit he/she needs only to redo the missing parts. Note: If written exams are not possible due to Covid-19 measures, the written test will be replaced by online oral exams.	

EE3125TU	Advanced Electronics for Robotics	5
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Modern robots make extensive use of many sensors and actuators and rely for their power on high-performance batteries that need proper care for safety and longevity reasons. Often, especially for space systems and swarm robots sent to harsh environments, the robots will be equipped with solar panels. The quality of operation of these robots strongly depends on the quality of the sensor readout systems and the power efficiency of the electronics that drive the actuators and convert the solar energy to energy stored in the batteries. For an optimal design of these robots, every member of the multi-disciplinary design team must be part of design discussions, bringing in their expertise domain and understanding the essence of the design problems and opportunities in the other disciplines. This course intends to enable the students to function like this for Analog Circuit Design and Power Systems Design by introducing a systematic design methodology for Analog Electronic Circuits and Power Electronics. The principles of power conversion are taught considering solar-panel/battery charging systems and motor drivers. The systematic analog circuit design principles are applied to negative feedback amplifiers for sensor applications. Still, they apply to other electronics circuits, like oscillators, filters, and other feedback systems like power converters. The course intends to provide insight to the students into how these systems can be optimally designed by using the proper models at the right time, applying for the correct order in the design, and what design rules exist and how they are applied. It introduces a method to systematically separate a complex design problem into less complicated orthogonal sub-problems that are easier to solve, resulting in the optimal global solution. This insight will enable the students to understand circuit and system designs created by expert designers and critically evaluate the performance and correctness of complex structures. It will allow the students to specify circuit and system designs correctly and contribute effectively to interdisciplinary discussions on design decisions with other experts. The course treats: For analog electronics, the lectures focus on a systematic way to design a negative feedback amplifier with an operational amplifier used as controller. Attention is paid to noise, bandwidth and drive capability requirements for the operational amplifier and the feedback network. A synthesis model for negative feedback amplifiers will be discussed that includes the influence of the source and load impedance. A bandwidth analysis method will be discussed that can be used to find the performance requirements for the operational amplifier and how to implement frequency compensation using the Phantom Zero method to obtain the required frequency characteristics. For power systems, the principle of operation, analytical modelling and design of basic dc-dc power converters are presented. Also, the particularities of the solar-panels, such as the Maximum Power Point Tracking (MPPT), and battery charging principles and management systems and motor drivers. The influence in the selection of the control methods for the power electronics will be investigated. All this knowledge will be combined in order to derive a suited operational solution for the application at hand. The study will be verified using computational simulation tools. During the lectures many practical demonstrations will be given.	
Study Goals	After the lectures, the student: is be able to select a correct topology for a negative feedback amplifier matching with the sensor specifications is able to identify flaws in an amplifier based on its Transmission-1 matrix is able to use the nullor to ease amplifier synthesis is able to use the asymptotic gain model is be able to do a noise analysis for an amplifier is able to do a quick bandwidth prediction for an amplifier is able to evaluate the frequency performance of an amplifier is able to apply Phantom Zero frequency compensation is able to use the proper device and system models at every design phase and avoid unnecessary complexity knows the optimal design order is able to see the structure in a complex amplifier design and identify the potential flaws understands the particularities and requirements of renewable energy source based on photovoltaics, battery charging systems and motor drivers is be able to identify, analyse, control, and specify appropriate power electronics and their respective circuit components to understand power electronics circuit design trade-offs: Power Efficiency vs. Weight vs. Size vs. Cost. to verify the designs using computational aid tools suited for circuit analysis.	
Education Method	Lectures, practical demonstrations during the lectures, homework assignment, problem solving sessions and computational exercise	
Assessment	The assessment divided into two parts. The first part covers the topics of Power Electronics, the second part covers the topics of Analog Electronics. The weighted sum (60%/40%) of the results of both parts is used to compose the grade.	
Co-Instructor	A. Shekhar	

EE3130TU	Marsrover project	5
Responsible Instructor	Dr.ing. M. Shahraki Moghaddam	
Responsible Instructor	C. Galuzzi	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The Mars Rover is based on a simple vehicle that is driven by several electric stepper motors which are controlled by an onboard FPGA. In this project a group of 4 students have the task to develop a Mars Rover. The students have the task to feed the robot with solar energy and to enhance the robot with sensing and localization abilities such that the robot is capable to perform certain predefined tasks. In order to have control over the movement and the sensors the FPGA has to be programmed in VHDL. All the basic knowledge and skills needed for this project are provided by the courses in the minor.	
Study Goals	Applying the knowledge and skills obtained during the courses in the minor	
Education Method	Project in groups of 4 students	
Assessment	Delivery of a functional Mars Rover	
Co-Instructor	Dr.ing. M. Shahraki Moghaddam	
Co-Instructor	C. Galuzzi	

ET3033TU	Circuit Analysis	3
Responsible Instructor	Prof.dr.ir. L. Abelmann	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	The course requires familiarity with handling the complex numbers algebra.	
Course Contents	Electrical circuits fundamentals / basic concepts and components: current and charge, voltage and current sources, electrical circuits, circuit symbols, direct current, alternating current, resistors capacitors and inductors, Ohm's law, Kirchhoff's laws, power dissipation in resistors, series and parallel connections. Measurement of voltages and currents: common wave forms, measuring techniques and equipment. Resistance and DC Circuits: Thévenin and Norton theorems, superposition, nodal analysis, mesh analysis, (solving) circuit equations. Capacitance and electric fields: capacitors and capacitance, electric field strength and flux density, series and parallel connected capacitors, capacitor I/V relationship, sinusoidal voltage and current, energy stored in a capacitor. Inductance and Magnetic Fields: electromagnetism, reluctance, inductance, self-inductance, series and parallel connections, inductor I/V relationship, sinusoidal voltage and current, energy stored in an inductor, mutual inductance, transformers. Alternating voltages and currents: resistance vs. reactance, impedance, phasor diagrams, complex notation. Power in AC circuits: power dissipation in resistive circuits, power in capacitors and inductors, power in circuits with resistance and reactance, active and reactive power, power factor correction, three phase systems, power measurement. Frequency characteristics of AC circuits: two-port networks, the decibel, frequency response, filter networks, Bode diagrams, RLC circuits and resonance, stray capacitance and inductance. Transient behaviour: Charging of capacitors and energising of inductors, discharging and de-energising, first order systems, generalised response, second order systems, higher order systems.	
Study Goals	The course aims at building up insight and analysis instruments in the area of linear electrical circuits and components. It will also establish a solid foundation for understanding, analysing and synthesising electronic circuits using non-linear active components in both the analogue and digital domains.	
Education Method	Interactive lecturing as a preparation for more detailed self-study and engagement with the content. Lectures are not necessarily completely covering all topics.	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276. Material in the Brightspace learning environment.	
Assessment	Homeworks + written exam In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Permitted Materials during Tests	The exam is a closed book exam. However, a single-sided self-handwritten A4 as a cheat sheet is permitted. Only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines.	
Judgement	The final grade depends for 10% on the homework and for 90% on the written exam.	
maximum aantal deelnemers	100	

ET3604LR	Electronic Circuits	3
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	Major topics: Electronics in information processing (sensor readout) Design principles and trajectory Signal processing functions Electronic design analysis Negative feedback amplifiers Operational amplifiers Components and networks Semiconductor diodes Power supplies and voltage regulators Semiconductor transistors and IC technology (basic) Oscillators DC motor basics	
Course Contents	This course is intended to provide gain insight, in a structured way, in the function of electronics as information-processing technology. The way of modelling will be discussed in relation to the physical characteristics of electronic components, circuits and systems. The course will provide the students with the skills to interpret circuit diagrams and measurement results and formulate specifications for electronic circuits that enable expert electronics designers to take proper design decisions when requested to design a high-performance application specific circuit. The students will learn the terminology that enables them to understand the explanation for these decisions by the experts (and potentially critically review them with respect to system level decisions). During the course many lecture demonstrations will be given, where the students both can observe the practical behavior of the test object and see what type of measurement equipment is used and how it is operated.	
Study Goals	After this course, the student -Is able to interpret electronic circuit diagrams and draw them according to standards -Is able to recognize and select electronic components -Is able to interpret and create Bode plots that describe the frequency behavior of an electronic circuit -Is able to analyze and design basic frequency filters -Is able to explain the basic behavior of semiconductor devices, like bipolar transistors, field effect transistors, diodes and special devices like thyristors. -Is able to determine the proper readout conditions for a sensor -Is able to determine the proper drive conditions for an actuator -Is able to interpret basic diode circuits like rectifiers -Is able to explain the basics of negative feedback systems and able to calculate and interpret the essential parameters -Is able to design a negative feedback amplifier using an Operational Amplifier (OpAmp) as active circuit that optimally meets the requirements -Is able to explain the basics of power circuits -Is able to name the different oscillator classes and explain the working principle the oscillators in each class -Is able to determine the class of a given oscillator -Is able to name the specific good and bad properties of each oscillator class -Is able to communicate with experts in electronic design and translate system specifications to specifications for the electronics in such a way that the expert can create an optimal design	
Education Method	Lecture. There is a mandatory practical with its own course code: ET3604LRP	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276 (older versions can very well be used for this course)	
Assessment	The exam is multiple-choice. During the lecture period, there will be one bonus test on oscillators. A pass grade gives a bonus point for the exam. The bonus point remains valid for the re-sit exam. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

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